ACL Milestone 1 Report

Formula 1 Podium Prediction Companion

Data engineering, pre-race feature engineering, predictive modelling and explainability for podium prediction on the Formula 1 World Championship dataset (1950-2024)

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Accompanying notebook	acl-ms1.ipynb (completed saved run; current audit and complete figure appendix follow section 12)

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1. Executive summary

We turned the 14 relational Formula 1 tables into a validated modelling table with exactly one row per driver per race (26,668 rows, 1950-2024), and predicted whether a driver finishes on the podium (`positionOrder <= 3`) with pre-race inputs and the historical grid convention in section 3.3.

- **Cleaning.** All `\N` sentinels were converted to missing values, dates and durations were typed, 85 shared-drive groups (91 extra rows) were consolidated to one row each, and all primary keys (unique, non-null) and 23 foreign-key relationships reconcile with zero orphans. Raw tables stay unchanged; cleaning works on copies.
- **Q1.** Many circuits tie at a 100% front-row-to-podium rate in both eras (25 before 2014, 21 from 2014), so ranking is driven by sample size; 21 of the 24 circuits present in both eras change rank. Grid and qualifying fronts rows differ in 6.9% of races but the conversion rate hardly changes (91.5% vs 91.1% of races).
- **Q2.** After giving home and away drivers the same grid-position distribution, the home podium rate is 13.84% versus 14.14% away (-0.30 percentage points): no home-country advantage is visible once starting position is controlled.
- **Q3.** Highest mechanical-retirement rate: Caterham (31.25%) in 2014-2021 and Alfa Romeo (12.64%) in 2022-2024; rates fall for most constructors in the later period.
- **Models.** Logistic regression (baseline), random forest and a two-hidden-layer neural network were trained on the same rows and the same 14 pre-race features. On the unseen 2022-2024 test seasons: random forest PR-AUC 0.729 / ROC-AUC 0.940 / F1 0.686; neural network 0.694 / 0.936 / 0.665; logistic regression 0.691 / 0.933 / 0.652 (no-skill PR-AUC is the 15.0% podium rate).
- **What matters.** Starting grid dominates every model (removing it costs 0.085-0.109 validation PR-AUC); standings and recent form add smaller amounts. Explanations describe what the models rely on and are **not** causal effects.

2. Task, data and pre-race cutoff

Data. Formula 1 World Championship 1950-2024 (Kaggle / Ergast), 14 relational CSV tables: circuits, constructors, constructor_results, constructor_standings, drivers, driver_standings, races, results, qualifying, lap_times, pit_stops, sprint_results, seasons and status.

Unit of analysis and target. One driver entering one race. Target `podium = 1` when `positionOrder <= 3` (podium rate 12.7% over all 26,668 rows). A binary problem; the exact finishing position is out of scope.

Prediction cutoff - after qualifying, before the race. Current-race results, status, points, laps, pit stops and post-race standings are future information and are never used as features. Round-r championship standings are recorded *after* round r, so the model uses the standing from the previous round of the same season (for round 1, the previous season's final standing). All history features (form, experience) are shifted so the current race is excluded.

3. Exploratory data analysis and data cleaning

3.1 EDA before cleaning

Each table was profiled (rows, columns, types, missing values including the literal `\N`), primary keys were tested for uniqueness and non-null values, and foreign keys were tested for orphans. The initial profile found no duplicate or null primary keys and no orphan foreign keys; the problems were placeholders, text-typed dates and durations, and repeated driver-race pairs in *results*.

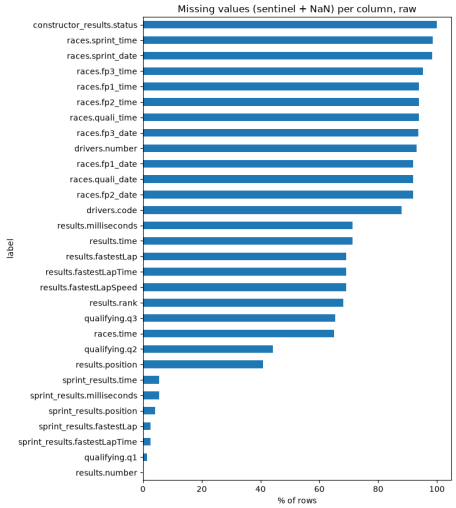


Figure 1. Missing-value shares (literal \N plus NaN) across the raw tables. Source: notebook cell 17. Results timing fields show extensive historical missingness; later cleaning decisions are audited separately.

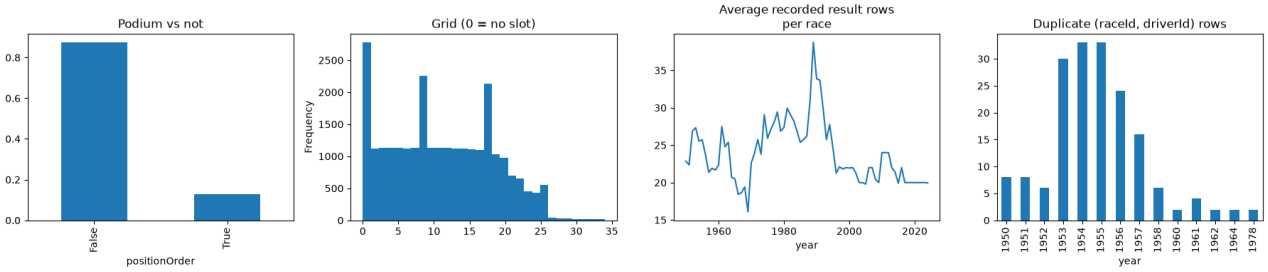


Figure 2. Raw results: podium balance, recorded grid distribution, mean rows per race, and repeated driver-race rows by season. Source: notebook cell 19.

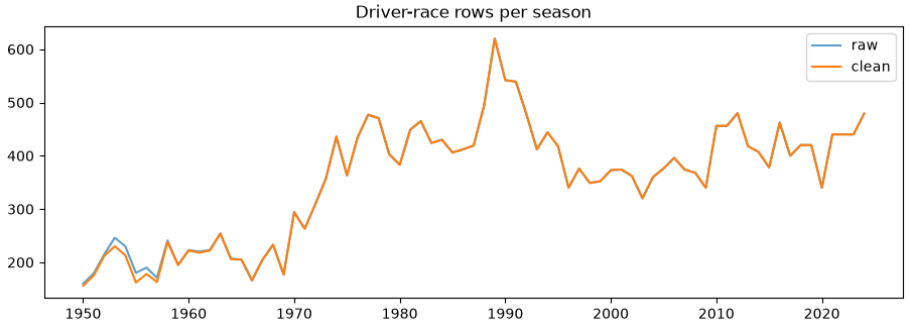


Figure 3. Raw car-entry records versus consolidated driver-race rows per season. Source: notebook cell 326. Historical multi-entry records explain the difference between the two lines.

3.2 Sentinel handling and semantic typing

- **Sentinels.** \N was read as a recognised missing value in every table (`na_values=['\N']`); for each column the count of \N before equals the count of NaN after (reconciliation table in the notebook). Where blanks carry meaning (constructor_results status) they were given an explicit label *ND* (no disqualification flag recorded) instead of an invented value.
- **Dates.** races.date and drivers.dob became datetime; the 11 session schedule columns of *races* (missing for all of 1950-2004, and before 2021 for practice/qualifying) were audited, the 88 gaps in 2021 recovered from official schedules, and then the columns were excluded from the all-years pipeline.
- **Durations.** lap_times.time, pit_stops.duration and qualifying q1/q2/q3 became numeric seconds (checked against the milliseconds columns); pit_stops.time became a clock offset; results.time and fastestLapTime were converted and cross-checked. One genuine mismatch was found among 1,128 comparable results (Russell, Belgium 2024: 4,797,566 ms vs official 4,797,040 ms) and corrected with the official result as evidence.
- **Position redundancy.** Numeric comparisons confirm redundancy (results r=0.999982; sprint r=1.000000). Both cleaned tables drop position. Keep positionText for outcome codes and positionOrder for the podium target. No negative position category encoding is applied; pair counts are in Appendix A.
- **Other table fixes (all evidence-based).** 'USA' to 'United States' in 11 circuits; one trailing-space/variant nationality (Colapinto) standardised; seven redundant and two missing constructor_results rows resolved with official evidence (12,625 to 12,620 rows); 10 qualifying constructor contradictions and 32 car-number disagreements corrected; 41 verified completed-lap totals, two 2009 driver-ID assignments and the 2001 Alesi/Trulli timing mix-up corrected; unresolved cases (17 lap-timing sequences) are kept visible and held out of timing analysis rather than invented.

Table	Before	After	Note
results	26,759 rows; 122,887 sentinels (18 original columns)	26,668 rows; 111,382 missing cells	Shared drives consolidated; position dropped; all six timing fields retained for auditing; excluded from model inputs
drivers	861 x 9	861 x 7	number (802 missing) and code (757 missing) dropped; nothing invented
races	1,125 x 18	1,125 x 7	Schedule columns excluded after audit (structural absence)
qualifying	q1/q2/q3 as text	numeric seconds	Remaining NaN: q1 156 (1.5%), q2 4,647 (44.3%), q3 6,865 (65.4%) - format/era, not errors
constructor_results	12,625	12,620	7 redundant removed, 2 verified rows reconstructed

3.3 Grain: exactly one row per (raceId, driverId)

Raw *results* has **85** driver-race groups (176 rows; **91** extra rows) with a repeated (raceId, driverId). All lie in 1950-1978, mostly 1953-1957 (shared drives where drivers swapped cars mid-race, plus a few withdrawal/non-start duplicates and Indianapolis 500 entries).

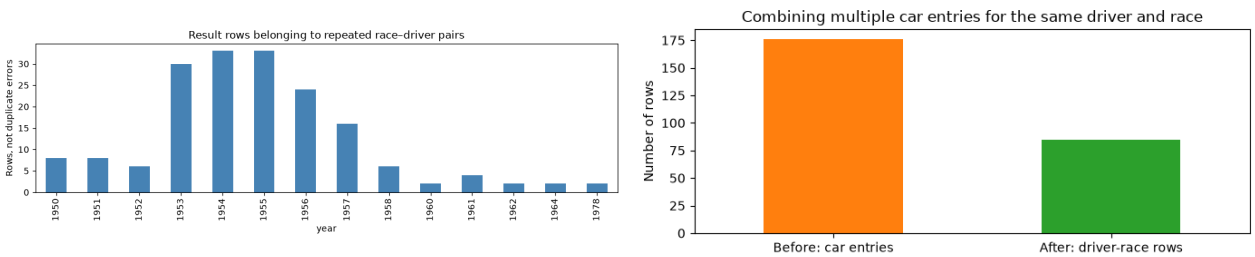


Figure 4. Left: raw rows in repeated driver-race groups by season (cell 126). Right: 176 raw entries become 85 unique driver-race records after consolidation (cell 320).

Rule applied: keep the entry with the **best (lowest) positionOrder** (ties: smallest resultId), keep that entry's other fields, and **sum the points** of all entries; the number of original entries is stored in *n_raw_rows*. **Why:** the prediction target is whether the driver finished on the podium, and the best recorded finish represents that outcome, so a driver who shared a podium car is credited with it; the other fields stay from one real car entry

rather than mixing details from two cars. We compared three candidate rules (lowest resultId, best positionOrder, smallest positive grid); lowest resultId keeps a grid slot that is not the best declared slot in 27 groups, best positionOrder in 56, smallest grid in 0, but only best positionOrder aligns the label with the credited result.

Documented limitation. Because the kept entry is chosen by finishing order, its grid is an outcome-dependent historical convention, not a reconstruction of the driver's original starting car (81 groups have different grids; the best finish retains the smallest grid in 25 of them). All 85 groups are in 1950-1978 (about 0.35% of training rows) and none are in validation or test. Assertions confirm that points, entry counts and best positionOrder are preserved for every group and that (raceId, driverId) is now unique.

3.4 Key validation

For all 14 cleaned tables the primary keys are unique and non-null (composite keys for lap_times (raceId, driverId, lap) and pit_stops (raceId, driverId, stop)); all 23 foreign-key relationships (e.g. results.raceId to races, results.driverId to drivers, results.constructorId to constructors, results.statusId to status, qualifying and sprint_results references, lap_times/pit_stops/standings references, constructor_results references) have **0 orphans before and after cleaning**. A final pass/fail gate verifies the allowed missingness: qualifying q1/q2/q3 and six results timing fields, documented codes, and unchanged raw tables/CSVs (fingerprints/hashes). The current null counts and position comparisons are in Appendix A.

3.5 EDA after cleaning - impact of the cleaning

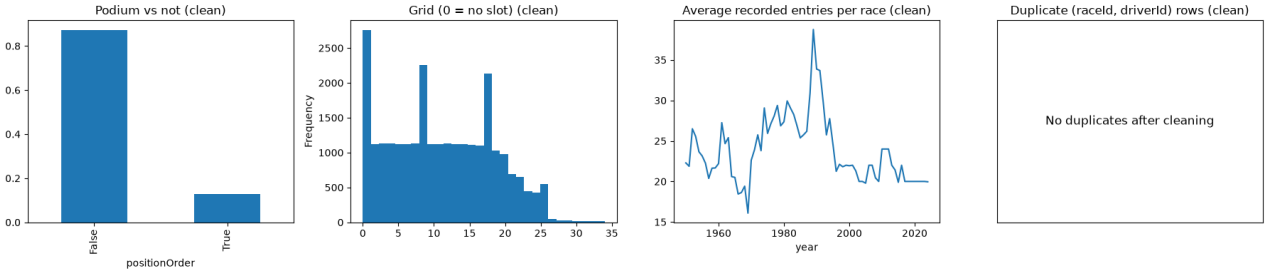


Figure 5. Clean results: podium balance, recorded grid distribution, mean driver-race rows per race, and a uniqueness check confirming no repeated driver-race keys. Source: notebook cell 418.

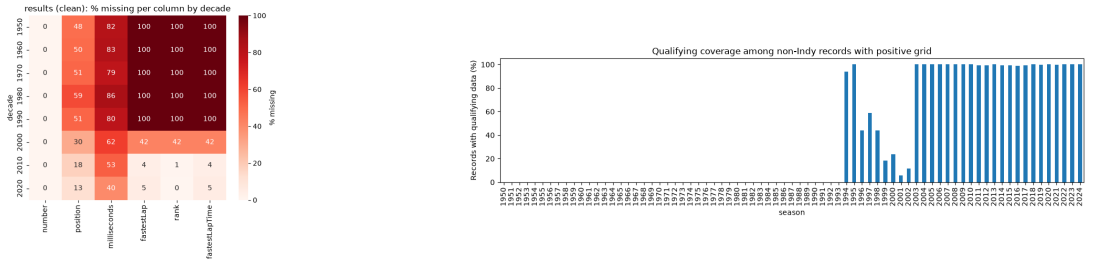


Figure 6. Left: missingness by decade at the section 3.11.3 audit, before position removal (cell 325). Right: qualifying coverage among positive-grid records (cell 335). Timing columns are retained; their nulls are not filled.

Grid zero (1,635 rows, none finished on the podium) is a recorded special code - mostly 'Did not qualify' (1,012), 'Did not prequalify' (330) and 'Withdrew' (163) - so it is retained and handled explicitly in modelling (section 5). The modelling population keeps all 26,668 cleaned rows; Questions 2 and 3 apply a separate recorded-activity screen described in section 4.

4. Data-engineering questions

All three answers use multi-hop joins on the cleaned tables (results, races, circuits, qualifying, drivers, constructors, status, lap_times, pit_stops). Grain is one driver-race row for every question; the pre-race cutoff matters only for Q1 (grid and qualifying are known before the start) because Q2 and Q3 are retrospective analyses of outcomes.

4.1 Question 1 - Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Method. Join results, races and circuits. A race counts as a success when **at least one** driver starting from grid slots 1 or 2 finishes in positions 1-3 (even if both do); the rate is successful races / eligible races, computed separately before 2014 and from 2014 onward. Grid (not qualifying position) defines the front row because the grid is where the car actually started after penalties. A race is eligible only if both slots 1 and 2 are recorded: 1,103 of 1,125 races (1,124 had at least one front-row record). Ties share a rank.

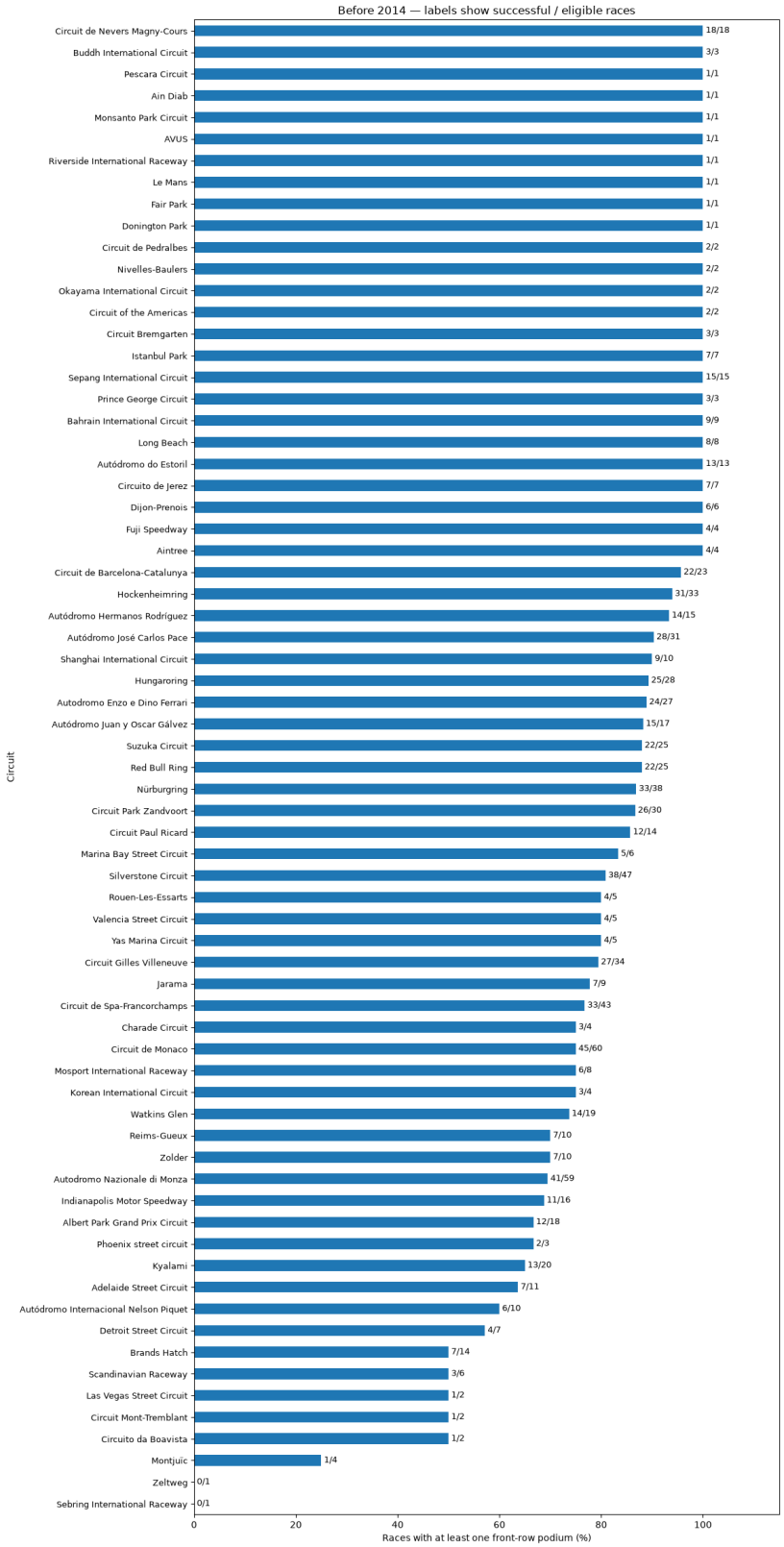


Figure 7. Question 1 - before 2014: share of races in which at least one front-row starter reached the podium, by circuit (labels = successful / eligible races).

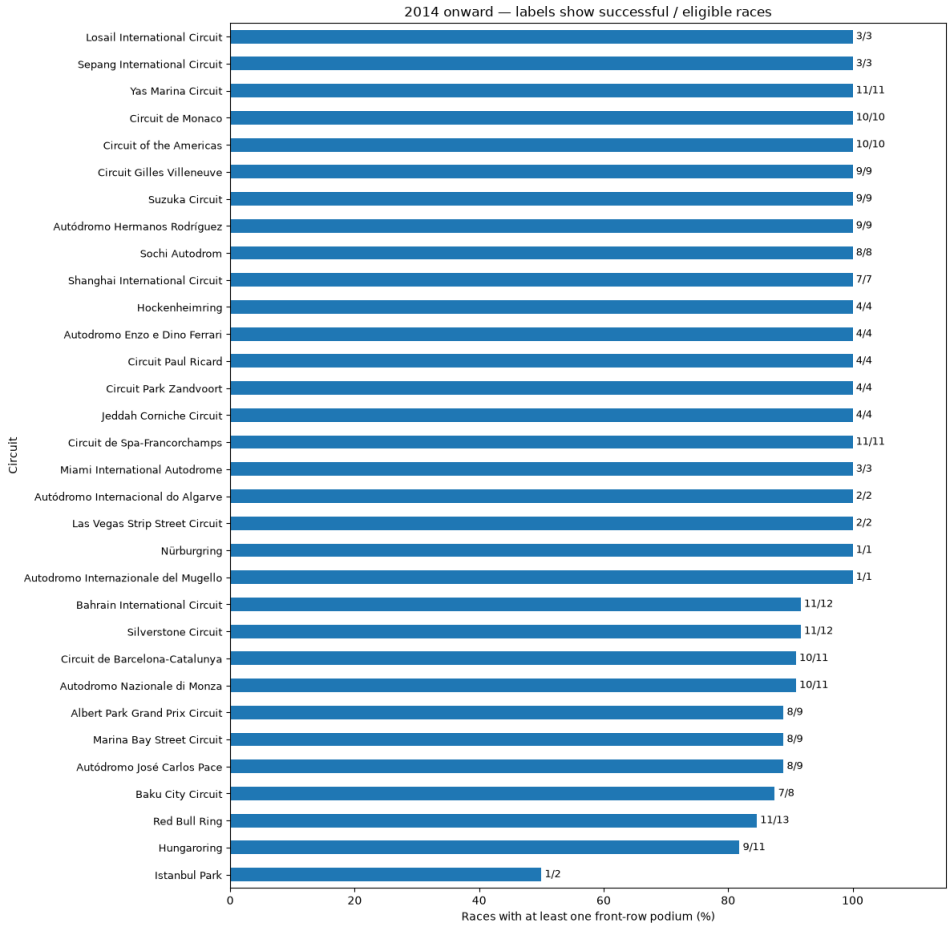


Figure 8. Question 1 - 2014 onward: same measure; many circuits sit at 100% with different sample sizes.

Answer. Before 2014, 25 circuits tie for the top rate of 100%; the largest samples are Magny-Cours (18/18), Sepang (15/15), Estoril (13/13), Bahrain (9/9) and Long Beach (8/8). From 2014, 21 circuits tie at 100%: Spa-Francorchamps (11/11), Yas Marina (11/11), Monaco (10/10), Circuit of the Americas (10/10), Gilles Villeneuve (9/9), Suzuka, Mexico City (9/9 each), and others. **The ranking changes:** of the 24 circuits that appear in both eras, 21 have a different within-period rank. Because so many circuits are tied at 100%, the numerator and denominator (not the rank alone) must be read: a 100% rate on 4 races is weak evidence compared with 100% on 18.

Qualifying position versus grid

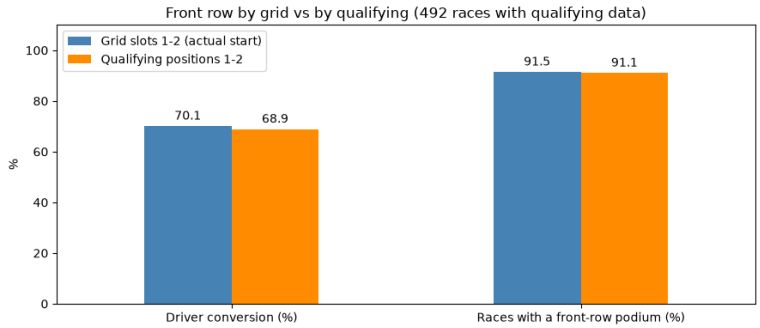


Figure 9. Front-row conversion measured with grid slots 1-2 versus qualifying positions 1-2 on the same 492 races (1994-2024).

- In **34 of 492** races (6.9%) the grid front row differs from the qualifying front row; **36 drivers** qualified P1/P2 but started elsewhere (e.g. Hamilton, Spain 2012: qualified P1, started 24th after an exclusion; Verstappen 2022 engine penalty).
- Driver-level conversion is 70.1% (grid) versus 68.9% (qualifying); race-level 91.5% versus 91.1%; only 2 races (Mexico 2019, Sao Paulo 2022) change outcome.

- **Missingness/cutoff:** qualifying positions exist only from 1994, so this comparison covers 1994-2024; the main ranking uses grid and so covers all years. Both quantities are known before the race.

4.2 Question 2 - Do drivers podium more often at home, controlling for starting position?

Method. Join results, races, circuits and drivers (grain: driver-race). A record is *home* when the driver's nationality maps to the circuit's country (every nationality mapped explicitly; dual nationalities match either country). To control for starting position we compare home and away records **at the same grid slot**, then give both groups the **same grid distribution** (weights = each grid slot's share of the combined records at slots observed in both groups). Population: records with recorded race activity (positive laps, or a lap-time record, or a pit-stop record) and a positive grid: 24,068 of 26,668 rows (2,528 excluded for no recorded activity; 72 more for non-positive grid). Exclusion is not a claim that a driver did not start.

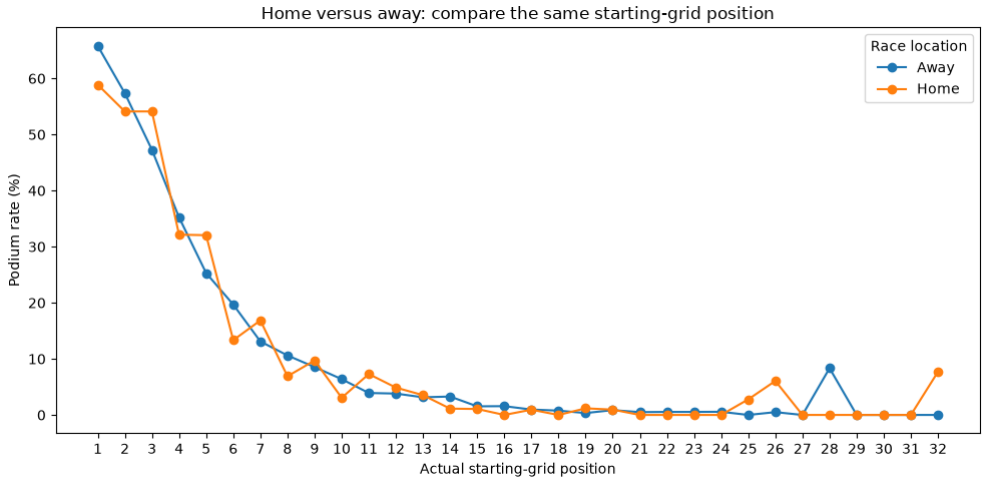


Figure 10. Question 2 - podium rate at each actual starting-grid position, home versus away.

Measure	Home	Away
Records / podiums (unadjusted)	2,224 / 266	21,844 / 3,129
Unadjusted podium rate	11.96%	14.32%
Grid-adjusted podium rate (same grid weights)	13.84%	14.14%
Home minus away (adjusted)	-0.30 percentage points	

Answer. No. Raw, home drivers podium slightly *less* often (11.96% vs 14.32%); after matching grid positions the gap shrinks to -0.30 points (13.84% vs 14.14%), i.e. no measurable home advantage in this population, and the curves overlap at almost every grid slot (differences at slots with few home records are noise). **Caveats:** this is a descriptive grid adjustment, not a significance test; driver ability, team strength and season are not controlled, so it is an association, not proof for or against a causal home effect.

4.3 Question 3 - Which constructors have the highest mechanical-retirement rate, 2014-2021 vs 2022-2024?

Method. Join results, races, constructors and status (grain: driver-race). Population: Grand Prix records with recorded activity (positive laps, or a matching lap-time or pit-stop record): 3,163 of 3,267 records in 2014-2021 (160 races) and 1,324 of 1,359 in 2022-2024 (68 races). Rate = mechanical-labelled retirements / records with recorded activity. Every observed status was mapped explicitly (mechanical, non-mechanical or finished/lapped); an unmapped label would stop the run. Constructor IDs stay separate across rebrands.

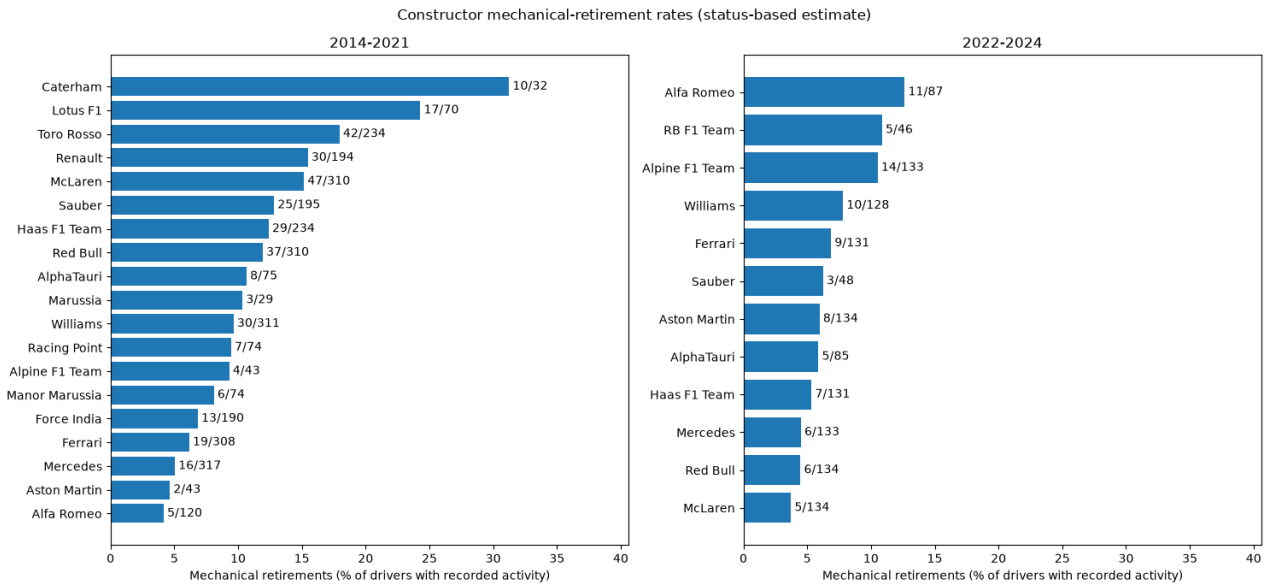


Figure 11. Question 3 - mechanical-retirement rate by constructor (labels = failures / recorded participants).

2014-2021	Rate	2022-2024	Rate
Caterham	31.25% (10/32)	Alfa Romeo	12.64% (11/87)
Lotus F1	24.29% (17/70)	RB F1 Team	10.87% (5/46)
Toro Rosso	17.95% (42/234)	Alpine F1 Team	10.53% (14/133)
Renault	15.46% (30/194)	Williams	7.81% (10/128)
McLaren	15.16% (47/310)	Ferrari	6.87% (9/131)

Answer. 2014-2021: **Caterham** (31.25%), then Lotus F1 (24.29%) and Toro Rosso (17.95%). 2022-2024: **Alfa Romeo** (12.64%), then RB F1 Team (10.87%) and Alpine (10.53%). Among the 11 constructors present in both periods, rates fall for seven (largest drops McLaren -11.43 pts, Red Bull -7.46, Haas -7.05, Sauber -6.57) and rise for Alfa Romeo (+8.48), Aston Martin (+1.32), Alpine (+1.22) and Ferrari (+0.70).

Assumptions to state openly. (i) The statuses 'Retired' and 'Damage' are counted as mechanical by a team decision although the labels alone do not prove component failure - this is broader than the brief's definition, which excludes crash/driver-caused failures. (ii) The denominator is drivers with recorded activity, not independently verified starters; a first-lap retirement with no lap or pit record is excluded. (iii) Small denominators (e.g. Caterham 32 records) make rates unstable. Sprint results are not used.

5. Feature selection and engineering

Candidate features were explored on all years (the notebook records that this exploration on all available years was confirmed acceptable by the TA; model fitting, hyper-parameters and thresholds never use test rows). Evidence came from outcome-group plots, rank correlations with the target, coverage by year, redundancy heatmaps and per-era stability, followed by controlled experiments (E1-E9).

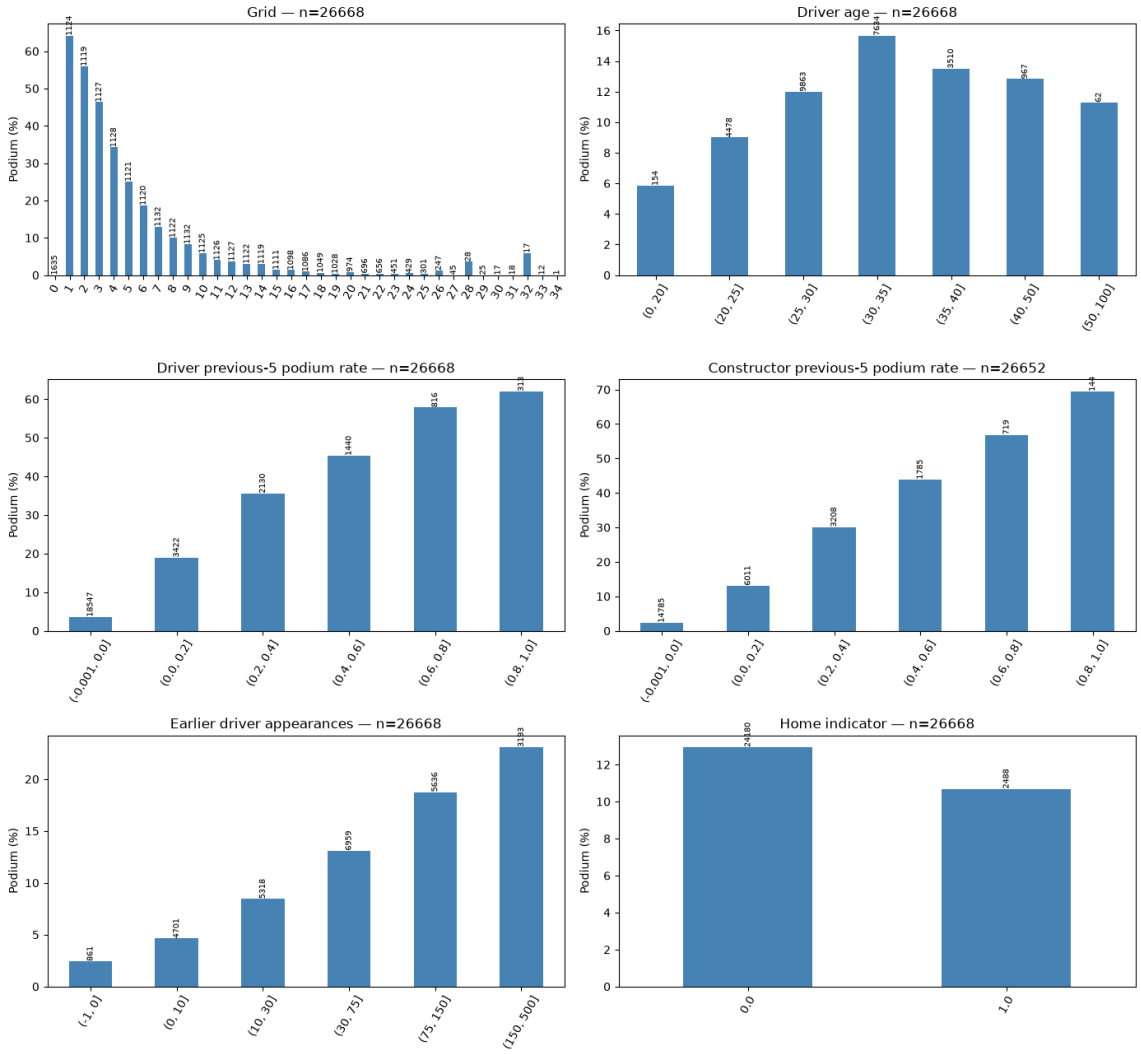


Figure 12. Podium rate by value of the main candidates (grid, age, recent driver form, recent constructor form, driver experience, home indicator) - counts above bars.

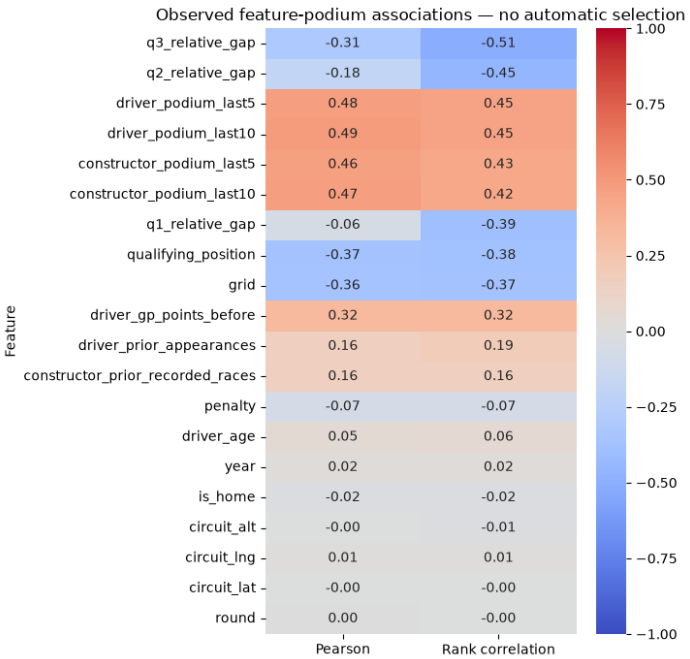


Figure 13. Pearson and rank correlations of each numeric candidate with podium (paired observed rows).

What the EDA showed. Grid has the clearest monotone relationship (podium rate 64.1% from pole, 55.9% P2, 46.4% P3, falling to under 1% from grid 19+; grid 0 never podiums). Driver form (rank correlation +0.45)

and constructor form (+0.43) rise steadily with recent success; age rises toward 30-35 then falls (non-linear despite Pearson +0.05); experience gives 2.4% podium rate at debut versus 23.1% above 150 appearances; home status is weak (-0.02). Five- and ten-race windows are almost interchangeable (Pearson 0.92 driver, 0.94 constructor). Qualifying gap features correlate strongly (Q3 rank -0.51) but only on the 14-39% of rows where they exist.

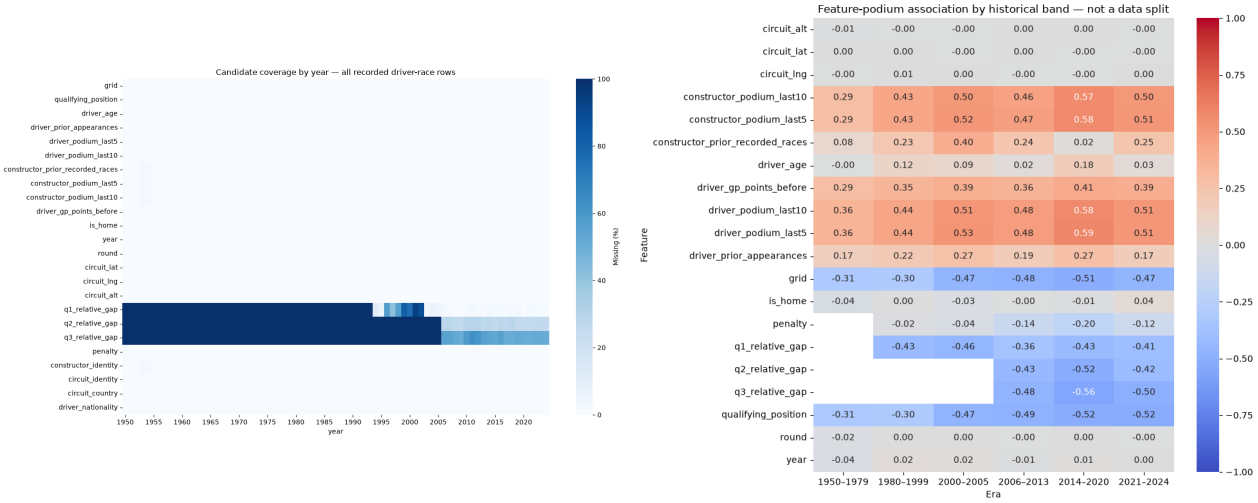


Figure 14. Left: candidate coverage by year (qualifying times missing before the 1990s, partial to 2005, formats change); right: feature-podium rank correlation by era.

5.1 Leakage-safe features built

Feature	How it is built	Why it is valid at the cutoff
Form: driver_podium_last5, constructor_podium_last5	Podium share over up to 5 earlier appearances / team races; 0 with a history count when no history	Only races dated before the current one; hand-checked on driver 1 (earlier outcomes recomputed independently) and team form shared by teammates
Standings: driver_standing_before, constructor_standing_before (+ unranked flags)	Position after the *previous round* of the same season; round 1 uses last season's final standing	Never this race's standing; verified by hand (Hamilton 2008) and by a date assertion on every row
driver_gp_points_before	Sum of the driver's GP points earlier in the same season	Earlier races only; sprint points excluded
Experience: driver_prior_appearances, constructor_prior_recorded_races	Count of earlier recorded appearances	Counts only earlier races (starts at 0, never decreases)
driver_age	(race date - birth date)/365.2425	Known from the calendar
Grid: grid, no_grid_slot	Recorded grid; a recorded 0 is placed after max raw grid (before consolidation) and flagged	Normal entries: grid known before the race. Shared-drive convention: section 3.3; all 85 groups retained.
Context: is_home, year	Nationality-country match; season	Known from the entry list and calendar

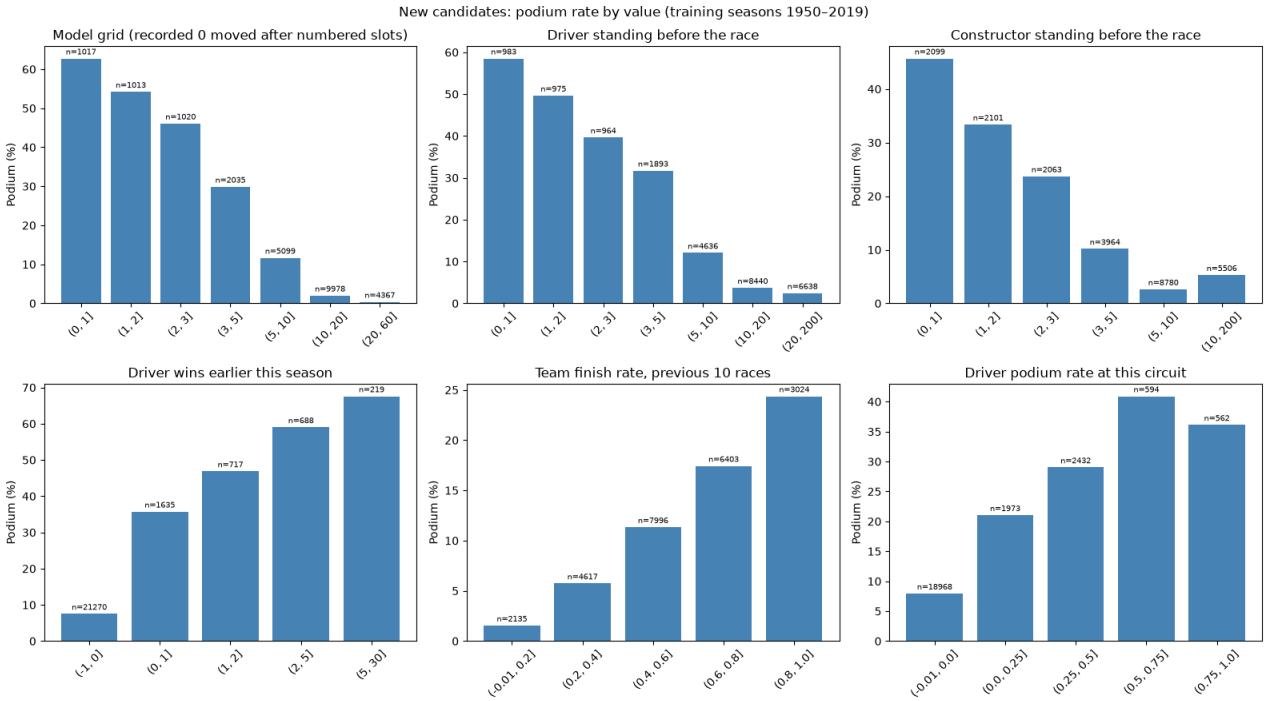


Figure 15. Podium rate by value of the additional candidates on training seasons only (grid with adjusted zero, standings, wins, team finish rate, driver circuit record).

5.2 Controlled feature experiments

Same fixed random forest each time, two pre-test checks (validation 2020-21, and an earlier check training to 2017/scoring 2018-19), three seeds each; PR-AUC is the main score. A change counts as *helps* / *hurts* above +/-0.003 average PR-AUC (a screening rule, not a significance test).

Exp.	Change	Features	Val PR-AUC	2018-19 PR-AUC	Avg change	Verdict
E1	Recorded grid + no-grid flag	10	0.7014	0.7217	-	reference
E2	Grid zero moved after numbered slots	10	0.7095	0.7236	+0.0050	Helps
E3	+ driver and constructor standings	14	0.7239	0.7318	+0.0113	Helps
E4	+ driver wins earlier this season	15	0.7297	0.7328	+0.0034	Helps
E5	+ team finish rate (10 races)	16	0.7278	0.7345	-0.0001	No real difference
E6	+ driver history at circuit	18	0.7276	0.7334	-0.0007	No real difference
E7	E6 - year, is_home, gp_points	15	0.7191	0.7312	-0.0054	Hurts
E8	Form over 10 races instead of 5	15	0.7078	0.7303	-0.0061	Hurts
E9	E3 - year, is_home	12	0.6899	0.7214	-0.0222	Hurts

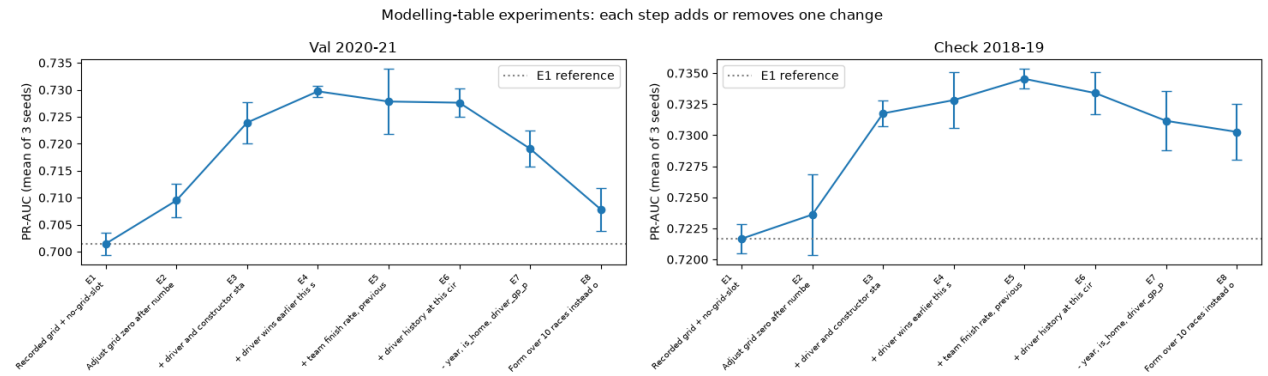


Figure 16. Modelling-table experiments: validation PR-AUC (mean of 3 seeds, bars = spread) as each change is added or removed.

Selected feature set: E3, 14 features - grid, no_grid_slot, driver_age, driver_prior_appearances, driver_podium_last5, constructor_prior_recorded_races, constructor_podium_last5, driver_gp_points_before,

is_home, year, driver_standing_before, constructor_standing_before, driver_unranked, constructor_unranked. E4 (adding season wins) also helped (+0.0034); the team chose the compact E3 set as a deliberate simplicity trade-off, did not claim E3 is more accurate, and keeps E4 as a documented extension (its benefit was only tested for the random forest). Ten-race form hurt, so five-race form was kept. Permutation importance on the fullest table (E6) showed grid far ahead (PR-AUC drop 0.337) then driver_standing_before (0.032).

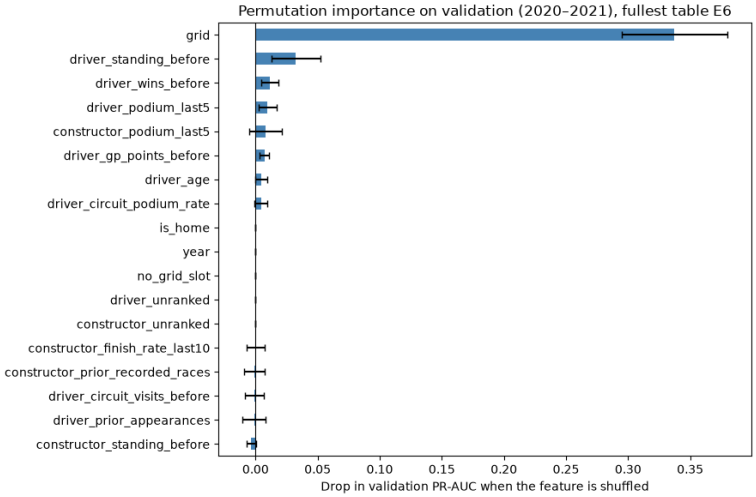


Figure 17. Permutation importance of the E6 candidates on validation (PR-AUC drop when shuffled, 20 repeats).

6. Splitting and preprocessing

6.1 Train / validation / test split

Set	Seasons	Rows	Podium rate	Use
Training	1950-2019	24,529	12.5%	Fit models and preprocessing
Validation	2020-2021	780	15.0%	Hyper-parameters, thresholds, early stopping, feature decisions
Test	2022-2024	1,359	15.0%	Final unbiased report only

Why chronological. Prediction always looks forward in time and features are built from history, so a random split would let later seasons inform earlier ones. Whole seasons (hence whole races) stay together - assertions verify that no race crosses sets and that dates strictly follow train < validation < test. The 2022 boundary also tests transfer into the new technical-regulation era, which makes the test harder; this does not imply the dates maximise performance. Test rows are excluded from fitting, learned preprocessing, hyper-parameter and threshold selection.

6.2 Effect of each preprocessing step

- **Median imputation (training rows only).** Only 64 training cells were missing (16 rows with ambiguous constructor identity in four constructor features); filled with training medians. Means/std barely move (e.g. constructor_podium_last5 mean 0.127 to 0.127) and observed values are verified unchanged.
- **StandardScaler (training rows only).** Every training feature ends at mean 0, std 1. Validation/test drift is visible after scaling - e.g. test driver_prior_appearances +1.41 std, constructor experience +1.18, gp points +1.24, year +1.85 - which measures era shift, not error.
- **Skewness.** 11 features have |skew| > 1 (e.g. driver_gp_points_before 4.97); log1p reduces it (to 0.76) but was shown only as an illustration and *not applied* because it would need its own validation comparison.
- **Encoding.** All 14 inputs are numeric, so no encoding is used; one-hot IDs would add 860 (driver), 209 (constructor) and 76 (circuit) sparse columns that the form and standing features already summarise.

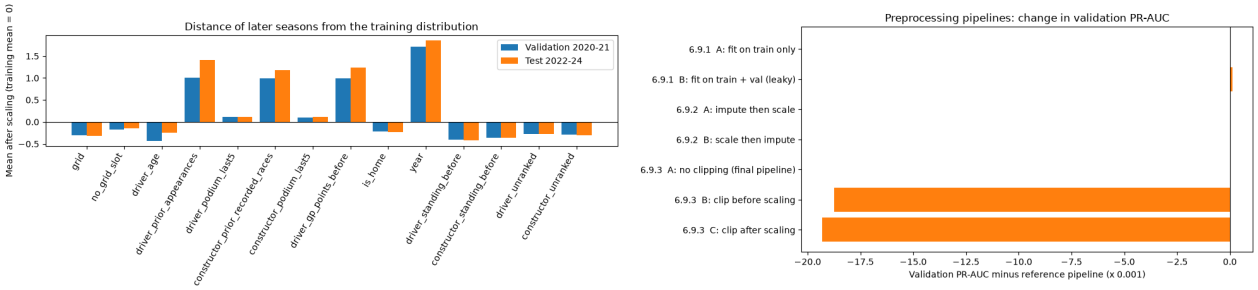


Figure 18. Left: distance of validation and test means from the training distribution after scaling. Right: preprocessing-order experiments - change in validation PR-AUC (x 0.001).

6.3 Preprocessing-order experiments

Logistic regression (C=1; scale-sensitive), same rows and features, threshold re-chosen on validation for each pipeline; only the preprocessing changes.

Experiment	Pipeline	Thr.	Val ROC-AUC	Val PR-AUC	Val F1	Val PR-AUC change
6.9.1 where to fit	A: train only (final)	0.32	0.92214	0.71407	0.6862	-
	B: train + validation (leaky)	0.29	0.92219	0.71418	0.6883	+0.0001
6.9.2 order	A: impute then scale (final)	0.32	0.92214	0.71407	0.6862	-
	B: scale then impute	0.32	0.92214	0.71407	0.6862	+0.0000
6.9.3 outlier clipping	A: no clipping (final)	0.32	0.92214	0.71407	0.6862	-
	B: clip 1-99% before scaling	0.37	0.92157	0.69531	0.6867	-0.0188
	C: clip after scaling	0.37	0.92158	0.69476	0.6867	-0.0193

Reading. Fitting on validation as well changes the score by only +0.0001 here, but it is still leakage for validation and impossible at prediction time, so train-only fitting is kept. Imputing before or after scaling gives identical scores (the median commutes with scaling; only four columns with 64 missing cells can differ). Clipping hurts PR-AUC by about 0.019 whether done before or after scaling - the long tails (e.g. many points, deep experience) carry signal - so **no clipping** is used. Final pipeline: fit on training only, impute then scale, no clipping.

7. Predictive modelling

All final models use the same rows (1950+ training, 24,529 rows), the same 14 features and the same chronological split. Hyper-parameters and the decision threshold (best validation F1) are chosen on training/validation only.

7.1 Models, how they work and their limitations

Model	How it works	Settings	Main limitations
Logistic regression (**baseline**)	Weighted sum of the scaled inputs through a sigmoid; weights show direction of association	C=10, no class weights (grid over 6 C x 2 class-weight options on validation)	Only a linear boundary in the inputs; the fall of podium chance over the first grid slots is approximated; no interactions unless built in
Random forest	Many bootstrap decision trees with random feature subsets; probability = average of leaf shares	max_depth 10, min_samples_leaf 30, max_features 0.6, 500 trees (grid of 64 settings on validation)	Cannot extrapolate (every test year is later than training years); probabilities not calibrated; fully grown default forest overfits (train PR-AUC 1.00 vs val 0.65)
Shallow neural network	Two hidden layers (128, 64 units; BatchNorm, ReLU, dropout 0.4/0.3), sigmoid output, Adam + binary cross-entropy, early stopping on validation loss (patience 8)	11,009 parameters; threshold 0.37	Harder to interpret; needs scaled inputs; sensitive to era shift; uncalibrated probabilities; non-deterministic training (3 seeds checked); early stopping uses validation so validation scores are slightly optimistic

7.2 Metrics and what is primary

Primary metric: PR-AUC - podiums are rare (about 12-15% of rows; no-skill level equals the podium rate), and PR-AUC directly measures how well real podium finishers are ranked above everyone else, which is more informative than ROC-AUC under imbalance. **Secondary:** ROC-AUC (overall separation, threshold-free) and F1 at the validation-chosen threshold (a yes/no call needs a cut-off), reported with precision and recall. Accuracy is also shown but not used for selection because 'never podium' already scores about 85-88%.

7.3 Model comparison - training, validation and unseen test

Model	Split	Thr.	ROC-AUC	PR-AUC	F1	Precision	Recall	Accuracy
Logistic regression	Train	0.32	0.9051	0.5590	0.5770	0.5248	0.6409	0.8823
	Validation	0.32	0.9222	0.7143	0.6862	0.6721	0.7009	0.9038
	Test	0.32	0.9334	0.6912	0.6523	0.5830	0.7402	0.8815
Random forest	Train	0.32	0.9325	0.6614	0.6212	0.5523	0.7098	0.8915
	Validation	0.32	0.9295	0.7247	0.6980	0.6449	0.7607	0.9013
	Test	0.32	0.9403	0.7291	0.6863	0.6015	0.7990	0.8904
Neural network	Train	0.37	0.9152	0.5909	0.5891	0.5272	0.6675	0.8833
	Validation	0.37	0.9312	0.7423	0.7012	0.6567	0.7521	0.9038
	Test	0.37	0.9364	0.6940	0.6653	0.5830	0.7745	0.8830

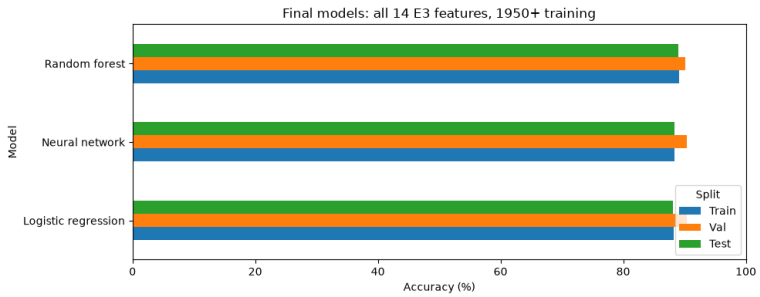


Figure 19. Final models (all 14 features, 1950+ training): accuracy on train, validation and test.

- **Validation** selects the neural network (PR-AUC 0.7423 vs 0.7247 random forest vs 0.7143 baseline), which is therefore the default inference model. The gaps are small relative to seed spread (network validation PR-AUC std about 0.005-0.011 across seeds).
- **Test** (2022-2024) ranks the random forest first (PR-AUC 0.7291, ROC-AUC 0.9403, F1 0.6863); the neural network and baseline are close (0.694, 0.691). Test scores describe performance and were not used to choose anything.
- **Does complexity earn its keep?** RF test AP exceeds LR by 0.0379; NN exceeds LR by 0.0028. On validation, NN and RF exceed LR by 0.0280 and 0.0104. The strong grid signal already captures much of the separation (test ROC-AUC > 0.93 for all three).
- **Train versus validation:** training AP is lower than validation AP. Training spans 1950-2019, with a different era mix and lower podium prevalence (12.5% vs 15.0%); scores alone do not establish why. From validation to test, NN AP falls 0.0483 and LR falls 0.0231; RF AP rises 0.0044. The era boundary is a plausible influence, not a demonstrated cause.

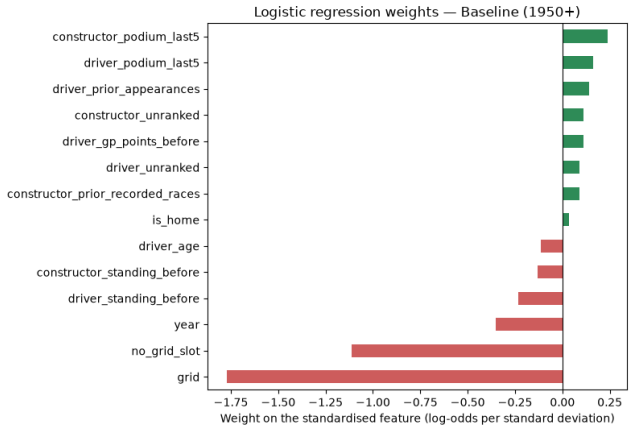


Figure 20. Logistic-regression weights (log-odds per standard deviation): grid -1.77 and no_grid_slot -1.11 dominate; associations only.

8. Experiments

8.1 Feature-group ablation (final models)

From the full 14-feature model each group was removed in turn (independent removals), the model was retrained with unchanged settings, and the threshold was fixed per model. Drop = full score minus score without the group; positive = removal hurt. The qualifying group is absent because the final feature set contains no qualifying-time columns (see the qualifying note below); grid, the post-qualifying starting position, is its own group.

Group removed (features)	NN val drop	NN test drop	RF val drop	RF test drop	LR val drop	LR test drop
Grid (2)	0.1085	0.1544	0.0849	0.1541	0.0868	0.1406
Championship standings (5)	0.0294	-0.0128	0.0230	0.0105	0.0112	-0.0046
Recent form (2)	0.0116	-0.0069	0.0035	0.0001	0.0193	0.0349
Experience and age (3)	0.0011	0.0055	-0.0051	0.0117	0.0027	-0.0040
Context: is_home, year (2)	0.0040	0.0075	0.0348	0.0063	0.0091	-0.0113

Validation PR-AUC of the full models: NN 0.7423, RF 0.7247, LR 0.7143. **Reading:** grid is by far the most important group for all three models (removal costs 0.085-0.109 validation PR-AUC and 0.14-0.15 on test). Standings matter for the network and forest on validation but their removal does not hurt on test, partly because their information overlaps with form and points. Recent form matters most for logistic regression, which cannot learn interactions. Experience/age is nearly redundant (RF even gains 0.005 on validation). Small differences come from one split and are not significance tests.

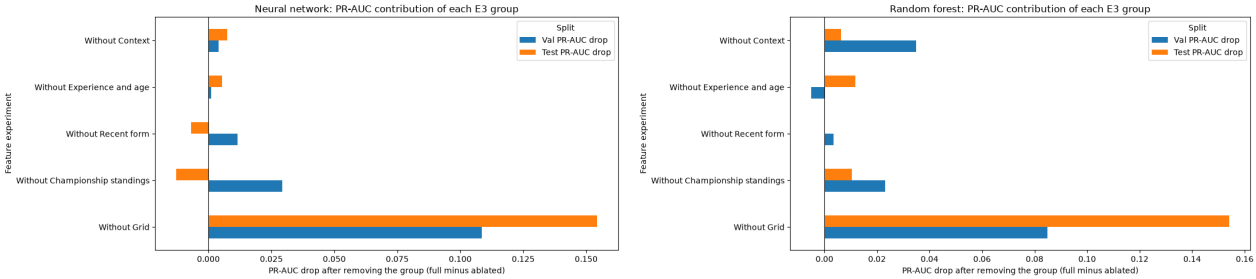


Figure 21. PR-AUC drop per removed group - neural network (left) and random forest (right).

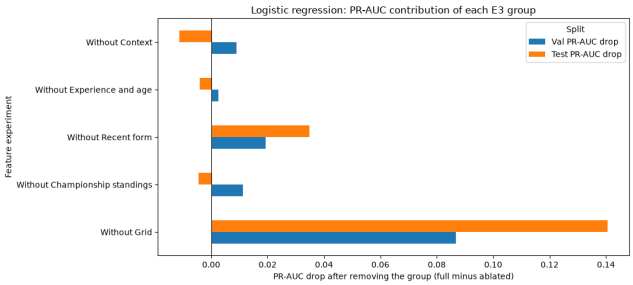


Figure 22. PR-AUC drop per removed group - logistic regression.

8.2 Training versions: all history versus modern seasons and qualifying times

To test whether recent seasons alone or qualifying lap times add information, each model was trained in four versions on identical validation and test sets: Baseline 1950+, 2003+ (when Q1 times exist for almost every driver), 2003+ with raw Q1, and 2003+ with Q1 relative to the fastest Q1 in that race.

Model	Version	Val PR-AUC	Test PR-AUC	Val ROC-AUC	Test ROC-AUC	Test F1
Neural network	Baseline 1950+	0.7423	0.6940	0.9312	0.9364	0.6653
	2003+	0.7415	0.6762	0.9290	0.9313	0.6214
	2003+ + q1	0.7322	0.6651	0.9272	0.9299	0.6295
	2003+ + q1 gap	0.7330	0.6743	0.9254	0.9305	0.6037
Random forest	Baseline 1950+	0.7247	0.7291	0.9295	0.9403	0.6863
	2003+	0.7309	0.7214	0.9293	0.9367	0.6720
	2003+ + q1	0.7192	0.7246	0.9281	0.9379	0.6706
	2003+ + q1 gap	0.7233	0.7228	0.9300	0.9382	0.6606
Logistic regression	Baseline 1950+	0.7143	0.6912	0.9222	0.9334	0.6523
	2003+	0.7083	0.7071	0.9259	0.9351	0.6437
	2003+ + q1	0.7122	0.7074	0.9257	0.9349	0.6464
	2003+ + q1 gap	0.7088	0.7067	0.9260	0.9350	0.6404

Reading. Adding raw or relative Q1 time does not improve validation PR-AUC for any model, and the 2003+ versions are not consistently better than using all history. This is why the final comparison keeps the common 1950+ / E3 setup for all three models. Random-forest tuning also mattered: the default forest reached validation PR-AUC 0.647 (train 1.000) versus 0.725 tuned. For the network, four architectures (A 128-64, B 32-16, C 256-128, D one layer of 64; three seeds each) gave mean validation PR-AUC 0.7434 / 0.7351 / 0.7420 / 0.7232 and test PR-AUC 0.6954 / 0.6982 / 0.6882 / 0.7056, so capacity changes had little effect; architecture A was kept on validation.

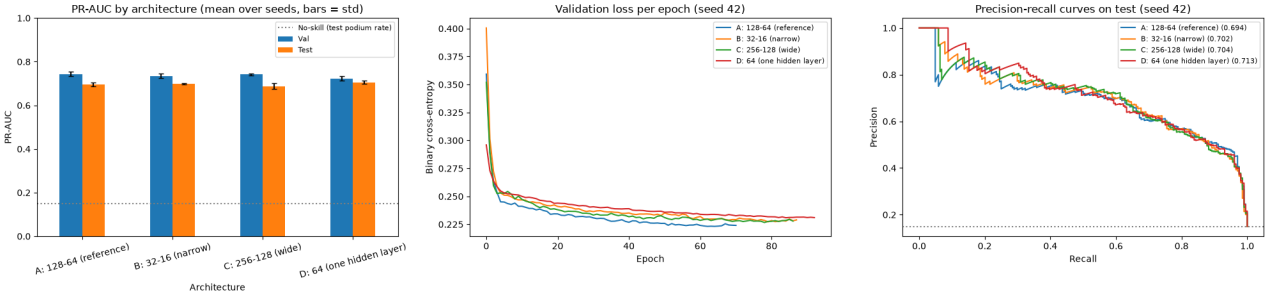


Figure 23. Architectures: validation/test mean PR-AUC; validation loss and test precision-recall curves (seed 42). Cell 593.

Modelling attempts shown in the notebook (more than the three required): four training versions x three model families; a default versus a 64-setting-grid tuned random forest; a 12-setting logistic-regression grid; four neural-network architectures x three seeds; nine feature-table experiments; three preprocessing-order experiments; and group ablations of all three models.

Qualifying features. The final 14-feature set uses the grid (the starting position set by qualifying and penalties) rather than qualifying lap times. Qualifying evidence is the separate 2003+ Q1 experiments above, which did not help; qualifying coverage is complete only from 2003 (patchy 1996-2002, none before 1994), which is the main reason Q1 times are not in

the all-years feature set. Consequently the ablation has no qualifying-time group.

9. Explainability (XAI)

All three final models (14 features, 1950+) were explained on the same 780 validation records, the same 100-record training background and the same 300-record sample for global SHAP. **Important: explanations describe what the models rely on to predict - they are not causal effects.** A feature being important does not mean it causes podium finishes (e.g. a high grid slot is a consequence of car and driver quality), and correlated features share importance.

9.1 Global: permutation importance

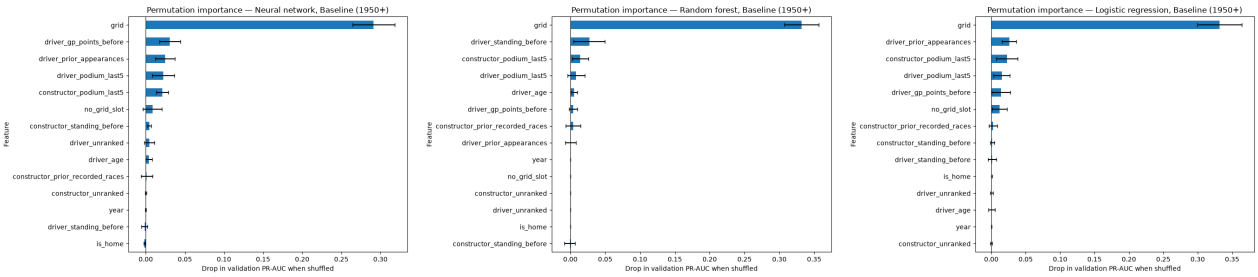


Figure 24. Permutation importance (validation PR-AUC drop when each feature is shuffled, 20 repeats): neural network, random forest, logistic regression.

Feature	Neural network	Random forest	Logistic regression
grid	0.2914	0.3319	0.3317
driver_gp_points_before	0.0308	0.0042	0.0140
driver_prior_appearances	0.0247	0.0003	0.0260
driver_podium_last5	0.0224	0.0083	0.0154
constructor_podium_last5	0.0213	0.0142	0.0229
driver_standing_before	-0.0015	0.0273	0.0014

Interpretation. Grid dominates all three models (shuffling it removes about 0.29-0.33 PR-AUC), then a handful of form, experience and standing features contribute 0.01-0.03 each. The forest leans on driver_standing_before while the network and linear model use points, experience and form instead - overlapping information that different models split differently. year and is_home show about zero because validation seasons lie beyond the training range, so shuffling year swaps values the models treat alike.

9.2 Global: SHAP

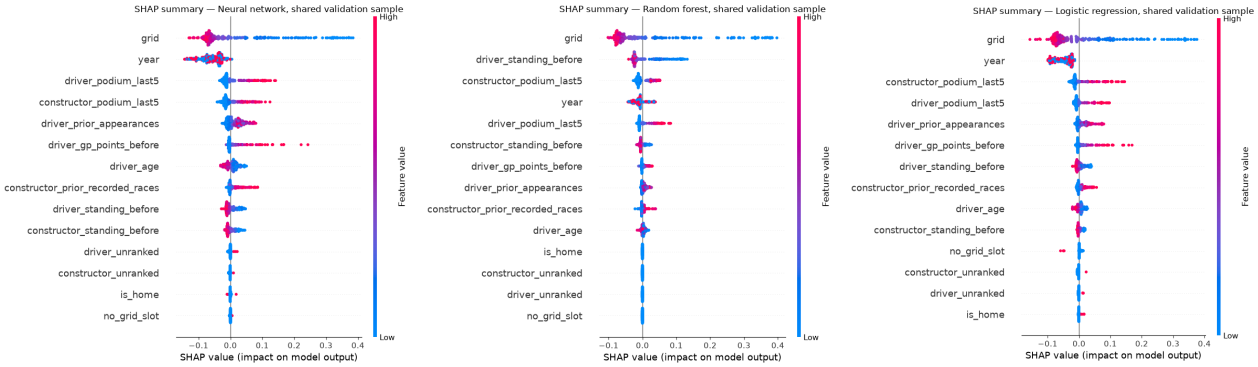


Figure 25. SHAP summary (beeswarm) on 300 validation records: neural network, random forest, logistic regression (colour = feature value, x = effect on podium probability).

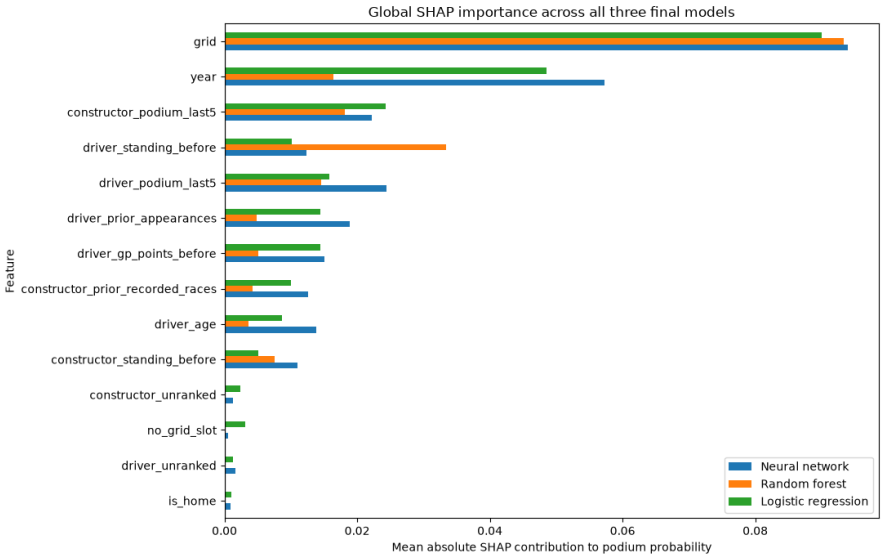


Figure 26. Mean absolute SHAP value per feature, all three models (probability units).

Interpretation. Grid again leads (mean |SHAP| 0.094 NN, 0.093 RF, 0.090 LR). Low grid numbers push the probability up and back-of-grid values push it down. Recent constructor and driver form, and standings (strongest for the forest, 0.034), follow. year has a large mean |SHAP| in the network (0.057) and logistic regression (0.049) but little in the forest (0.016): relative to a background of training seasons, validation years are out-of-range values, so this reflects era shift (the logistic weight on year is -0.35) rather than a meaningful effect of calendar year, and is a reason for caution when extrapolating to new seasons. SHAP and permutation importance measure different quantities (probability contribution versus PR-AUC loss) and are not directly comparable.

9.3 Local: one prediction (local SHAP and LIME)

The same record is explained for every model: the first validation record in dataset order, selected before looking at outcomes or probabilities - **Lewis Hamilton, 2020 Austrian Grand Prix** (grid 5, 250 prior appearances, previous-5 podium rate 0.80, constructor form 0.70, standing 1st before the race). It is illustrative, not representative.

Model	Podium probability	Threshold	Predicted	Actual	Local SHAP base value	LIME local R-squared
Neural network	0.5610	0.37	Podium	No podium	0.1514	0.675
Random forest	0.4685	0.32	Podium	No podium	0.1381	0.742
Logistic regression	0.4194	0.32	Podium	No podium	0.1352	0.674

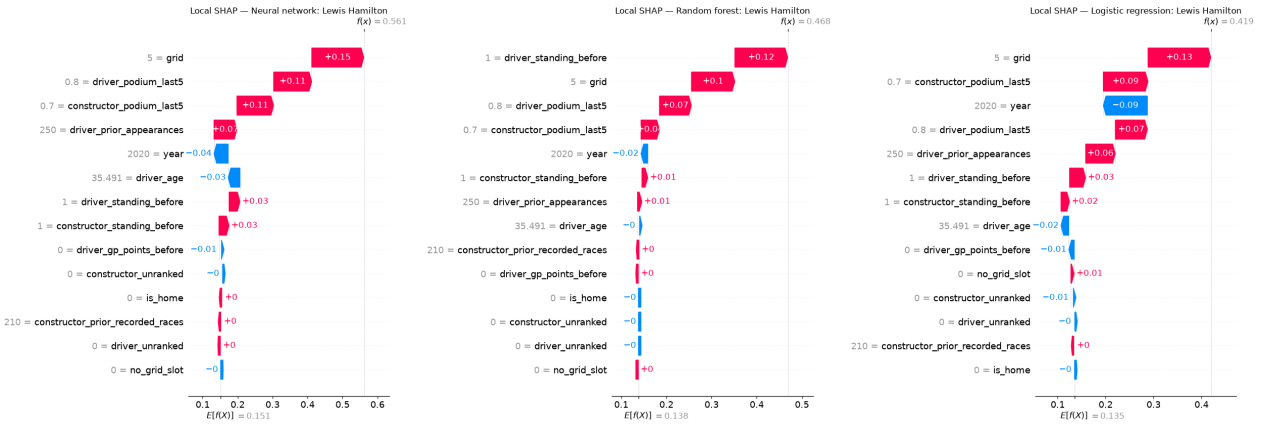


Figure 27. Local SHAP waterfall for the same record: neural network, random forest, logistic regression.

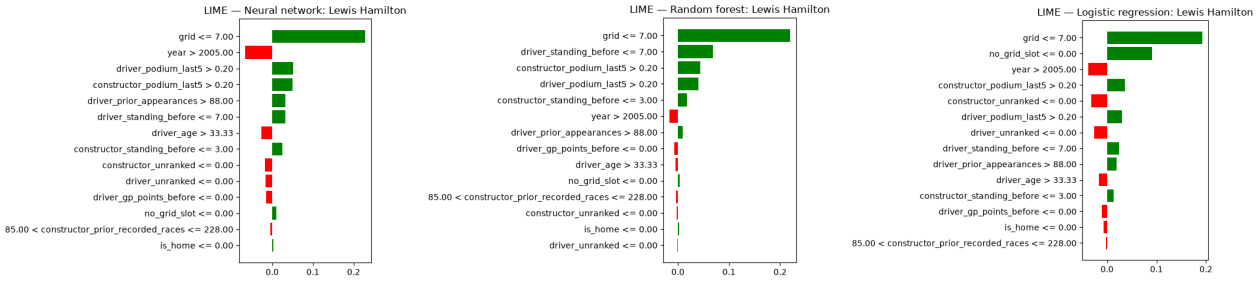


Figure 28. LIME explanation for the same record: neural network, random forest, logistic regression.

Interpretation. Local SHAP contributions sum exactly to each model's probability (checked). For the network the largest positive contributions are grid (+0.149), driver_podium_last5 (+0.109) and constructor_podium_last5 (+0.105); for the forest, driver_standing_before (+0.117), grid (+0.096) and driver_podium_last5 (+0.071). LIME agrees on the leader in every model - the condition grid <= 7 (+0.228 NN, +0.220 RF, +0.193 LR) - followed by form and standing conditions; its local fit R-squared is 0.67-0.74, so its weights are a rough local approximation and not an exact decomposition. All three models predicted a podium (the real outcome was no podium), a reminder that these are probabilities, not guarantees.

10. Inference function

`predict_podium(record)` takes a dictionary or Series of the 14 unscaled E3 features (grid already in its modelling form), orders them, fills missing inputs with the training-fitted median imputer, applies the fitted scaler (network/logistic regression), and returns the podium probability, the saved validation-chosen threshold, a podium/no-podium prediction and which inputs were filled. Unknown names, invalid text, infinite values and records with no observed inputs are rejected; nothing is re-fitted. It uses the neural network, the model with the best validation PR-AUC.

Example	Missing inputs	Probability	Threshold	Prediction	Actual
A. Complete test record (Hamilton, 2022 Bahrain GP)	none	0.5975	0.37	Podium	Podium
B. Same record, driver_age omitted and driver_podium_last5 = NaN	driver_age, driver_podium_last5	0.3620	0.37	No podium	Podium

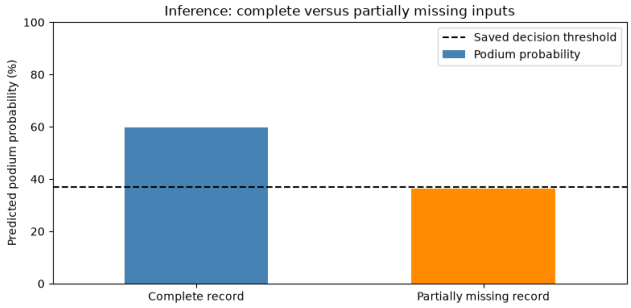


Figure 29. Inference demonstration: predicted probability for the complete and the partially-missing record.

Example B shows the imputer supplied driver_age = 29.58 and driver_podium_last5 = 0.0 (training medians, not recovered facts); hiding the two inputs lowered the probability by 23.55 percentage points and flipped the decision. This one example illustrates the mechanism and does not establish general robustness; the notebook asserts that example A matches the evaluation pipeline.

11. Known limitations

- **Historical qualifying coverage drift.** Qualifying records start in 1994, about 95% coverage in 1994-95, patchy 1996-2002, near complete from 2003; q1/q2/q3 are missing for 1.5%/44.3%/65.4% of qualifying rows because of format and elimination. The final feature set therefore uses grid, not qualifying times.
- **Era sensitivity and distribution shift.** Rules, scoring, field size and reliability change over 75 years. After scaling, test seasons sit well above the training mean for experience, points and year; trees cannot extrapolate and the network may not. NN/LR test PR-AUC drops; RF rises slightly. The test period

(2022-2024) is a new technical era.

- **Small evaluation sets.** Validation has 780 rows (2 seasons) and test 1,359 rows (3 seasons); differences between models need caution: no confidence intervals quantify split uncertainty. One split, no intervals or significance tests. Seed variation is reported for architecture trials; NN training can vary.
- **Multi-entry (shared-drive) rows.** The kept entry is chosen by finishing order, so its grid is an outcome-dependent convention for 85 historical groups (all in 1950-1978, all in training; masked constructor identity in 16 groups).
- **Recorded-activity population (Q2/Q3).** Records without positive laps, lap-time or pit-stop evidence are excluded (2,528 in Q2), which may drop genuine zero-lap starters; results describe recorded participants.
- **Mechanical-retirement labels (Q3).** 'Retired' and 'Damage' counted as mechanical by assumption (broader than the brief's definition); small constructor samples.
- **Q1 ties.** Many circuits at 100% with small samples; rank differences among tied circuits are not meaningful.
- **Feature exploration on all years.** EDA used all available seasons (TA-approved per the notebook); fitting, tuning and thresholds used training/validation only. E4 (season wins) improved the forest but was not adopted.
- **No calibration.** Probabilities rank drivers but are not measured chances; the 0.32-0.37 thresholds were tuned for validation F1.
- **Unresolved data issues.** 17 lap-timing/lap-total cases remain unresolved and are held out of timing analysis; they do not enter the pre-race features.
- **Explanations are not causal.** Importance, SHAP and LIME describe model reliance on correlated features; grid, form, standing and year are entangled.

12. Requirement cross-reference

Requirement in the brief	Where satisfied (report / notebook section)
Initial EDA before and after cleaning with plots	Report 3.1-3.5 / notebook 2.1-2.8, 3.15.2
Sentinels (\N), semantic typing of dates and durations	Report 3.2 / notebook 3.1-3.14, 3.15
One row per (raceId, driverId), duplicate rule and why	Report 3.3 / notebook 3.11.1-3.11.3, 5.2
PK unique and non-null, FK reconcile, raw immutable	Report 3.4 / notebook 2.5, 3.15.1 (gate)
Three data-engineering questions with answers, figures, grain/missingness/cutoff	Report section 4 / notebook section 4
Feature choices justified with plots/correlations/experiments	Report 5 / notebook 5, 6.4-6.6
Appropriate train/validation/test split, justified	Report 6.1 / notebook 6.2
Preprocessing effects and at least two preprocessing-order experiments	Report 6.2-6.3 (three experiments) / notebook 6.8-6.9
Two statistical models and a shallow neural network on the same table; baseline	Report 7 (LR baseline, RF, NN) / notebook 7-9, 11
ROC-AUC, PR-AUC, F1 (plus precision, recall, accuracy) on train, validation and test; primary/secondary justified	Report 7.2-7.3 / notebook 6.7, 11.5
Thresholds and hyper-parameters chosen without the test set	Report 6.1, 7 / notebook 6.7, 8.3, 9.2
At least three distinct modelling attempts	Report 8.2 note / notebook 7-9, 11.2.1
Feature-group ablation	Report 8.1 / notebook 10.2, 11
Permutation importance, SHAP, local SHAP/LIME, interpreted, not causal	Report 9 / notebook 12
Inference function on a complete and a partially-missing record	Report 10 / notebook 11.6
Known limitations	Report 11

Appendix A. Current-run audit and report alignment

This revision uses the saved **acl-ms1.ipynb** inspected on 19 October 2026. Of 357 code cells, 356 have saved execution counts; the edited setup cell is unexecuted. There are 101 PNG figures and no saved errors. These checks establish the state of the submitted file; this report revision did not retrain models or independently execute another Run All.

The main report retains its three data-engineering answers, feature evidence, chronological split, model comparison, preprocessing experiments, ablations, XAI interpretations and complete/partial inference demonstrations. Cleaning descriptions and counts have been corrected to the current code. Appendix B reproduces **all 101 plotted outputs from the current saved run**, including the detailed cleaning diagnostics and model plots that were absent from the earlier report.

A.1 Position-column redundancy: results and sprint results

Compare only paired numeric observations. Convert positionText to numeric temporarily for the comparison; keep its original retirement and outcome codes. Thus these correlations describe classified numeric positions, not a numeric encoding of all race outcomes. High correlation indicates redundancy, not perfect equality of every field.

Table / numeric comparison	Pairs	Pearson r	Spearman rho	Agreement	Mismatches
results: position vs positionText	15,796	1.000000	1.000000	100%	0
results: position vs positionOrder	15,796	0.999982	0.999981	99.9177%	13
results: positionText vs positionOrder	15,798	0.999982	0.999981	99.9177%	13
sprint: position vs positionText	345	1.000000	1.000000	100%	0
sprint: position vs positionOrder	345	1.000000	1.000000	100%	0
sprint: positionText vs positionOrder	345	1.000000	1.000000	100%	0

Decision in the executed code: drop **position** from results_clean and sprint_results_clean. Retain positionText for outcome codes and positionOrder for the podium label. The results audit covers 26,668 rows (10,872 missing position values); the sprint audit covers 360 rows (15 missing position values). The numeric subset alone is sufficient for the redundancy decision; it does not justify dropping positionText, whose nonnumeric codes carry separate information.

Notebook locations: **Results: compare position columns and remove redundancy** (cell 366) and **Sprint results: compare position columns and remove redundancy** (cell 401). Cell numbers throughout this appendix are one-based positions in the saved file.

A.2 Remaining null values after cleaning

The final cleaning gate passes while explicitly allowing the following missing values. Results retains all six timing-related fields; they are useful for retrospective auditing and are excluded from the pre-race model inputs. There is no general rule to replace these nulls with -1.

Clean table	Column	Null count	Share of rows
qualifying	q1	156	1.49%
qualifying	q2	4,647	44.28%
qualifying	q3	6,865	65.42%
results	time	18,988	71.20%
results	milliseconds	18,988	71.20%
results	fastestLap	18,416	69.06%
results	rank	18,158	68.09%
results	fastestLapTime	18,416	69.06%
results	fastestLapSpeed	18,416	69.06%

Results has **111,382 actual null cells** across 26,668 rows and 23 columns; qualifying has 11,668 across 10,494 rows. The other 12 cleaned tables have no actual NaN values in the final audit. Raw results had 122,887 literal \N sentinels across 26,759 rows and 18 columns; a sentinel count and an actual NaN count are different audit measures.

No NaN does not mean every value is observed. Existing explicit unavailable-value conventions remain: two unresolved results car numbers use -1, and unavailable sprint timing values use the documented -1 code. These codes are distinct from nulls and do not enter the selected 14-feature model. Four of the six initially absent results car numbers were recovered from evidence; two remain unknown.

A.3 Integrity and population

All 14 cleaned tables pass unique, non-null primary-key checks; all 23 foreign-key checks have zero unmatched references. Raw-table and CSV preservation checks pass. The final allowed-null audit above is from notebook cell 415; the before/after structural audit is cell 413.

All **26,668** cleaned driver-race records enter modelling, including the **85** consolidated multi-entry groups required by the team/TA convention. Consolidation replaces 176 repeated raw car-entry rows with 85 unique driver-race rows, removing 91 extra raw records; it does not discard those 85 drivers from modelling. The split remains training 24,529, validation 780 and test 1,359.

A.4 Feature validity and important assumptions

Historical multi-entry grid convention

For repeated (raceId, driverId) groups, the kept entry has the best positionOrder (smallest resultId on a tie); points are summed and n_raw_rows records the original entry count. All 85 consolidated groups remain in training. In 16 groups with ambiguous constructor identity, constructor history features remain missing until training-fitted imputation.

The retained grid is selected through the finishing-order rule, so it is an **outcome-dependent historical convention**. It must not be described as a verified reconstruction of the driver's original starting car. The convention is retained to follow the requested population requirement; all affected groups are in 1950-1978 and none are in validation or test. The broad pre-race claim therefore has this explicit exception.

Grid-zero handling in notebook section 5.9

Keep no_grid_slot = 1 for an original grid of zero. Replace that grid with **maximum positive raw grid for the same race + 1**. Compute the maximum from all raw race entries before driver-race consolidation. This prevents a large recorded grid from disappearing when a different entry is retained for a shared driver. The adjusted value is an ordering convention after numbered slots, not an invented known starting position. Recorded positive grid values are retained.

Mechanical-retirement interpretation

The professor clarification counts Puncture, Tyre, Tyre puncture, Fire and Safety belt as mechanical; Refuelling is non-mechanical. The current team decision also counts the ambiguous labels **Retired** and **Damage** as mechanical. This broadens the brief's component-failure definition and is disclosed as a limitation; "unmapped" status values trigger a code check rather than silently receiving a class. The Q3 rates in section 4 use this convention.

Metrics, selection and interpretation

PR-AUC in this notebook is **average precision** (sklearn average_precision_score), not trapezoidal integration of a precision-recall curve. It is the primary selection metric because podiums are uncommon. ROC-AUC is secondary; F1, precision and recall use validation-chosen thresholds, with accuracy reported as an additional measure. The neural network is the default inference model by validation PR-AUC (0.7423). The random forest has the best observed test PR-AUC (0.7291); the test score is not used to change the selected default.

Explanations describe model reliance and are not causal effects. Seed variability and era shift limit comparisons. Saved execution counts and absence of recorded errors are not independent proof that a future clean-environment run will reproduce identical neural-network values.

A.5 Neural-network architecture comparison

Notebook section 11.2.1 compares four shallow architectures on the same 14 features and fixed chronological rows, over seeds 42, 43 and 44. All configurations have one or two hidden layers. Values below are means across the three seeds; standard deviations refer to PR-AUC across those seeds, not uncertainty over future seasons.

Architecture	Hidden layers	Parameters	Mean epochs	Val AP (SD)	Test AP (SD)
A: 128-64 (reference)	2	11,009	73.33	0.7434 (0.0106)	0.6954 (0.0078)
B: 32-16 (narrow)	2	1,217	76.00	0.7351 (0.0092)	0.6982 (0.0039)
C: 256-128 (wide)	2	38,401	64.67	0.7420 (0.0045)	0.6882 (0.0138)
D: 64 (one hidden layer)	1	1,281	97.00	0.7232 (0.0093)	0.7056 (0.0079)

Architecture	Val ROC-AUC	Test ROC-AUC	Val F1	Test F1
A: 128-64	0.9308	0.9359	0.7053	0.6620
B: 32-16	0.9279	0.9338	0.7014	0.6547
C: 256-128	0.9291	0.9331	0.7021	0.6624
D: 64	0.9273	0.9357	0.7022	0.6616

Selection: A has the highest mean validation PR-AUC and remains the reference. C adds parameters without improving that validation mean. D is smaller and has the highest mean test PR-AUC, but choosing it on that basis would use the held-out test for selection. The small seed samples do not establish statistical superiority of A over C.

These three-seed means are separate from the single final fitted model in section 7.3. The final neural-network test PR-AUC of 0.6940 is therefore not expected to equal the architecture mean of 0.6954.

A.6 Report coverage after revision

Deliverable	Evidence retained or added
Before/after EDA, typing, grain and integrity	Sections 3.1-3.5; A.1-A.3; all cleaning plots in Appendix B
All three joined data-engineering questions	Section 4: answers, figures, populations, missingness and assumptions
Feature validity and chronological split	Sections 5-6; A.4 historical grid exception and raw-grid fallback
Three comparable models and all required metrics	Section 7.3: train/validation/test ROC-AUC, AP, F1, precision, recall
Ablations and preprocessing-order experiments	Sections 6.3 and 8; corresponding plots in Appendix B
Global/local XAI and inference	Sections 9-10: permutation, SHAP, local SHAP, LIME, complete/partial record
All notebook plots in the report	Appendix B: 101 current saved plotted outputs, with notebook-cell provenance

Submission links, external accessibility and upload status were not tested by this report edit. The notebook and source CSV files were left unchanged.

Appendix B. Complete current-run figure collection

This appendix contains every PNG plotted output saved in the current notebook, in notebook order: **101 figures**. Figures are labelled B1-B101 and cite the notebook cell and its nearest section heading. It supplements the interpreted figures in the main report; it does not add new model runs or new experiments.

Multiple plots from one cell are individually numbered. Very tall plots are divided into contiguous top-to-bottom image segments to keep labels readable; segment boundaries contain no omitted rows. Repeated charts are retained here because the brief asks for all plots in both the notebook and the report. Some plots show intermediate states before a documented later cleaning decision; interpret them with the cited notebook stage and the final audit in Appendix A.

Figure	Notebook	Section
B1	Cell 17	2.6 Missing Data Visualization
B2	Cell 19	2.7 Target Distribution & Race Diagnostics
B3	Cell 35	3.1.4 Circuits: Validation & Analysis
B4	Cell 45	Step 2 — Find out what an empty constructor status means
B5	Cell 51	Step 4 — Compare before and after
B6	Cell 52	Step 4 — Compare before and after
B7	Cell 71	Step 11 — Compare before and after and check the final table
B8	Cell 82	Step 15 — Add the two reviewed rows and check before and after
B9	Cell 92	Placeholder handling and before/after evidence
B10	Cell 98	Nationality inspection and correction
B11	Cell 110	Additional audit — Understand missing numbers and codes
B12	Cell 112	Intermediate missing-value treatment and number typing
B13	Cell 126	Repeated result pairs — Show the evidence before choosing a rule
B14	Cell 134	3.3.5 Handling missing driver-profile numbers and codes
B15	Cell 143	3.4.1 Lap Times: Initial Auditing & Placeholder Detection
B16	Cell 160	Unusual-value investigation
B17	Cell 170	See the cross-table problem before making changes
B18	Cell 172	Missing-record coverage — Before 1996 and within the covered period
B19	Cell 175	3.4.5 Lap Times: Edge-Case Investigation & Final Validation
B20	Cell 186	Before-cleaning inspection
B21	Cell 205	3.5.4 Pit Stops: Sequence Auditing, Edge Cases & Final Validation
B22	Cell 207	Describe coverage without declaring absent records wrong
B23	Cell 253	3.9.3 Qualifying: Cross-Table Consistency & Edge Cases
B24	Cell 255	Resolve the ten qualifying–race constructor-ID contradictions
B25	Cell 257	Verify the car numbers, then correct only confirmed disagreements
B26	Cell 260	3.9.4 Qualifying: Missing Data Coverage & Final Validation

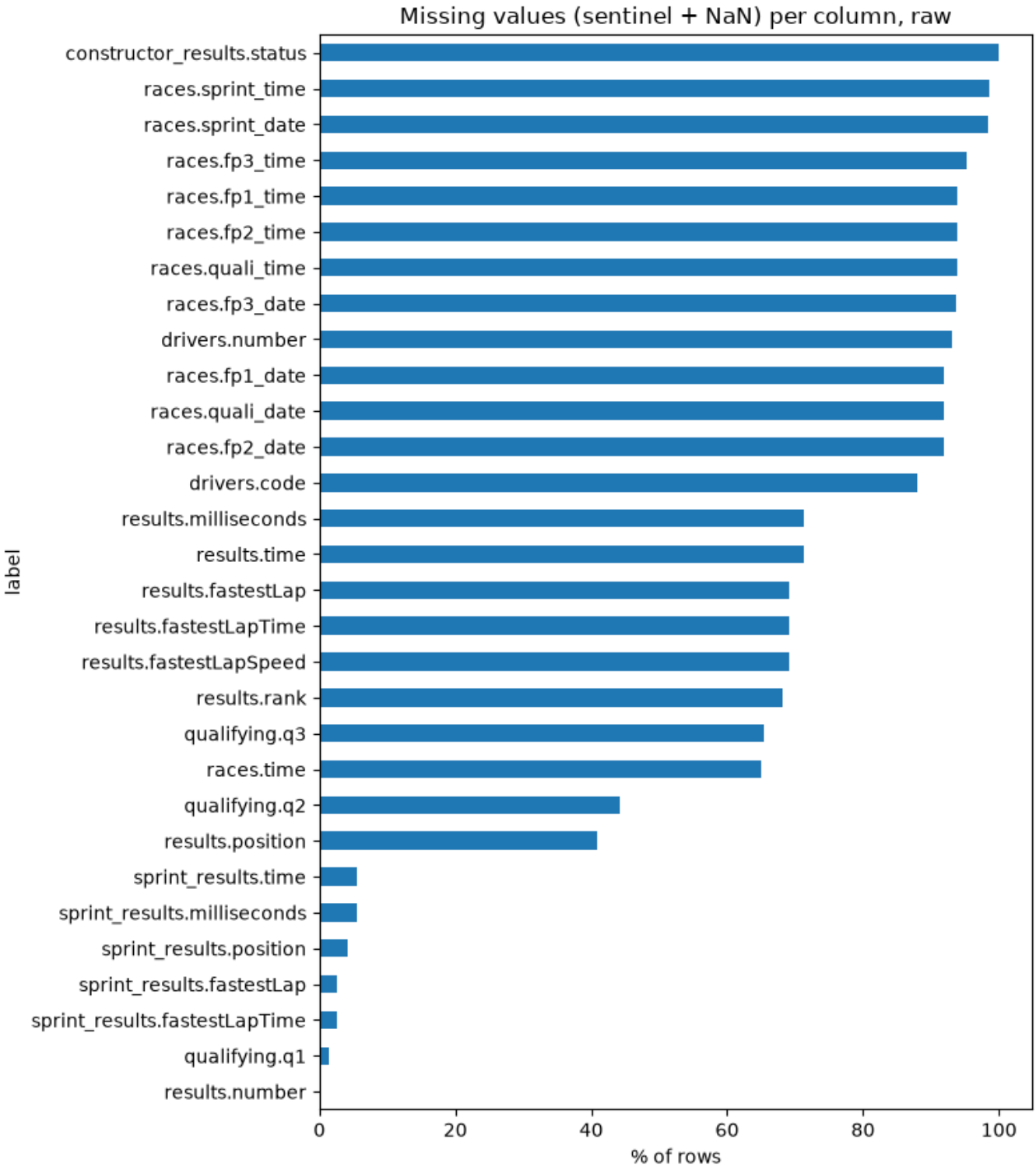
Figure	Notebook	Section
B27	Cell 260	3.9.4 Qualifying: Missing Data Coverage & Final Validation
B28	Cell 267	3.10.1 Races: Initial Auditing & Time Conversions
B29	Cell 269	1. Show the existing cleaning before and after
B30	Cell 277	4. Schedule relationships and missing-value coverage
B31	Cell 279	5. Recover the missing 2021 scheduled times using official evidence
B32	Cell 284	Apply only the verified corrections to the clean copy
B33	Cell 288	3.10.6 Historical coverage and race-feature selection
B34	Cell 289	3.10.6 Historical coverage and race-feature selection
B35	Cell 290	3.10.6 Historical coverage and race-feature selection
B36	Cell 292	Decision: exclude schedule columns from the final clean race table
B37	Cell 304	3.11.2 Results: Initial Auditing & Time Conversions
B38	Cell 312	Verify the discovered mismatch using the official result
B39	Cell 317	3.11.3 Results: Shared Drive Consolidation & Integrity Checks
B40	Cell 320	3.11.3 Results: Shared Drive Consolidation & Integrity Checks
B41	Cell 325	3.11.3 Results: Shared Drive Consolidation & Integrity Checks
B42	Cell 326	3.11.3 Results: Shared Drive Consolidation & Integrity Checks
B43	Cell 335	3.11.4 Results: Grid, Indianapolis and qualifying-coverage diagnostics
B44	Cell 344	Correct 41 verified completed-lap totals
B45	Cell 357	Check external evidence before changing an ID
B46	Cell 361	8. Final before-and-after checks
B47	Cell 366	Results: compare position columns and remove redundancy
B48	Cell 387	Sprint timing: investigate missing values and fill with an unavailable code
B49	Cell 399	3.13.4 Feature Distribution Analysis
B50	Cell 401	Sprint results: compare position columns and remove redundancy
B51	Cell 418	3.15.2 Target Class & Feature Sanity (Post-Cleaning)
B52	Cell 437	Question 1 calculation: at least one front-row driver on the podium
B53	Cell 437	Question 1 calculation: at least one front-row driver on the podium
B54	Cell 440	Question 1 — Qualifying position versus starting grid
B55	Cell 455	Combine the comparisons fairly
B56	Cell 468	Read the ranking and compare periods
B57	Cell 480	Show how the history features work
B58	Cell 482	5.4 Coverage before relevance
B59	Cell 484	5.5 Relationships with podium — descriptive evidence

Figure	Notebook	Section
B60	Cell 485	5.5 Relationships with podium — descriptive evidence
B61	Cell 488	5.6 Check overlapping information and historical stability
B62	Cell 488	5.6 Check overlapping information and historical stability
B63	Cell 502	Check the new columns
B64	Cell 502	Check the new columns
B65	Cell 513	6.4 Modelling-table experiments: which features help?
B66	Cell 515	6.5 Which single features matter: permutation importance
B67	Cell 527	6.8.1 Imputation (median, fitted on training rows only)
B68	Cell 531	6.8.2 Scaling (`StandardScaler`, fitted on training rows only)
B69	Cell 534	6.8.3 Skewness
B70	Cell 537	6.8.4 Engineered features: range, distribution and the past-only check
B71	Cell 554	6.9.4 Summary and chosen order
B72	Cell 560	7.2 Train the four candidate versions for comparison
B73	Cell 562	7.3 Compare the four versions
B74	Cell 570	8.4 Fit the tuned forests and compare the experimental versions
B75	Cell 576	9.3 Fit the tuned models and compare the experimental versions
B76	Cell 577	9.3 Fit the tuned models and compare the experimental versions
B77	Cell 581	10.1 Best experimental version of each model
B78	Cell 583	10.2 Feature-group ablation
B79	Cell 590	11.2 Neural network: final training and ablation
B80	Cell 590	11.2 Neural network: final training and ablation
B81	Cell 593	11.2.1 Neural network: comparing four architectures
B82	Cell 598	11.3 Random forest: final training and ablation
B83	Cell 598	11.3 Random forest: final training and ablation
B84	Cell 601	11.4 Logistic regression: final training and ablation
B85	Cell 601	11.4 Logistic regression: final training and ablation
B86	Cell 603	11.5 Compare the three full-feature final models
B87	Cell 609	Example B — the same record with two inputs missing
B88	Cell 613	12.1 Global explanation: permutation importance for each model
B89	Cell 613	12.1 Global explanation: permutation importance for each model
B90	Cell 613	12.1 Global explanation: permutation importance for each model
B91	Cell 615	12.2 Global explanation: SHAP for each model
B92	Cell 615	12.2 Global explanation: SHAP for each model

Figure	Notebook	Section
B93	Cell 615	12.2 Global explanation: SHAP for each model
B94	Cell 617	12.3 Local SHAP: the same driver–race prediction for all models
B95	Cell 617	12.3 Local SHAP: the same driver–race prediction for all models
B96	Cell 617	12.3 Local SHAP: the same driver–race prediction for all models
B97	Cell 619	12.4 Local LIME: the same prediction for each model
B98	Cell 619	12.4 Local LIME: the same prediction for each model
B99	Cell 619	12.4 Local LIME: the same prediction for each model
B100	Cell 621	12.5 Compare and interpret all three models
B101	Cell 621	12.5 Compare and interpret all three models

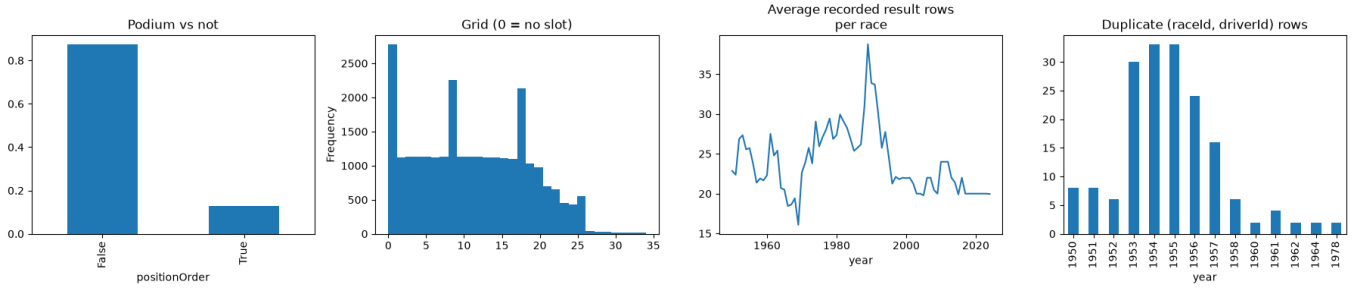
B1. 2.6 Missing Data Visualization

Notebook cell 17 | executed output | source: acl-ms1.ipynb



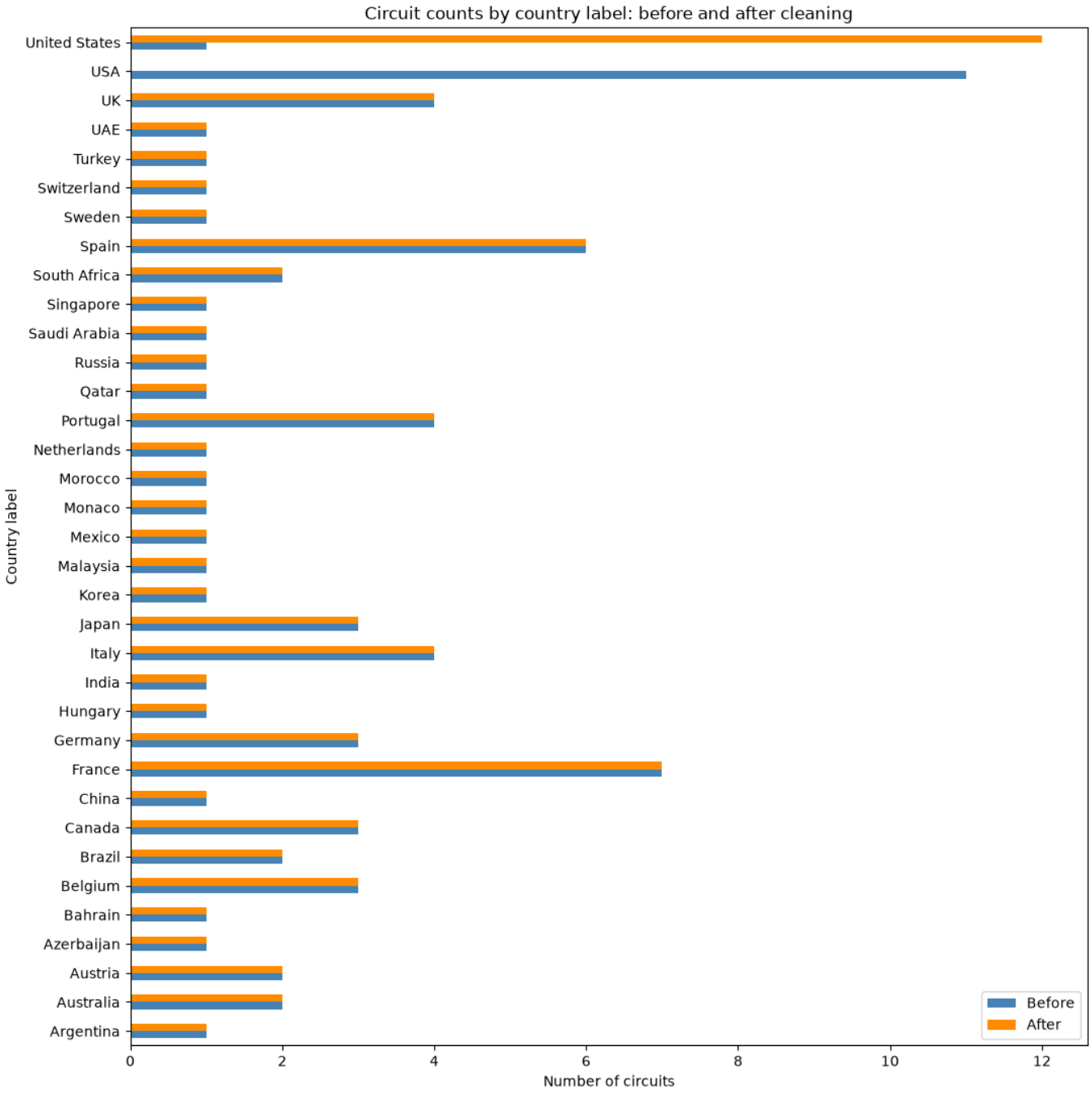
B2. 2.7 Target Distribution & Race Diagnostics

Notebook cell 19 | executed output | source: acl-ms1.ipynb



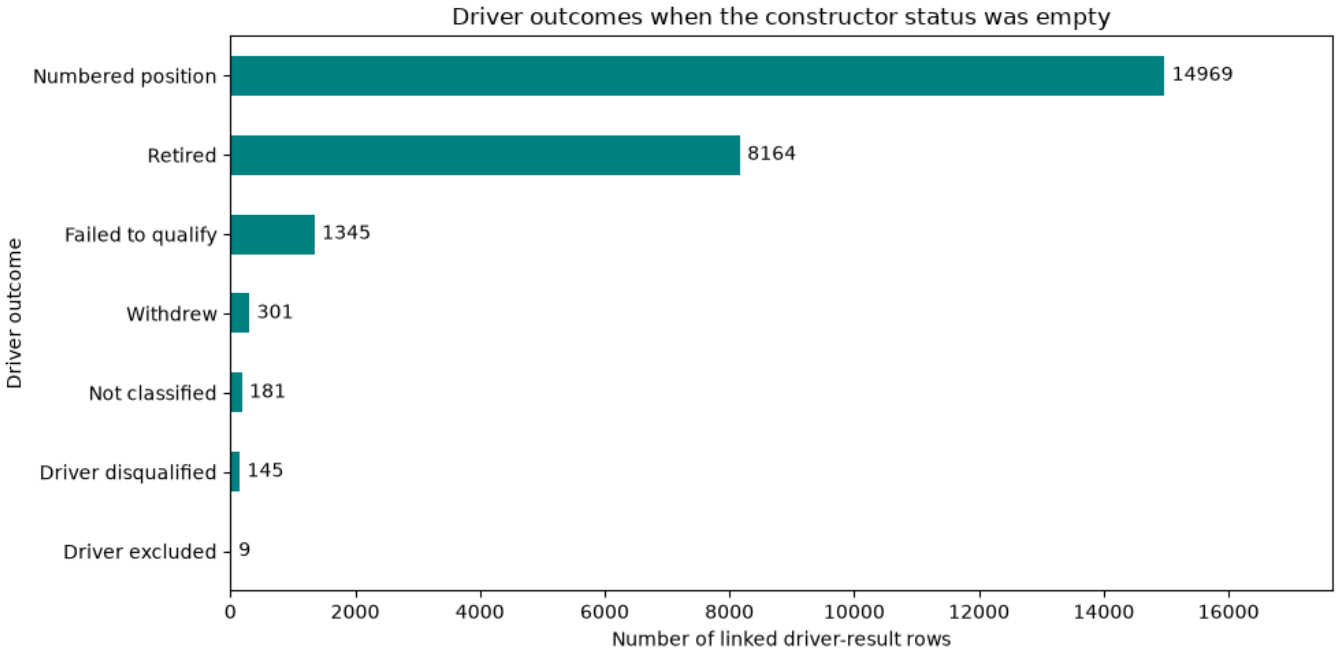
B3. 3.1.4 Circuits: Validation & Analysis

Notebook cell 35 | executed output | source: acl-ms1.ipynb



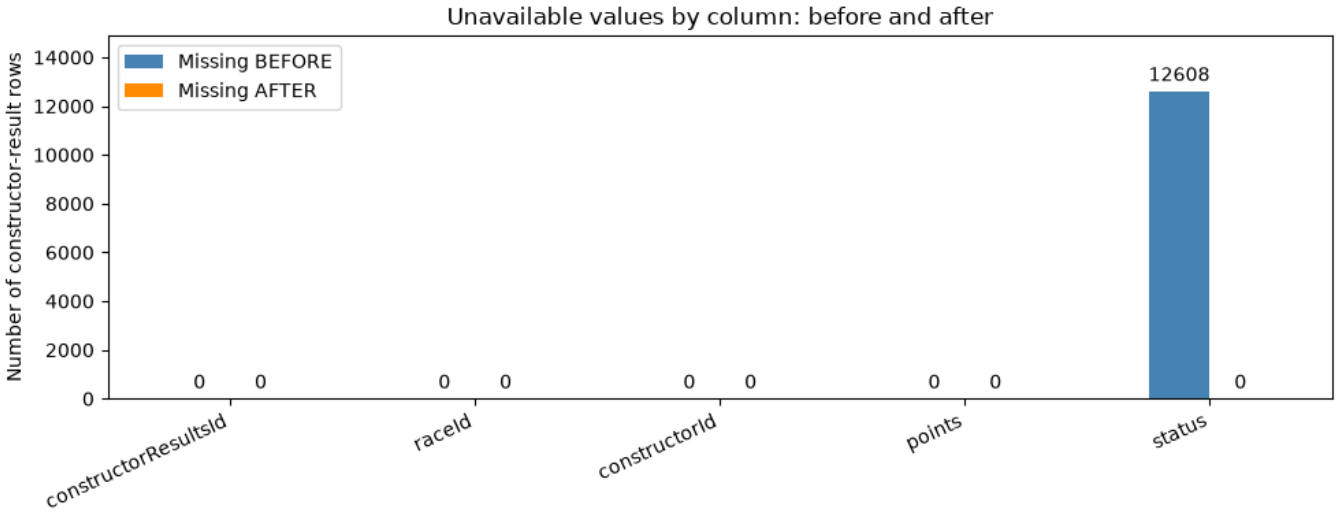
B4. Step 2 - Find out what an empty constructor status means

Notebook cell 45 | executed output | source: acl-ms1.ipynb



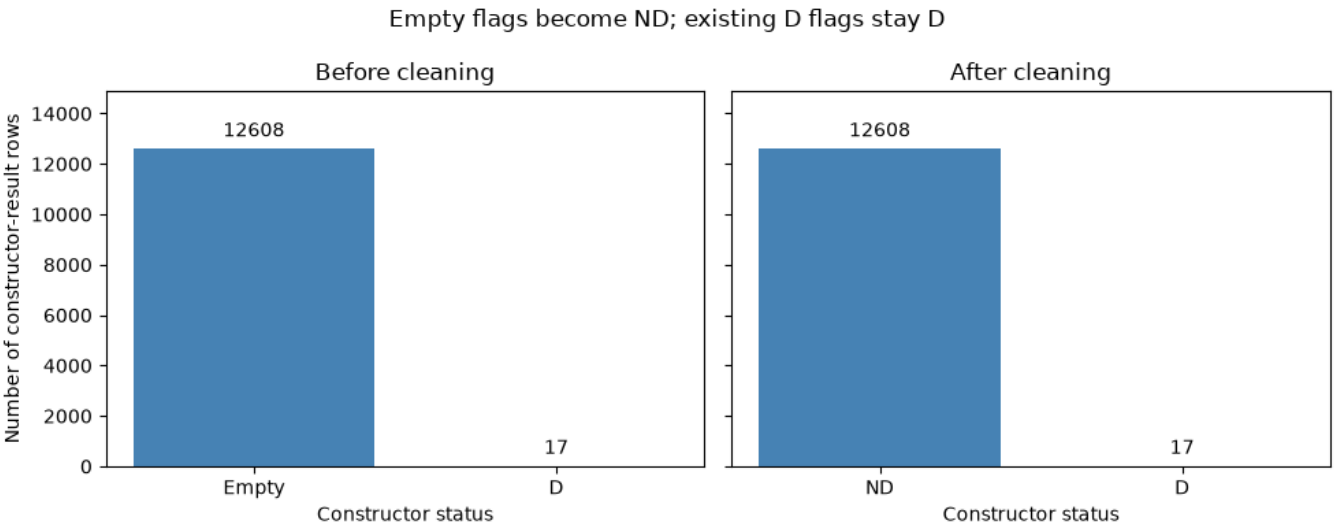
B5. Step 4 - Compare before and after

Notebook cell 51 | executed output | source: acl-ms1.ipynb



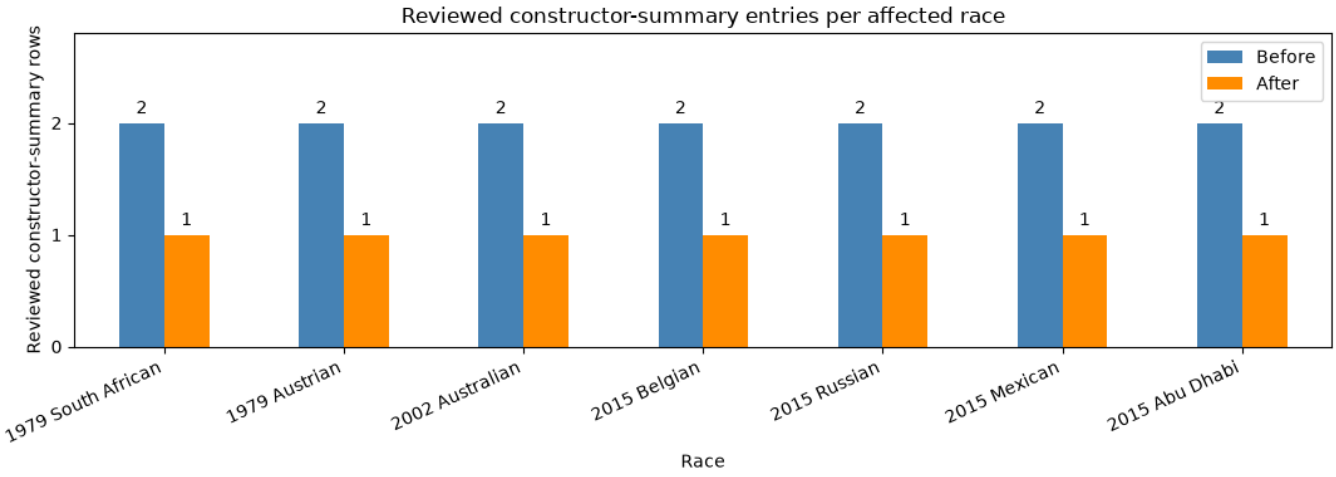
B6. Step 4 - Compare before and after

Notebook cell 52 | executed output | source: acl-ms1.ipynb



B7. Step 11 - Compare before and after and check the final table

Notebook cell 71 | executed output | source: acl-ms1.ipynb



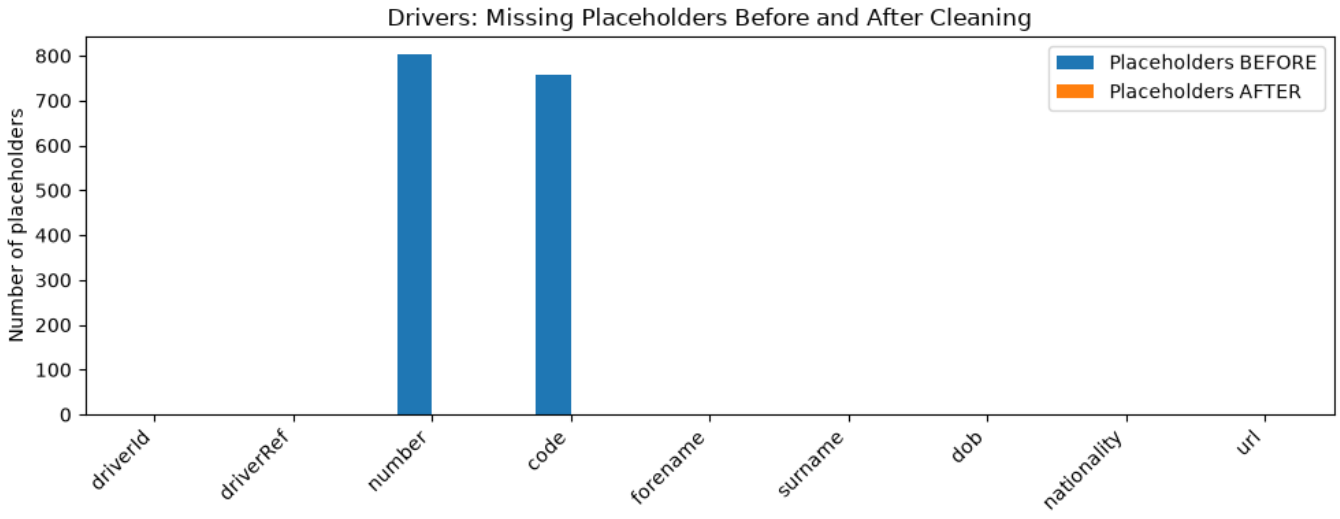
B8. Step 15 - Add the two reviewed rows and check before and after

Notebook cell 82 | executed output | source: acl-ms1.ipynb



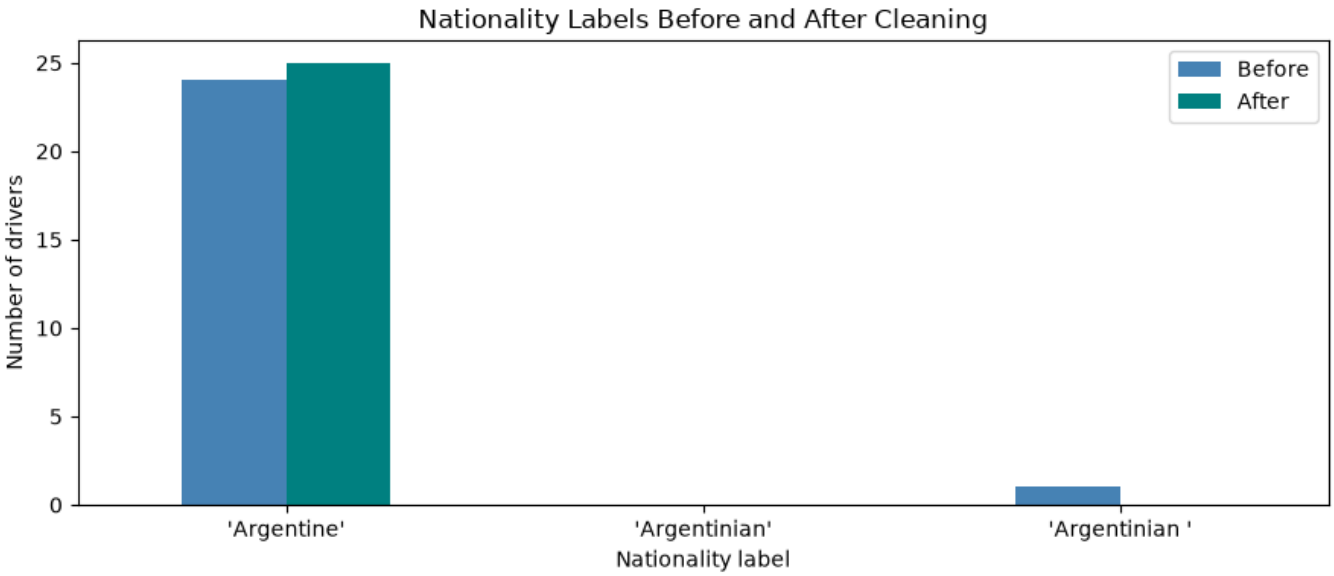
B9. Placeholder handling and before/after evidence

Notebook cell 92 | executed output | source: acl-ms1.ipynb



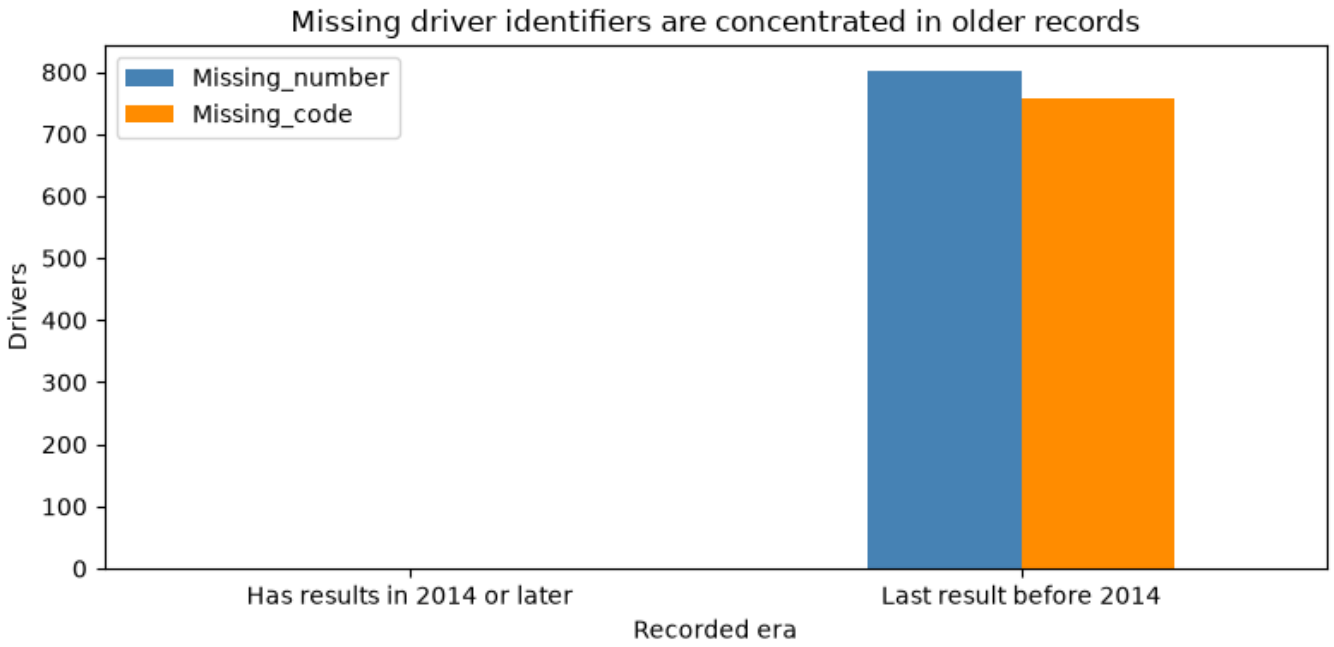
B10. Nationality inspection and correction

Notebook cell 98 | executed output | source: acl-ms1.ipynb



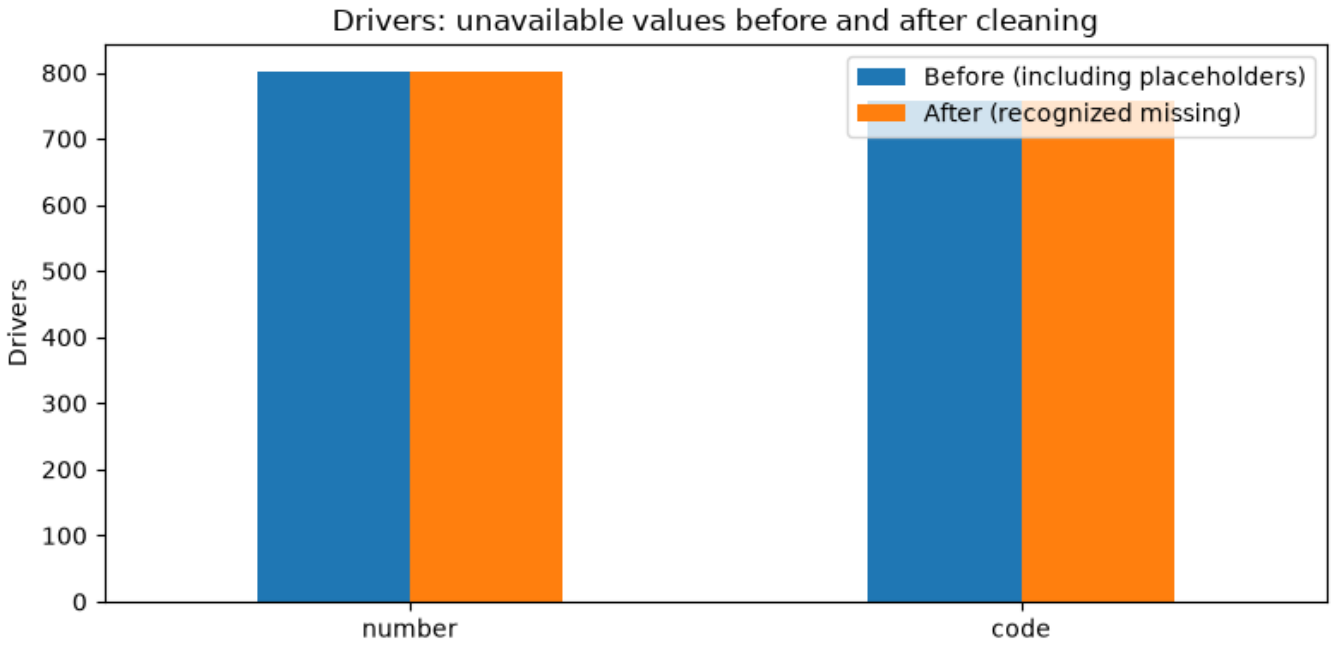
B11. Additional audit - Understand missing numbers and codes

Notebook cell 110 | executed output | source: acl-ms1.ipynb



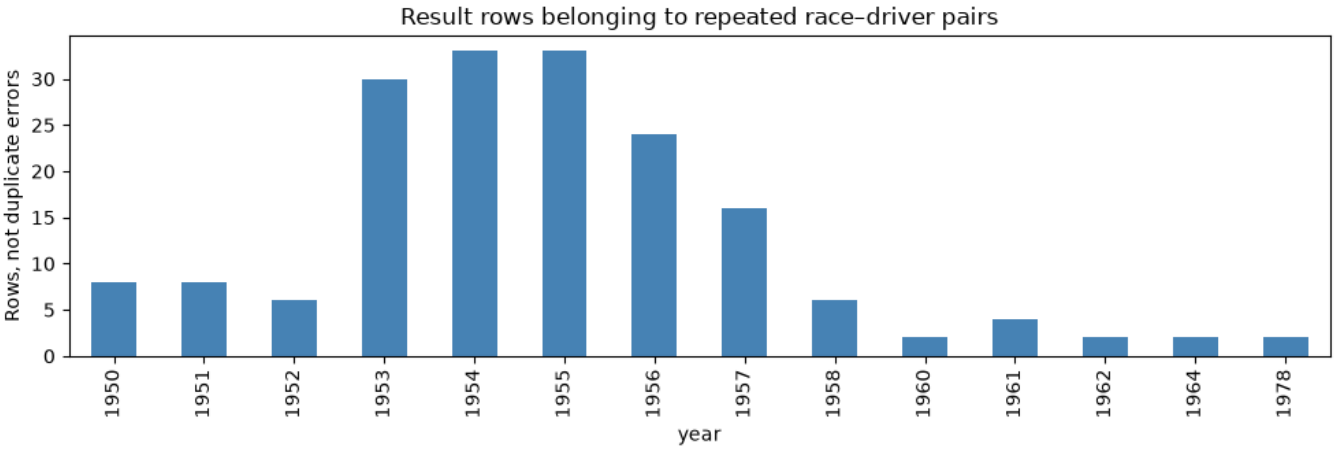
B12. Intermediate missing-value treatment and number typing

Notebook cell 112 | executed output | source: acl-ms1.ipynb



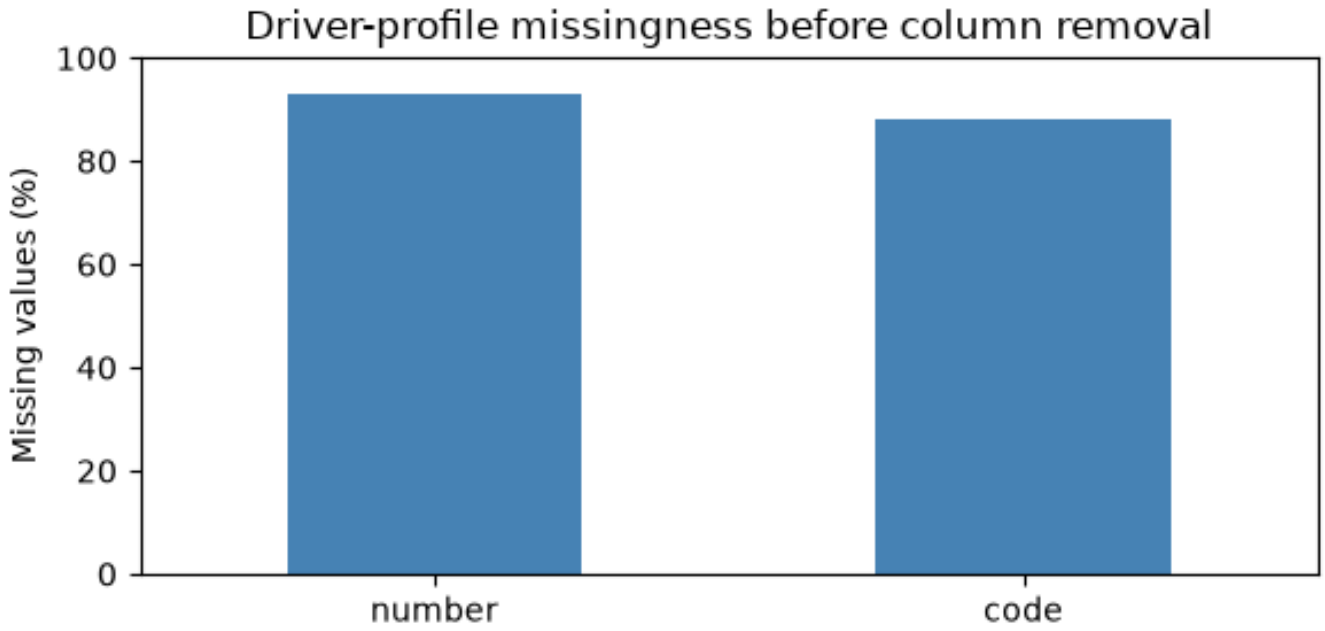
B13. Repeated result pairs - Show the evidence before choosing a rule

Notebook cell 126 | executed output | source: acl-ms1.ipynb



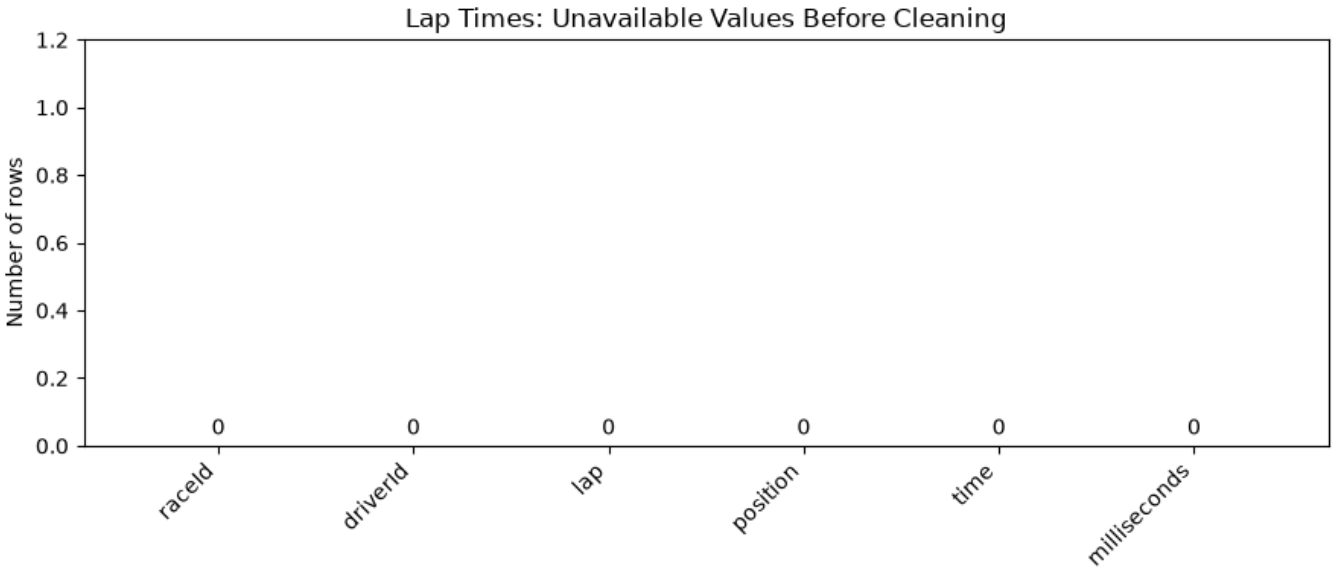
B14. 3.3.5 Handling missing driver-profile numbers and codes

Notebook cell 134 | executed output | source: acl-ms1.ipynb



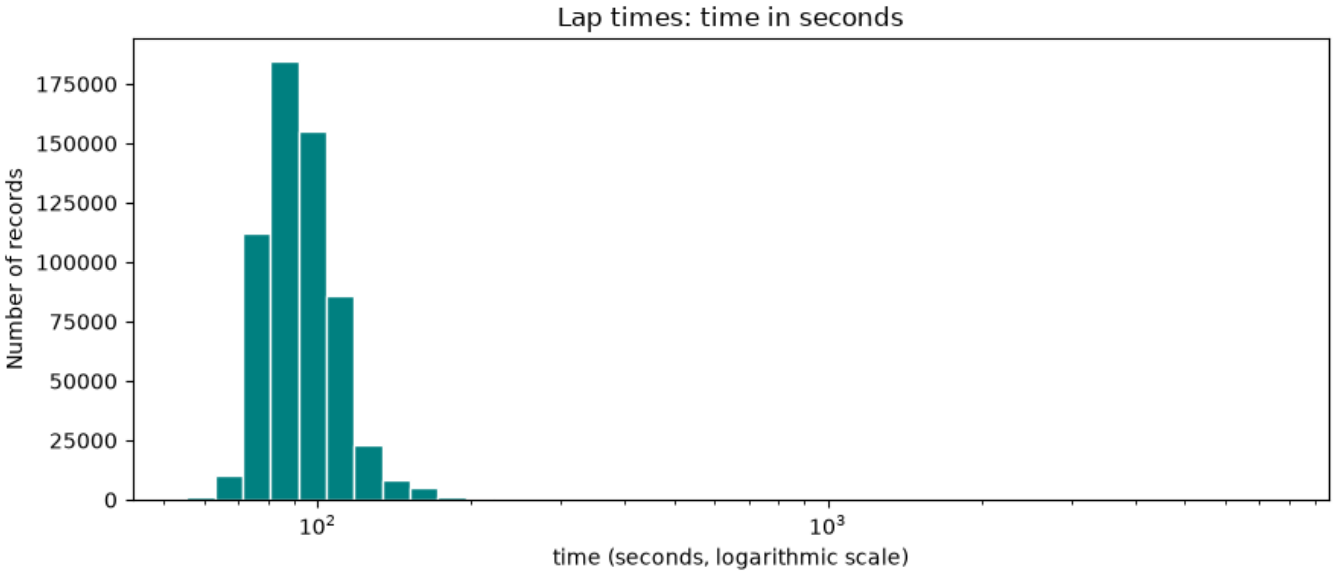
B15. 3.4.1 Lap Times: Initial Auditing & Placeholder Detection

Notebook cell 143 | executed output | source: acl-ms1.ipynb



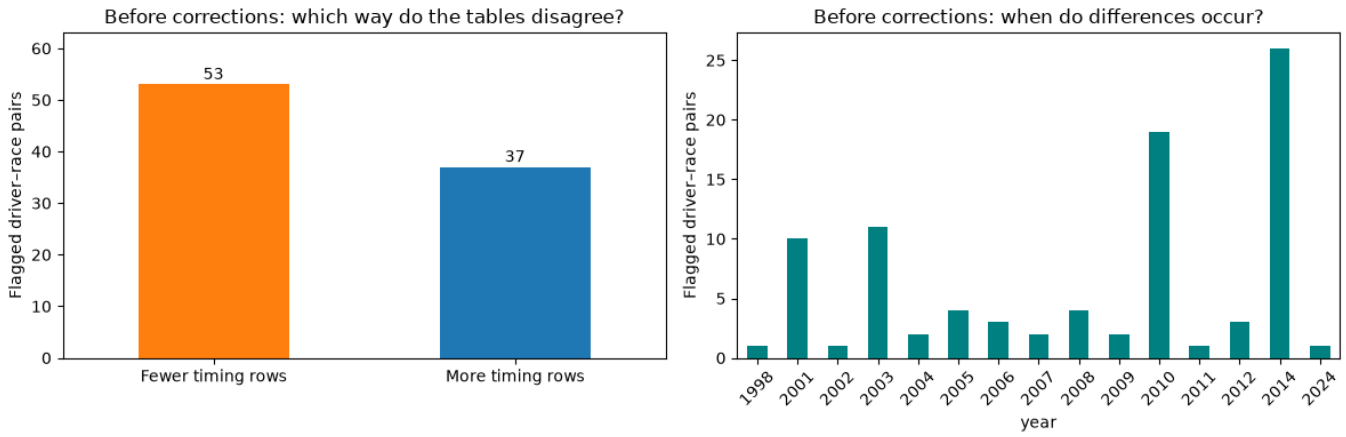
B16. Unusual-value investigation

Notebook cell 160 | executed output | source: acl-ms1.ipynb



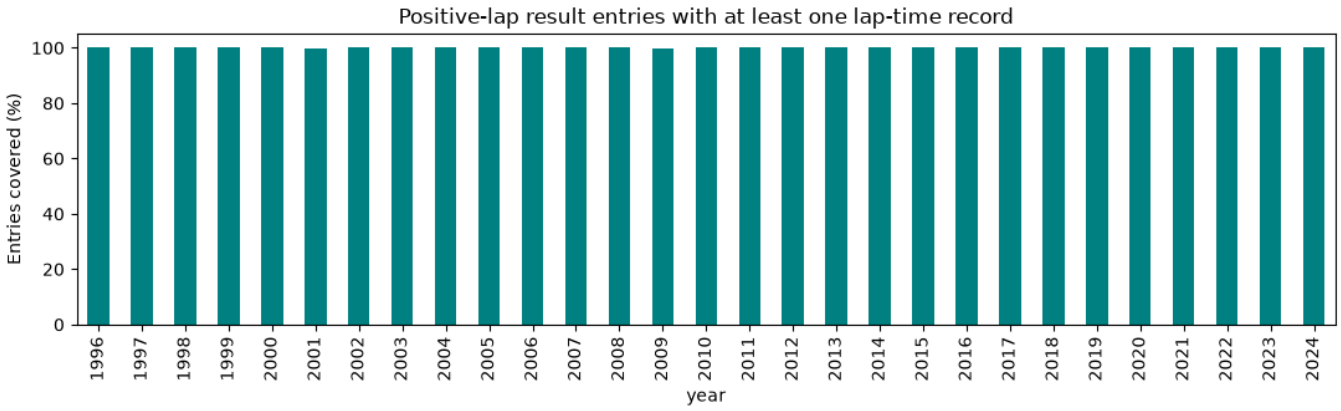
B17. See the cross-table problem before making changes

Notebook cell 170 | executed output | source: acl-ms1.ipynb



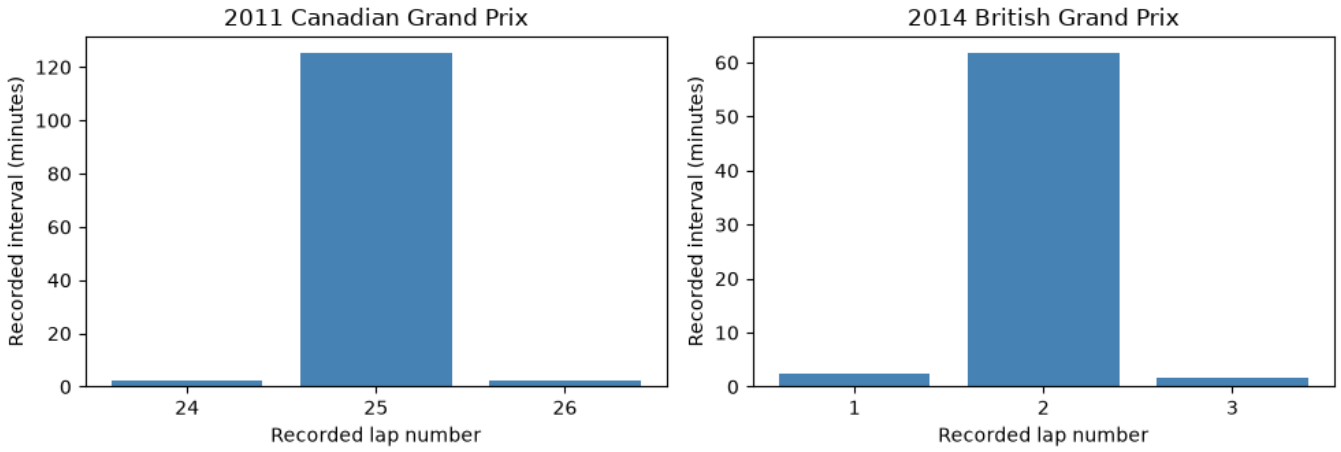
B18. Missing-record coverage - Before 1996 and within the covered period

Notebook cell 172 | executed output | source: acl-ms1.ipynb



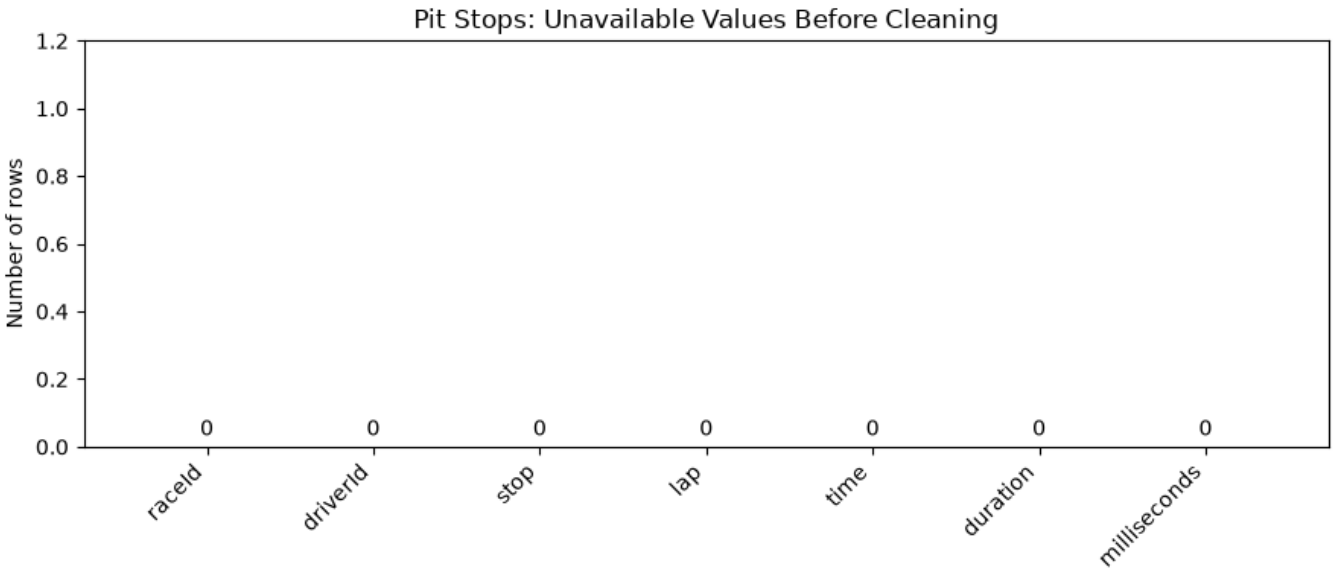
B19. 3.4.5 Lap Times: Edge-Case Investigation & Final Validation

Notebook cell 175 | executed output | source: acl-ms1.ipynb



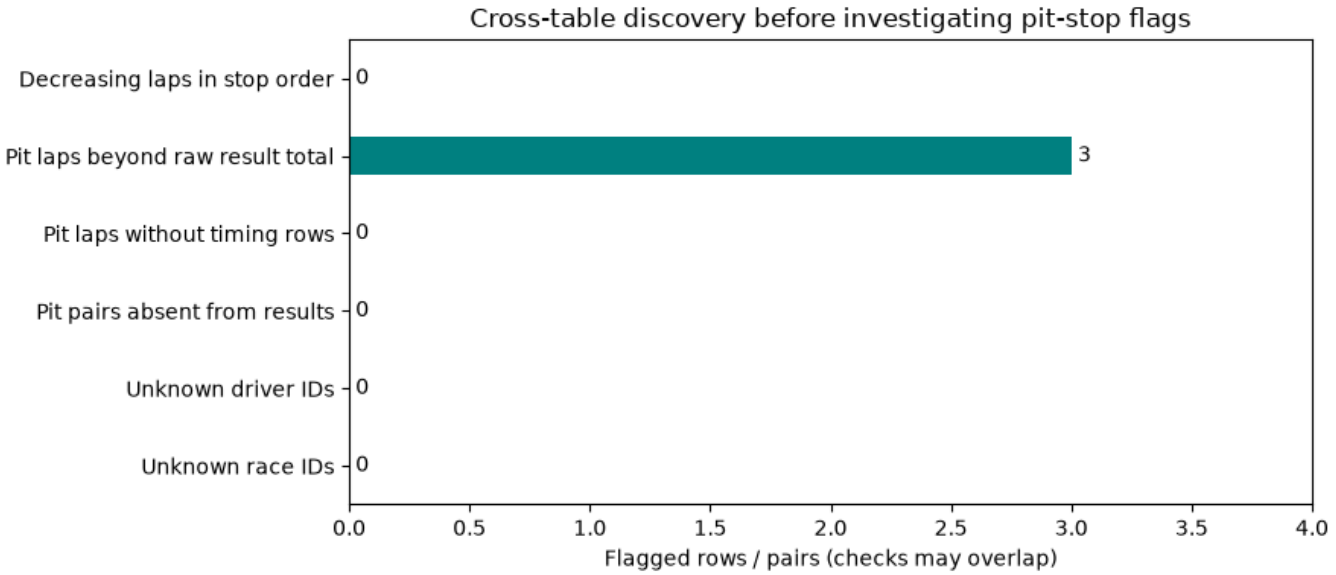
B20. Before-cleaning inspection

Notebook cell 186 | executed output | source: acl-ms1.ipynb



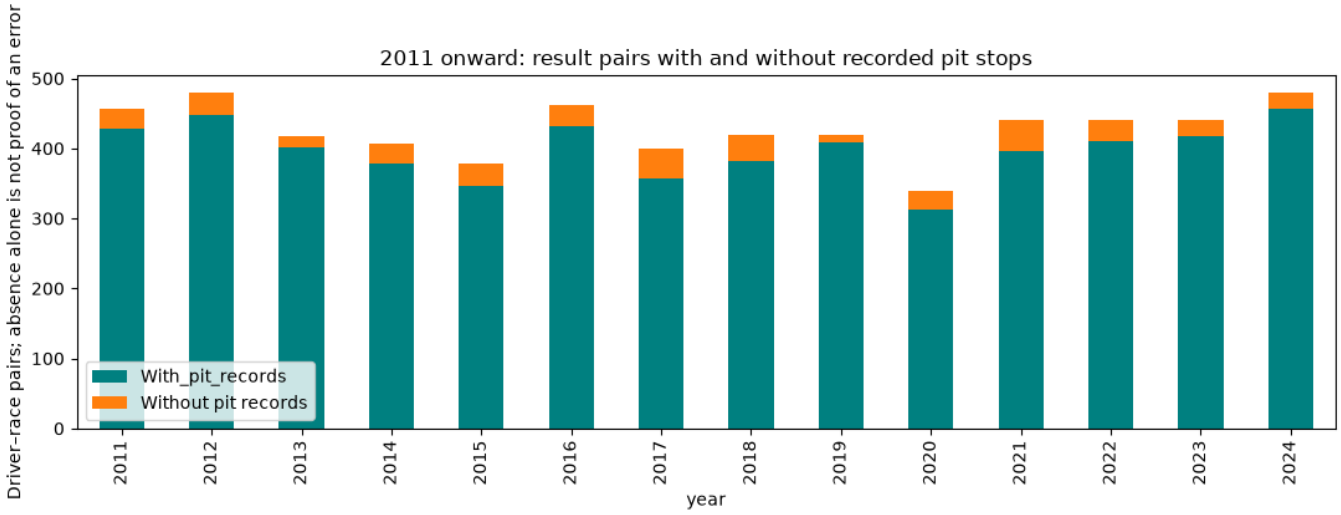
B21. 3.5.4 Pit Stops: Sequence Auditing, Edge Cases & Final Validation

Notebook cell 205 | executed output | source: acl-ms1.ipynb



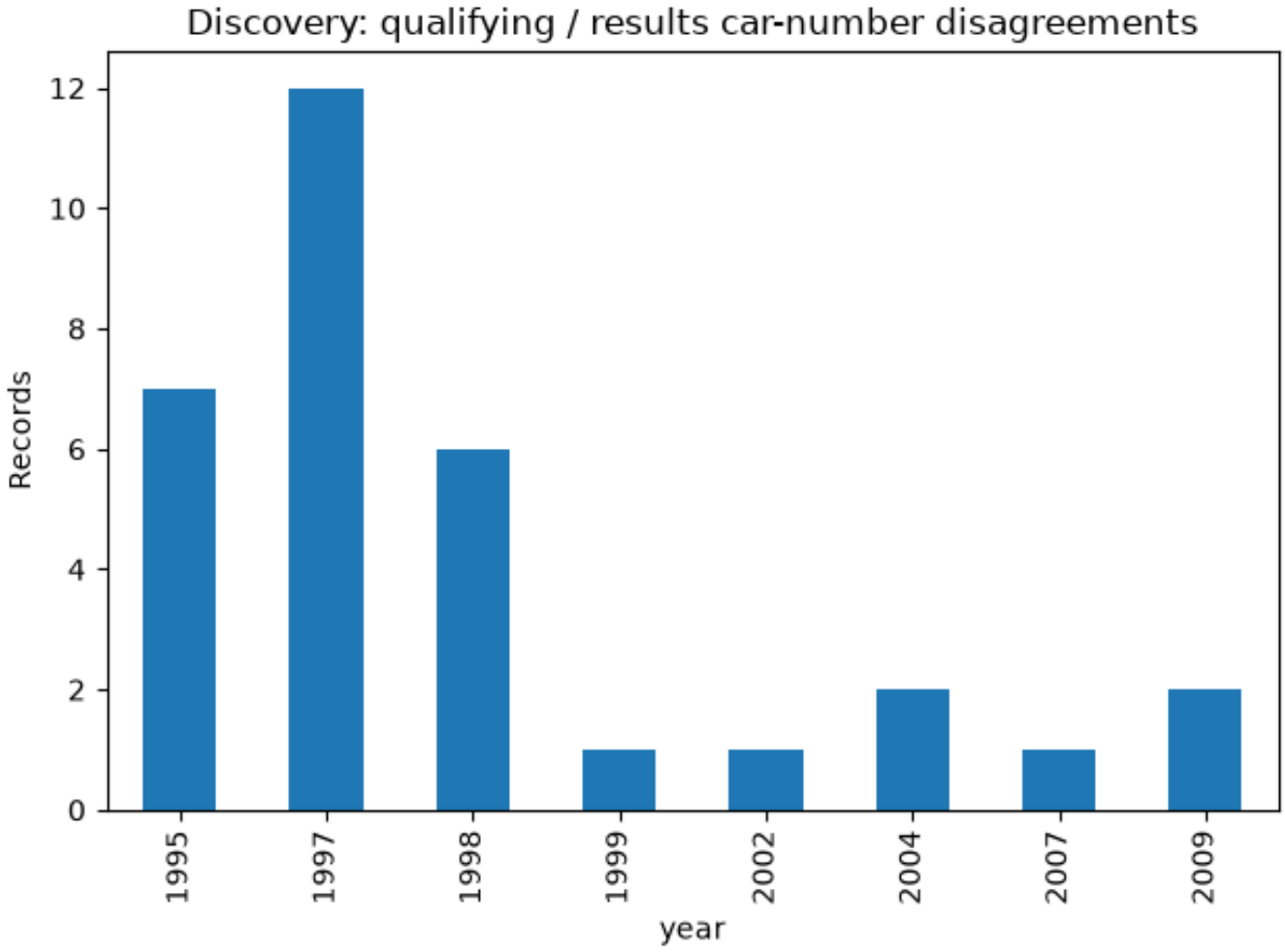
B22. Describe coverage without declaring absent records wrong

Notebook cell 207 | executed output | source: acl-ms1.ipynb



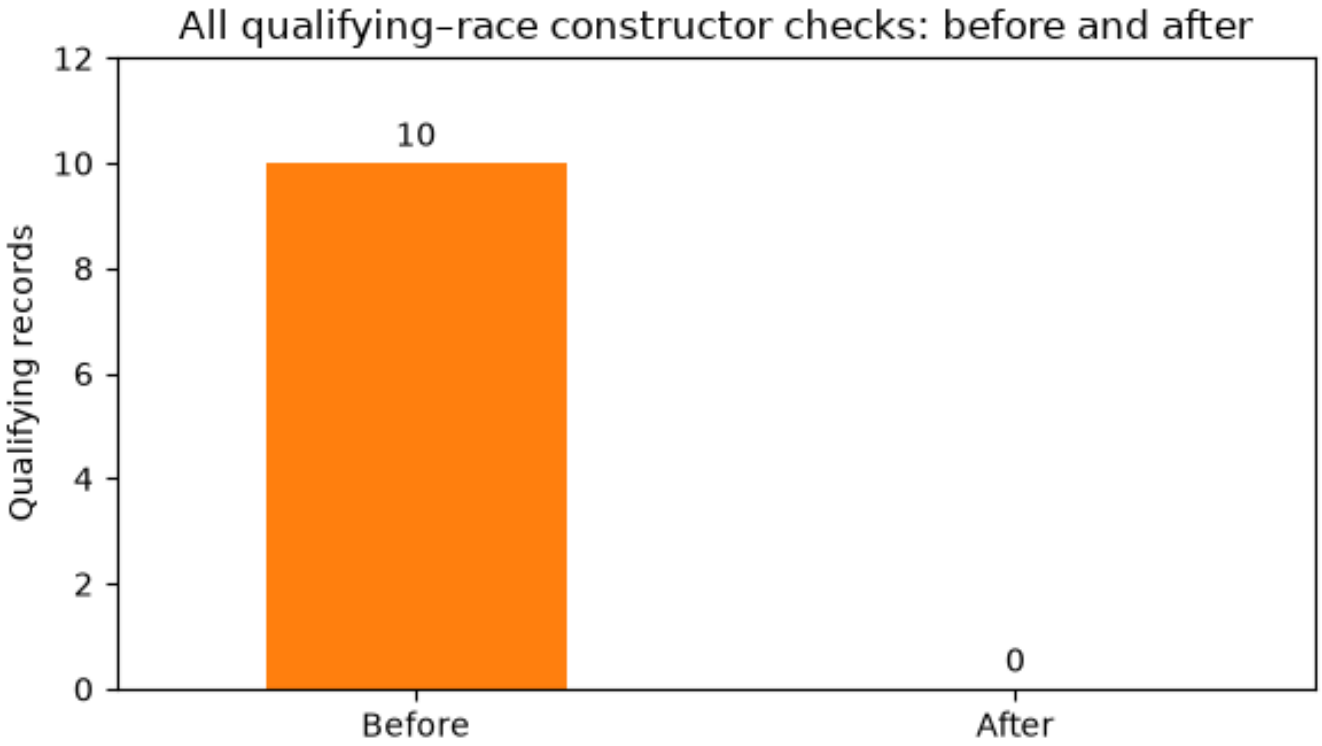
B23. 3.9.3 Qualifying: Cross-Table Consistency & Edge Cases

Notebook cell 253 | executed output | source: acl-ms1.ipynb



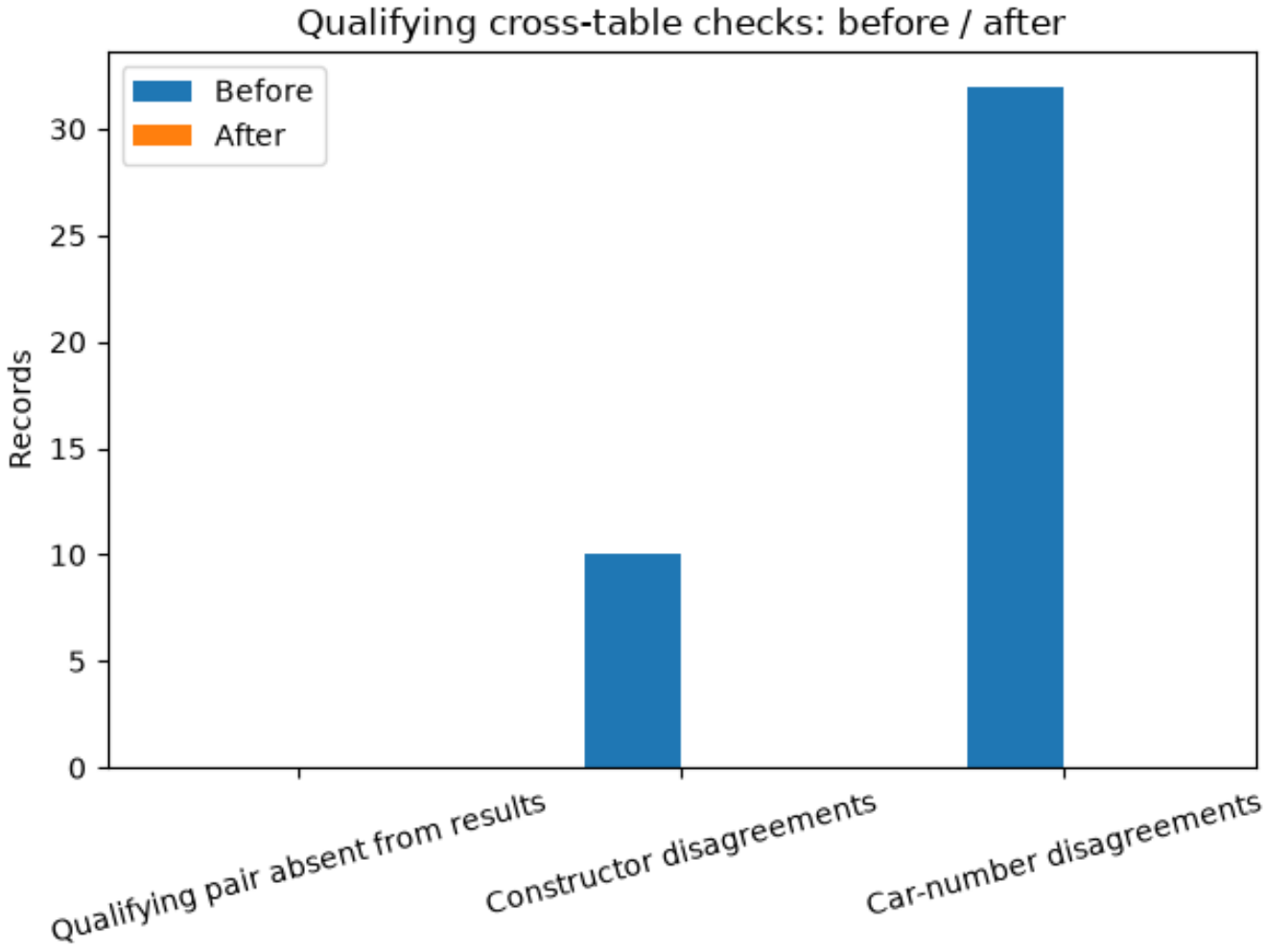
B24. Resolve the ten qualifying-race constructor-ID contradictions

Notebook cell 255 | executed output | source: acl-ms1.ipynb



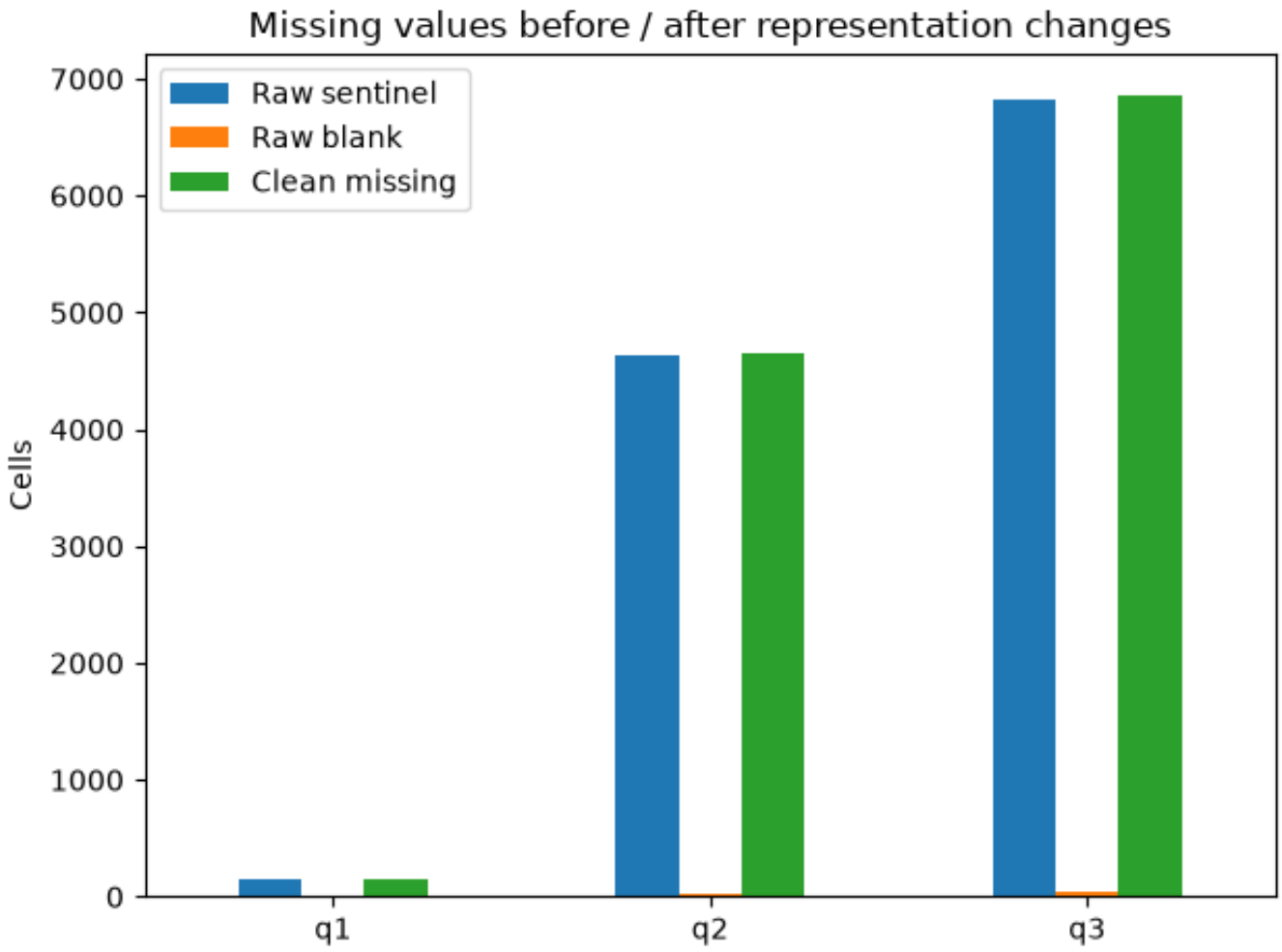
B25. Verify the car numbers, then correct only confirmed disagreements

Notebook cell 257 | executed output | source: acl-ms1.ipynb



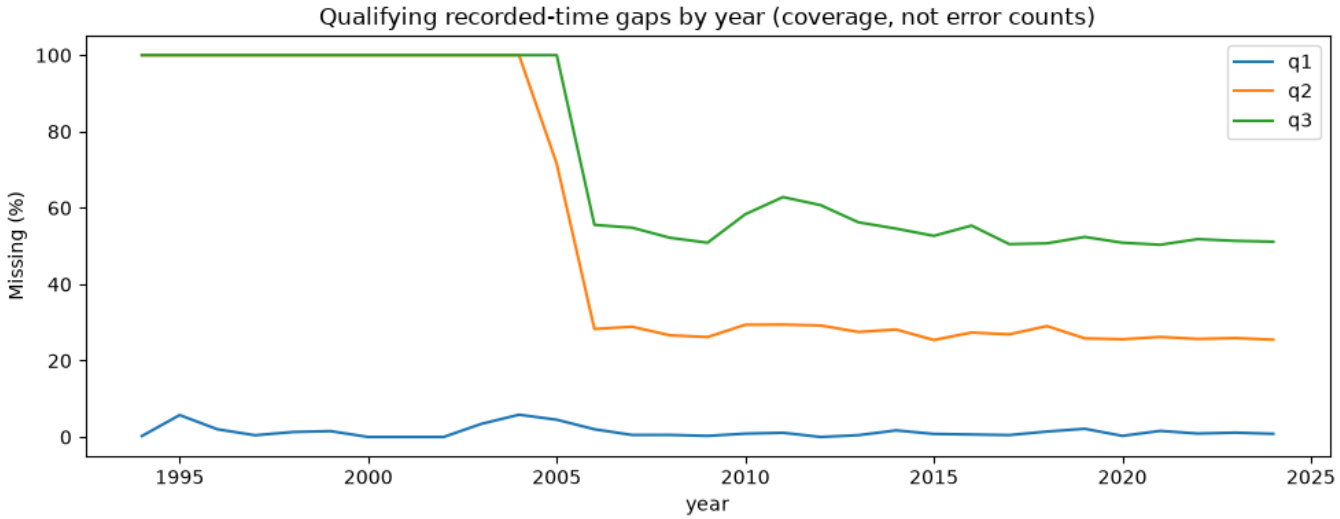
B26. 3.9.4 Qualifying: Missing Data Coverage & Final Validation

Notebook cell 260 | executed output | source: acl-ms1.ipynb



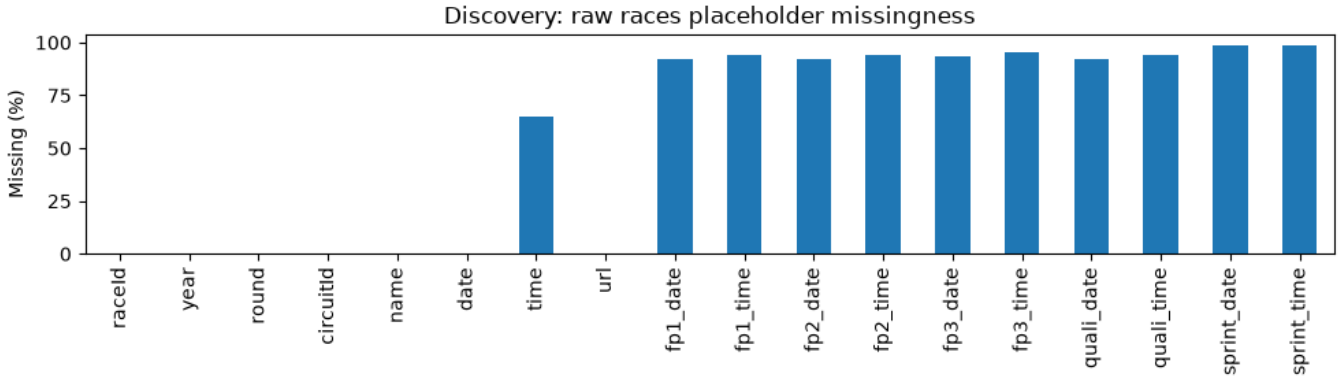
B27. 3.9.4 Qualifying: Missing Data Coverage & Final Validation

Notebook cell 260 | executed output | source: acl-ms1.ipynb



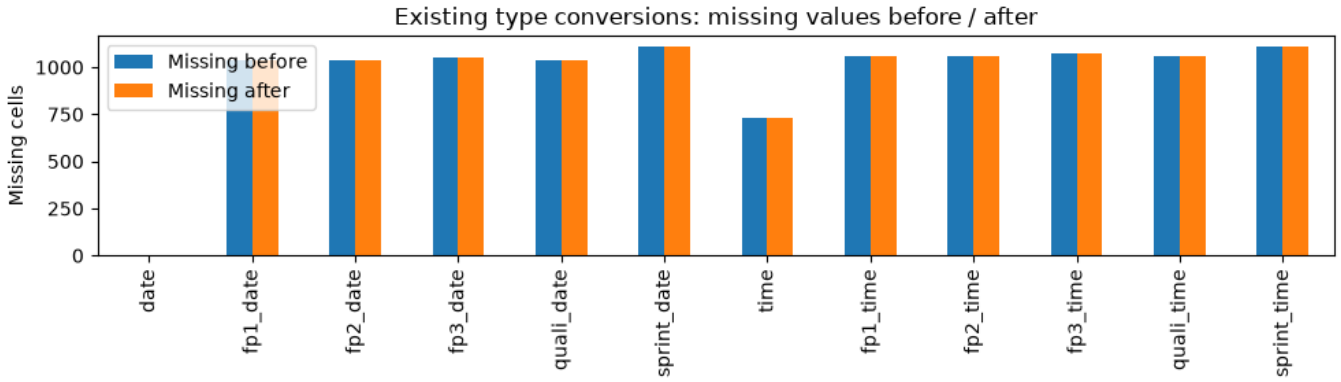
B28. 3.10.1 Races: Initial Auditing & Time Conversions

Notebook cell 267 | executed output | source: acl-ms1.ipynb



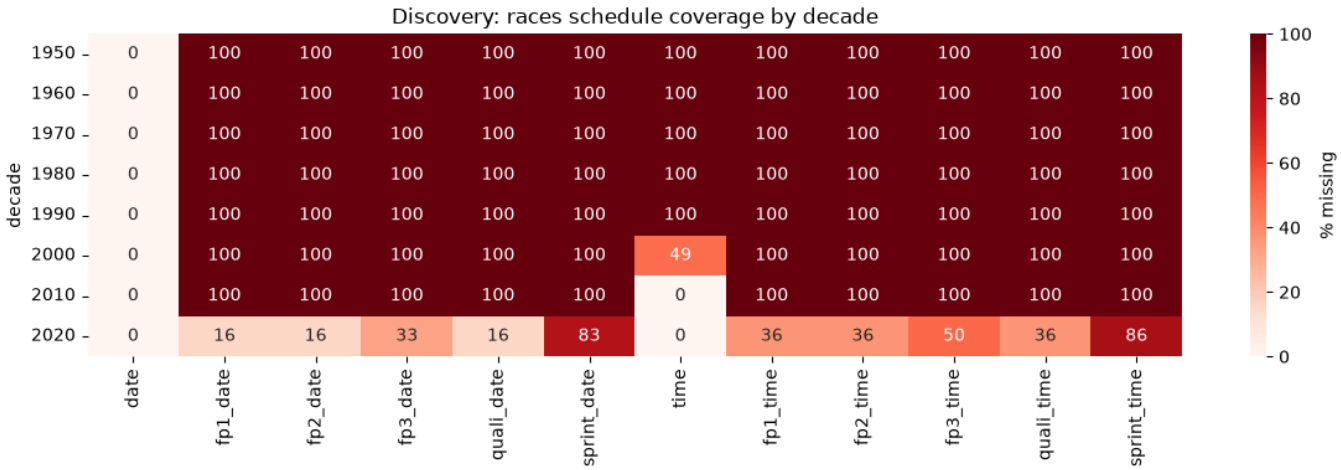
B29. 1. Show the existing cleaning before and after

Notebook cell 269 | executed output | source: acl-ms1.ipynb



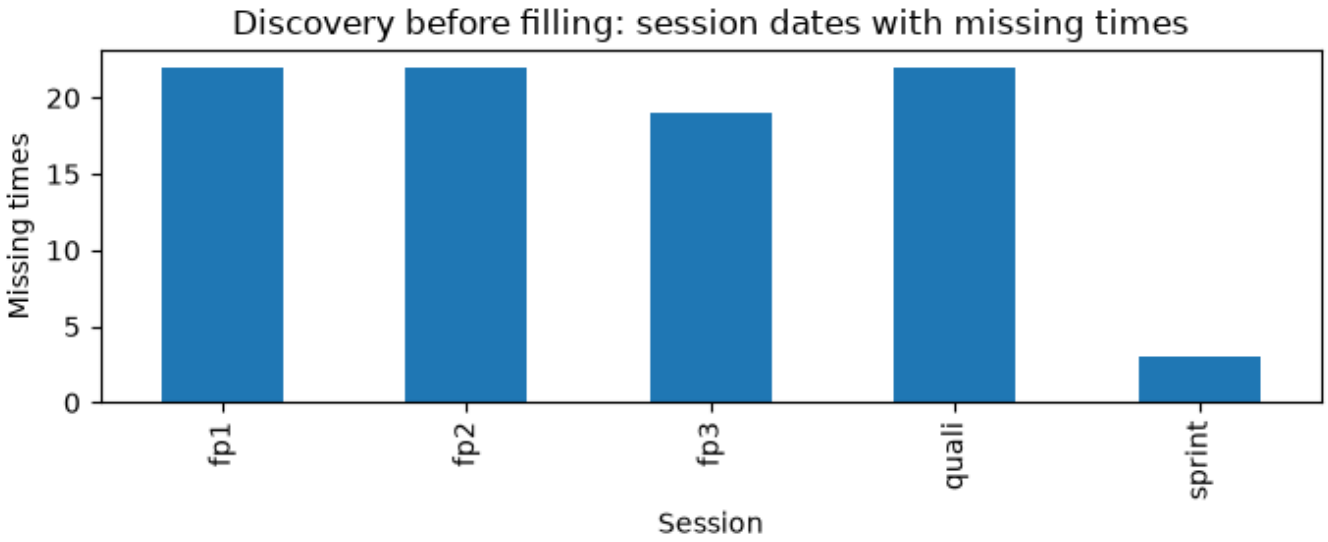
B30. 4. Schedule relationships and missing-value coverage

Notebook cell 277 | executed output | source: acl-ms1.ipynb



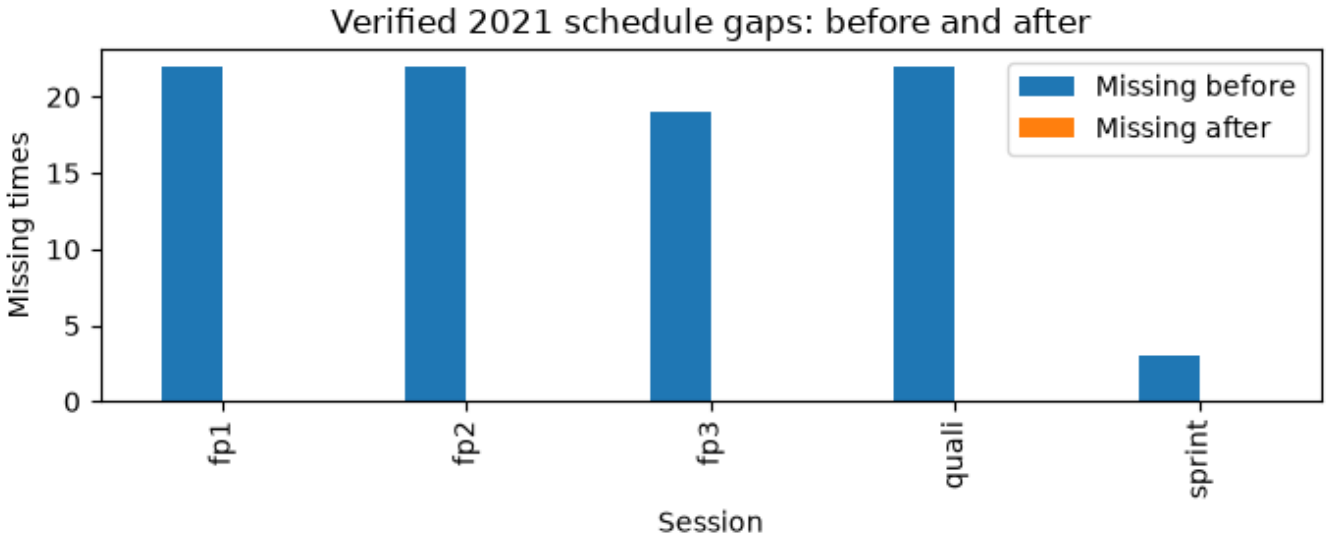
B31. 5. Recover the missing 2021 scheduled times using official evidence

Notebook cell 279 | executed output | source: acl-ms1.ipynb



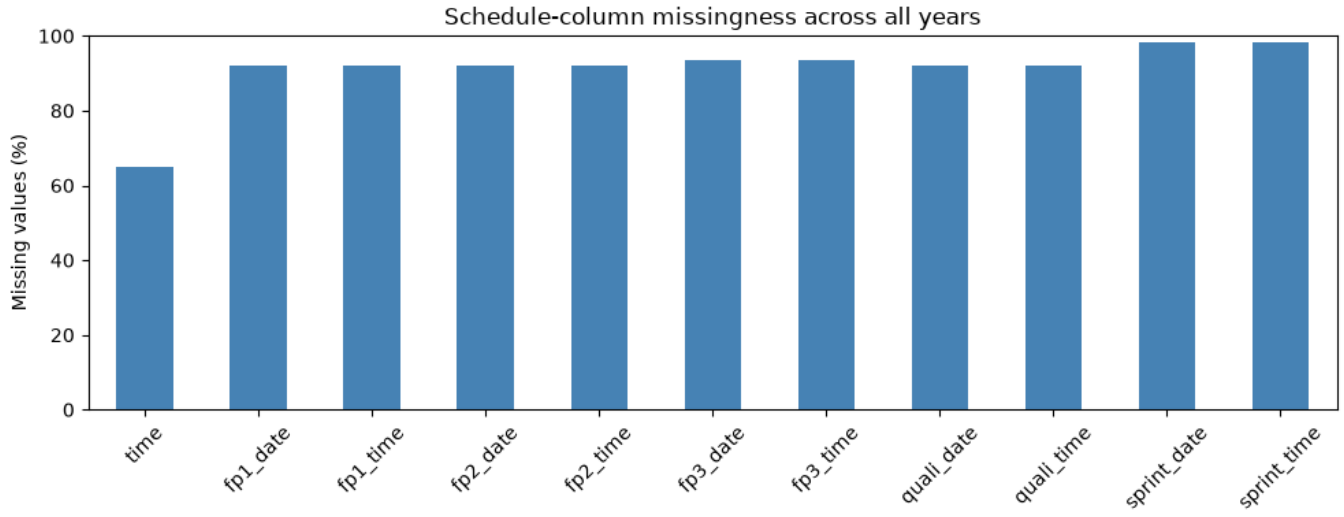
B32. Apply only the verified corrections to the clean copy

Notebook cell 284 | executed output | source: acl-ms1.ipynb



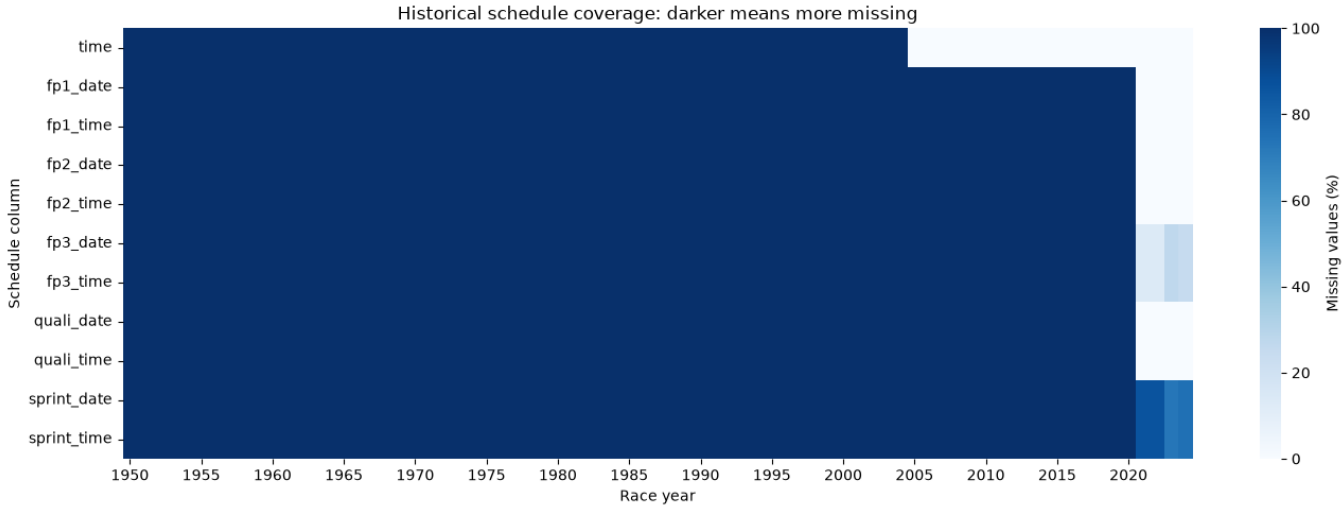
B33. 3.10.6 Historical coverage and race-feature selection

Notebook cell 288 | executed output | source: acl-ms1.ipynb



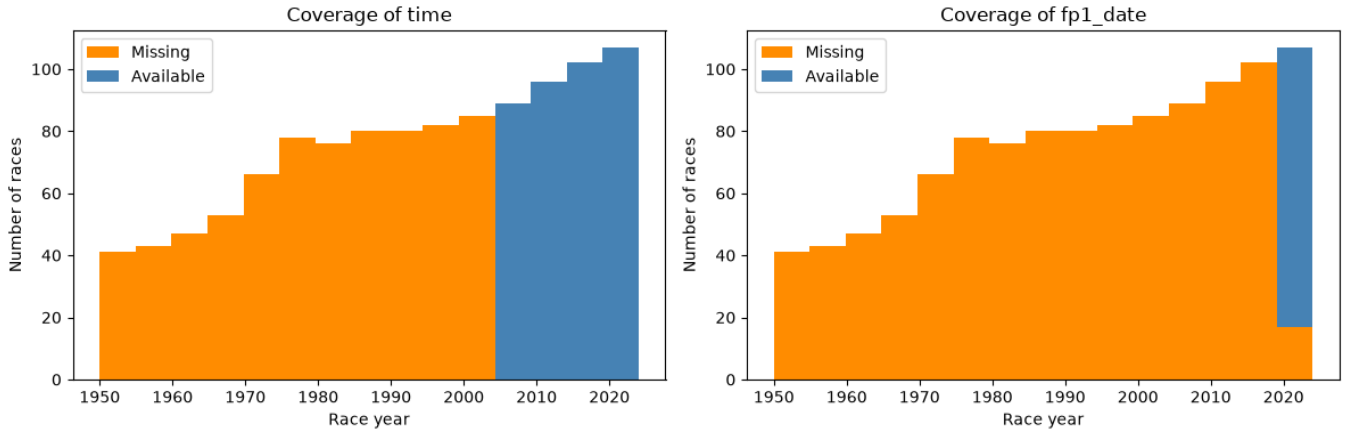
B34. 3.10.6 Historical coverage and race-feature selection

Notebook cell 289 | executed output | source: acl-ms1.ipynb



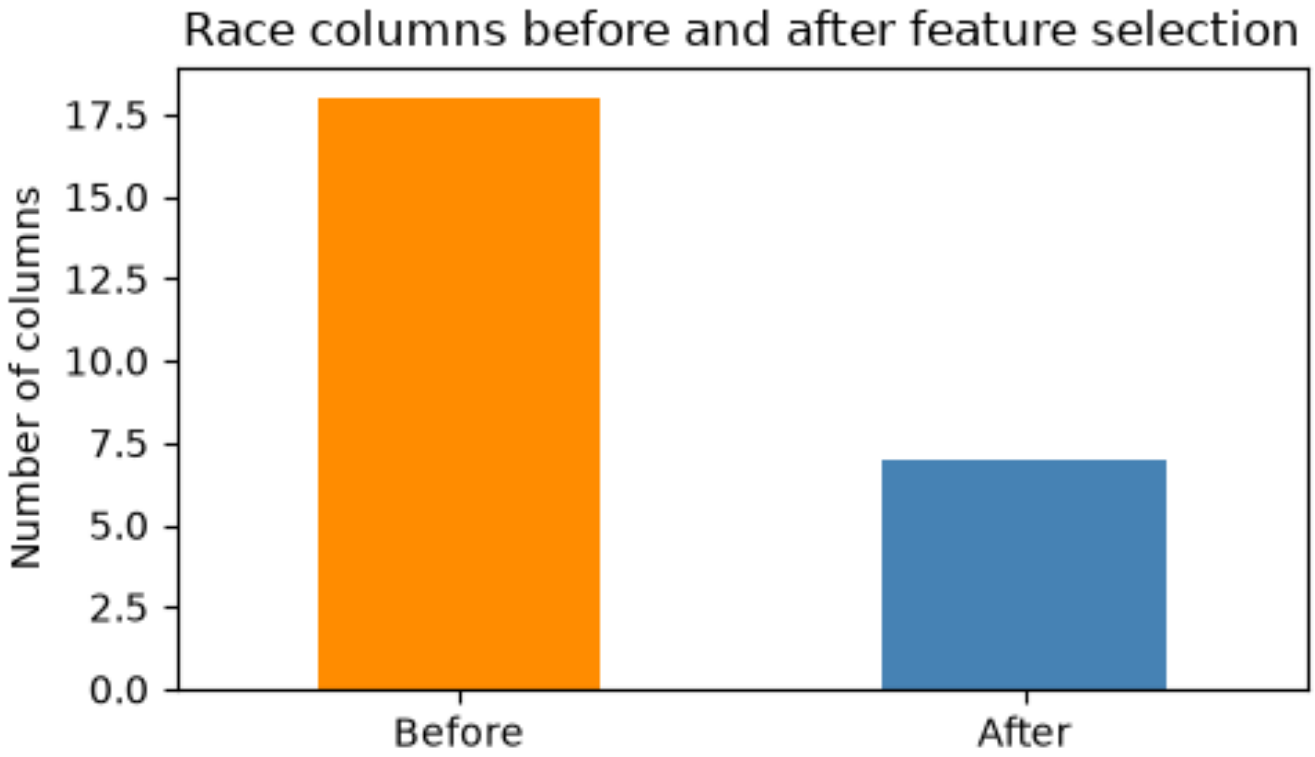
B35. 3.10.6 Historical coverage and race-feature selection

Notebook cell 290 | executed output | source: acl-ms1.ipynb



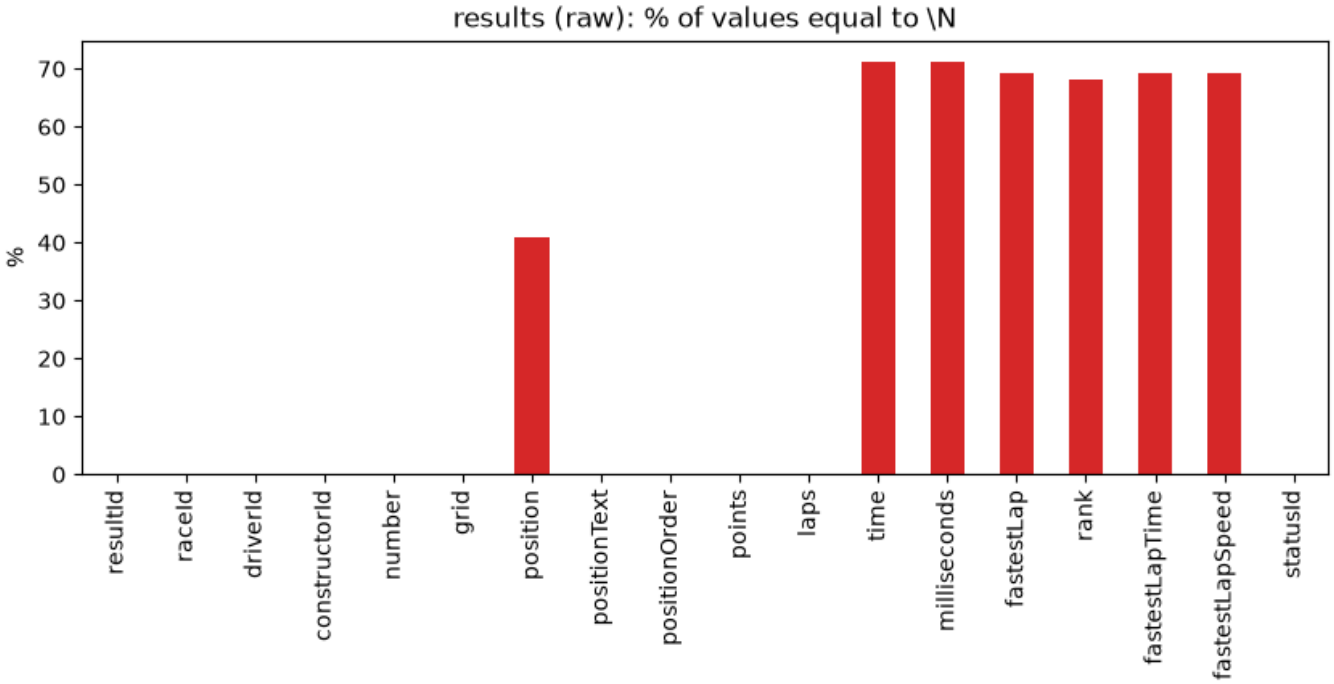
B36. Decision: exclude schedule columns from the final clean race table

Notebook cell 292 | executed output | source: acl-ms1.ipynb



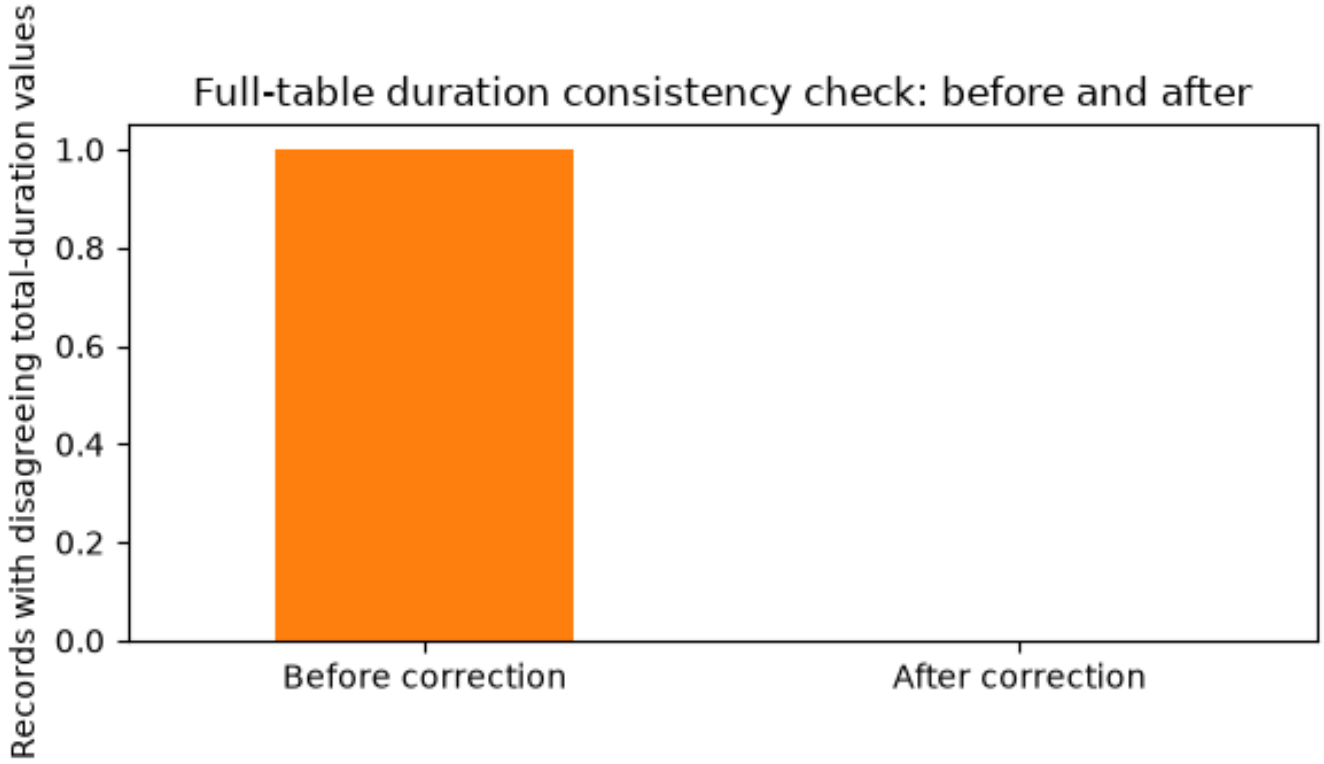
B37. 3.11.2 Results: Initial Auditing & Time Conversions

Notebook cell 304 | executed output | source: acl-ms1.ipynb



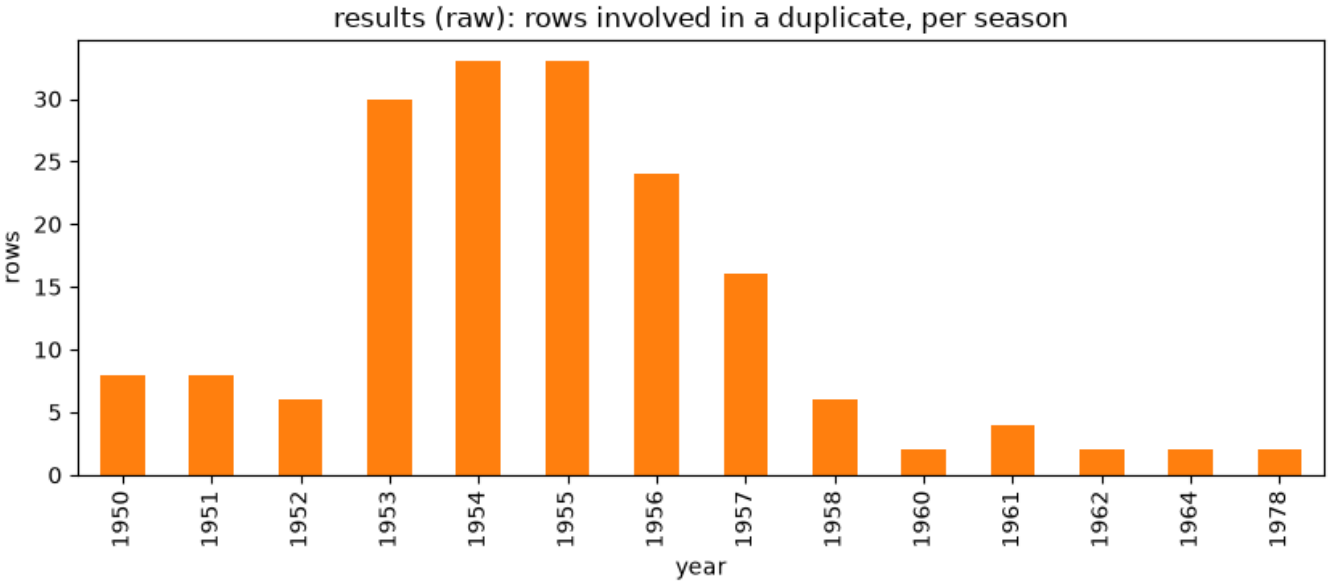
B38. Verify the discovered mismatch using the official result

Notebook cell 312 | executed output | source: acl-ms1.ipynb



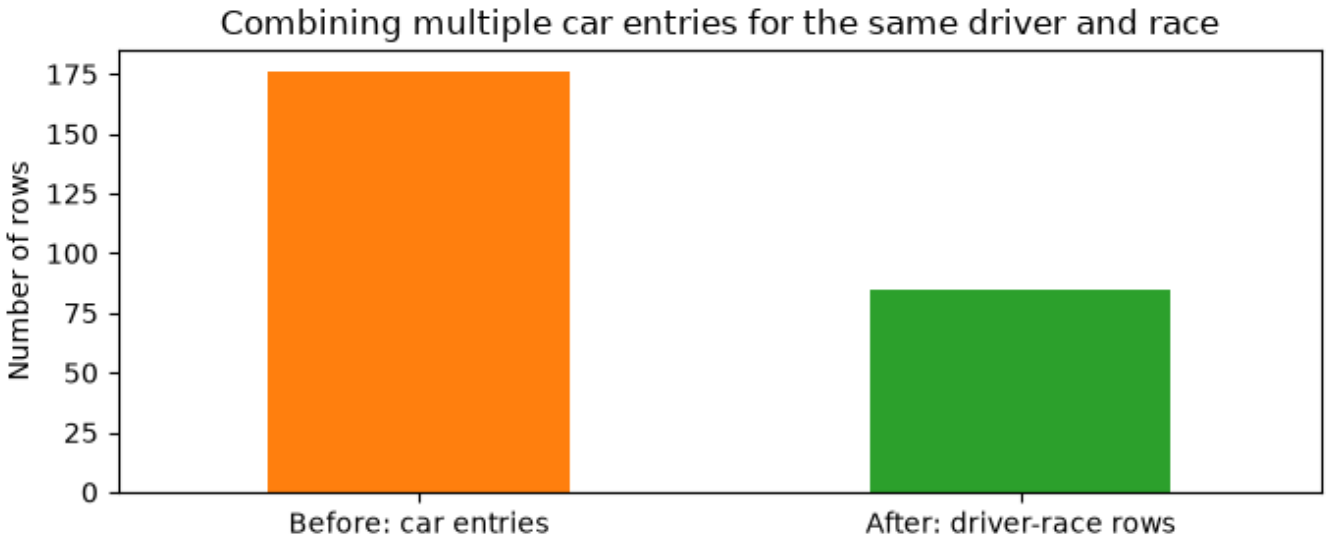
B39. 3.11.3 Results: Shared Drive Consolidation & Integrity Checks

Notebook cell 317 | executed output | source: acl-ms1.ipynb



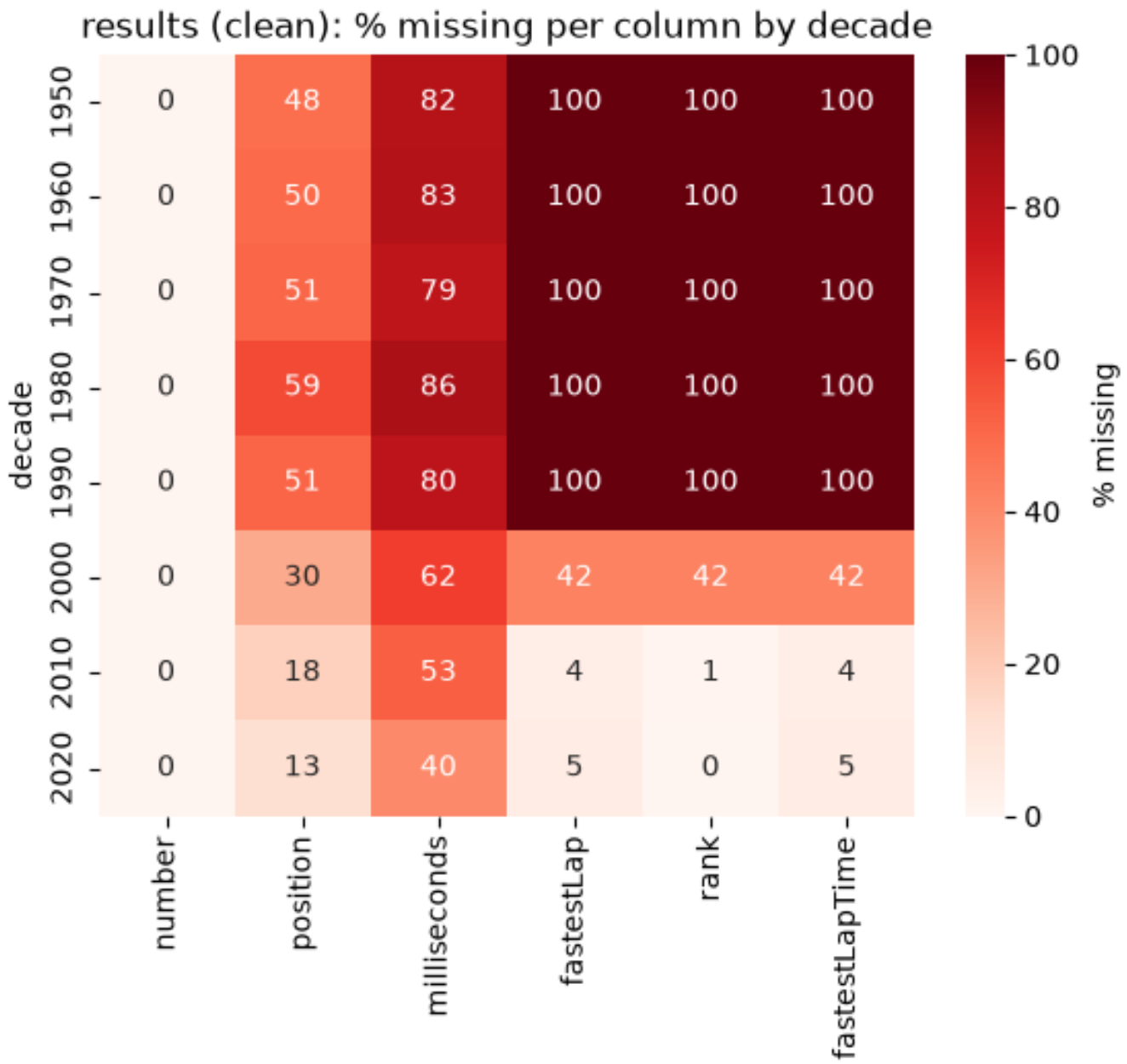
B40. 3.11.3 Results: Shared Drive Consolidation & Integrity Checks

Notebook cell 320 | executed output | source: acl-ms1.ipynb



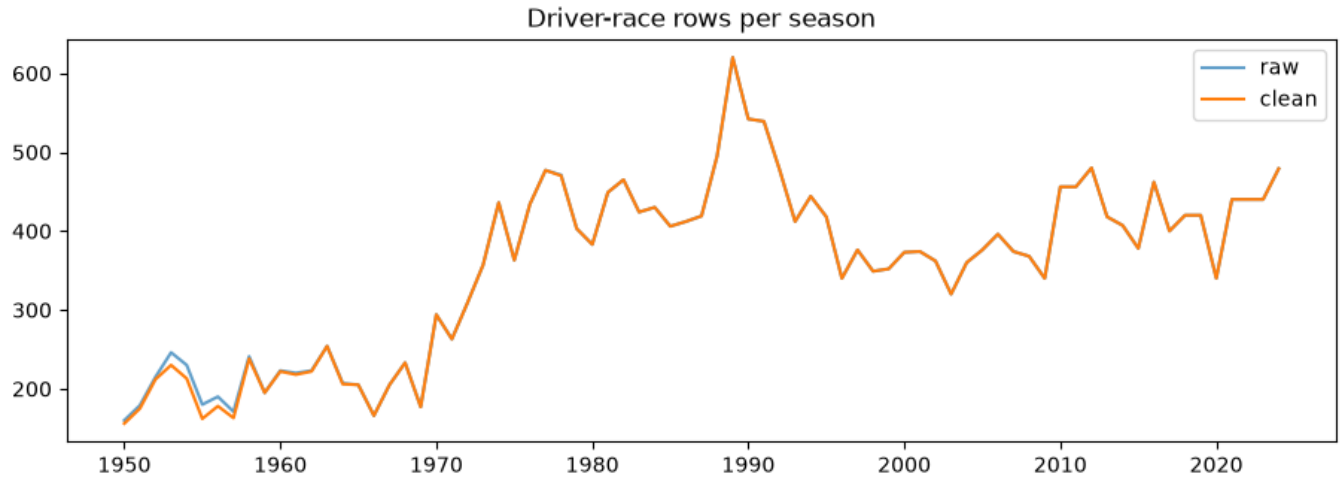
B41. 3.11.3 Results: Shared Drive Consolidation & Integrity Checks

Notebook cell 325 | executed output | source: acl-ms1.ipynb



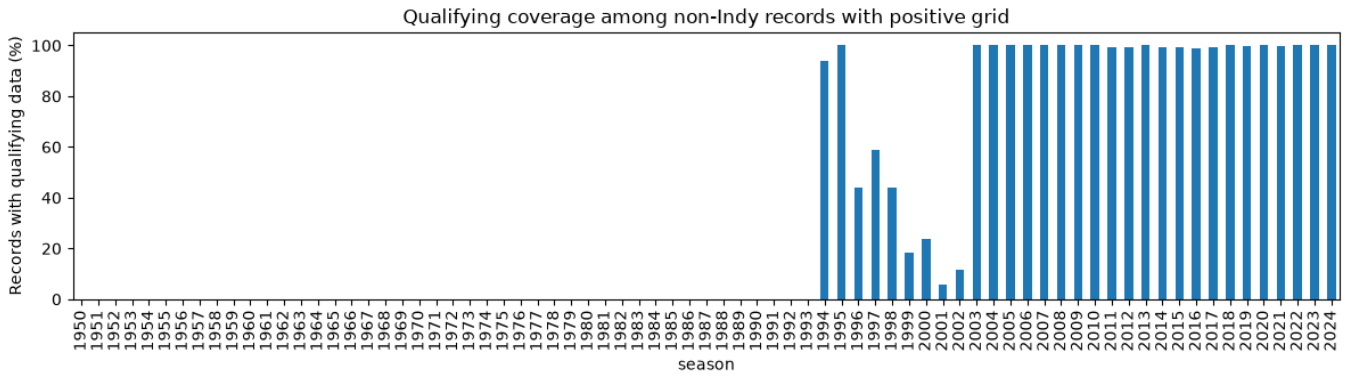
B42. 3.11.3 Results: Shared Drive Consolidation & Integrity Checks

Notebook cell 326 | executed output | source: acl-ms1.ipynb



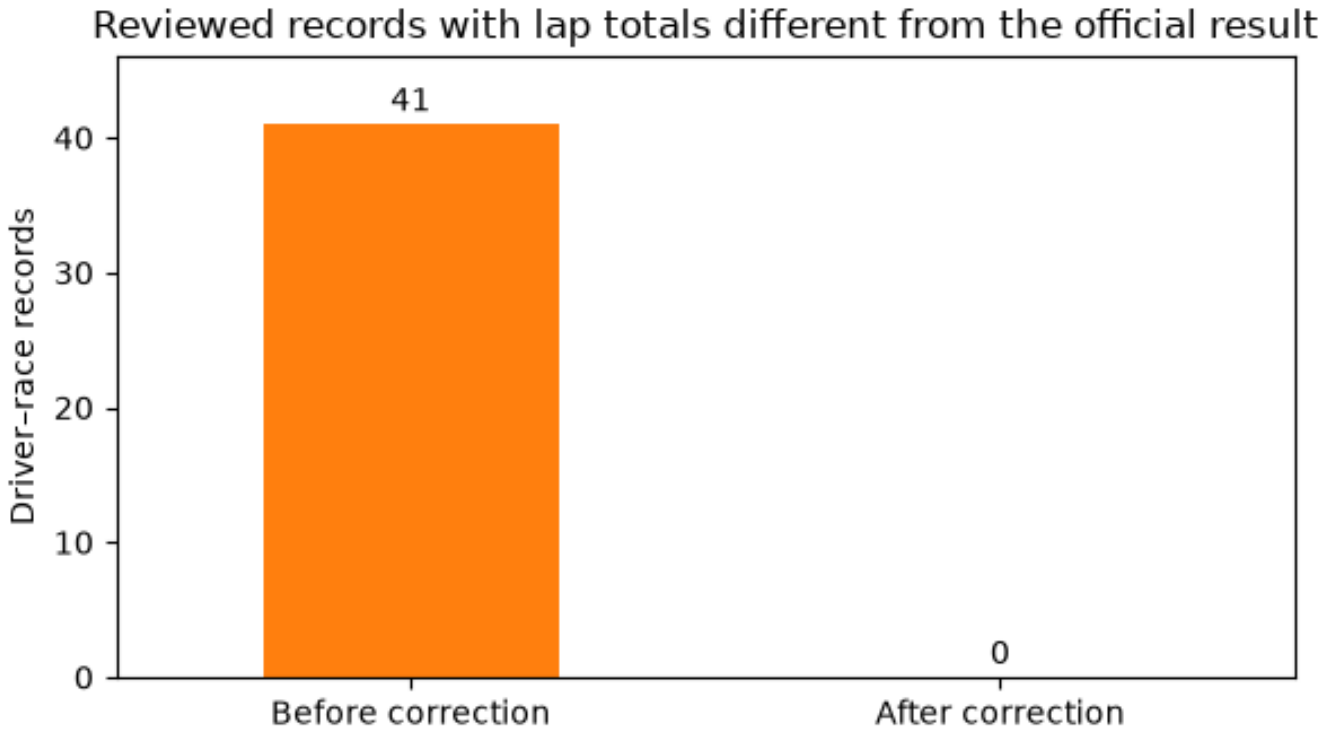
B43. 3.11.4 Results: Grid, Indianapolis and qualifying-coverage diagnostics

Notebook cell 335 | executed output | source: acl-ms1.ipynb



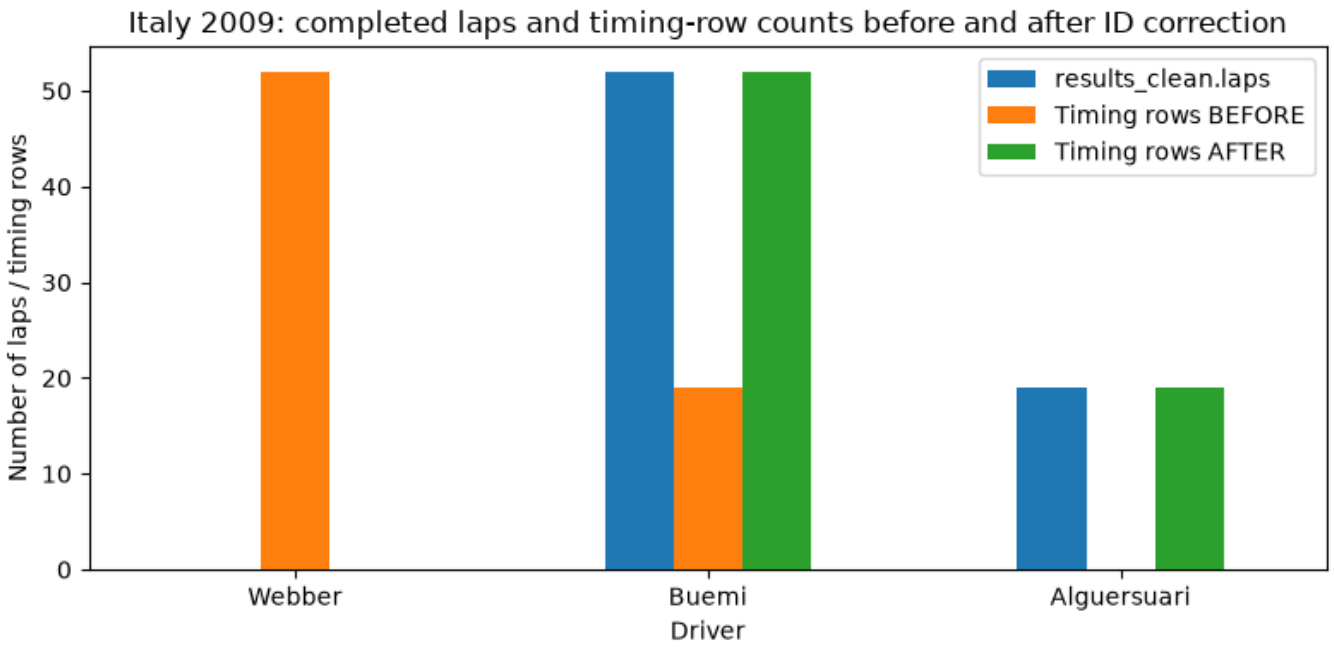
B44. Correct 41 verified completed-lap totals

Notebook cell 344 | executed output | source: acl-ms1.ipynb



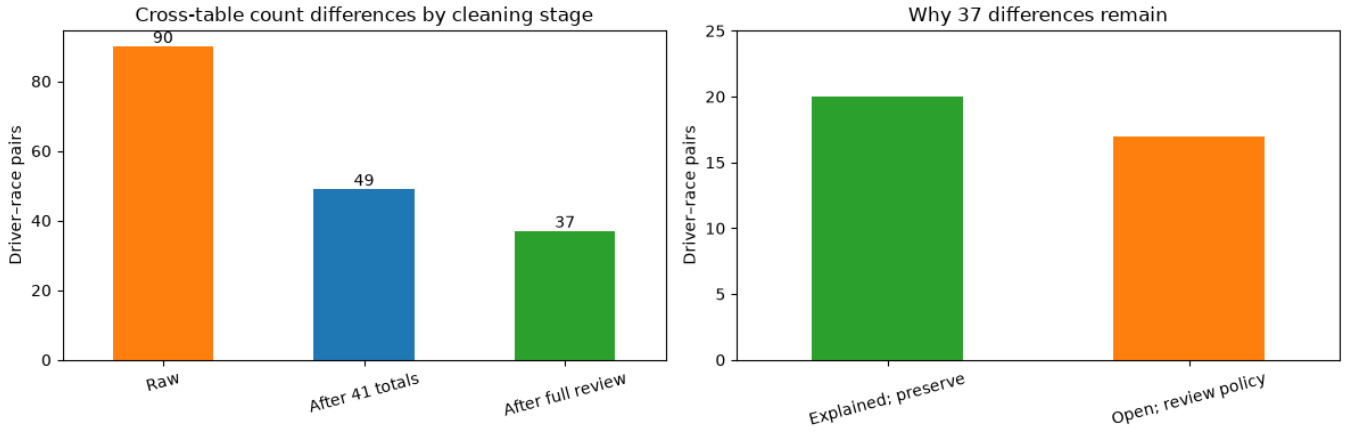
B45. Check external evidence before changing an ID

Notebook cell 357 | executed output | source: acl-ms1.ipynb



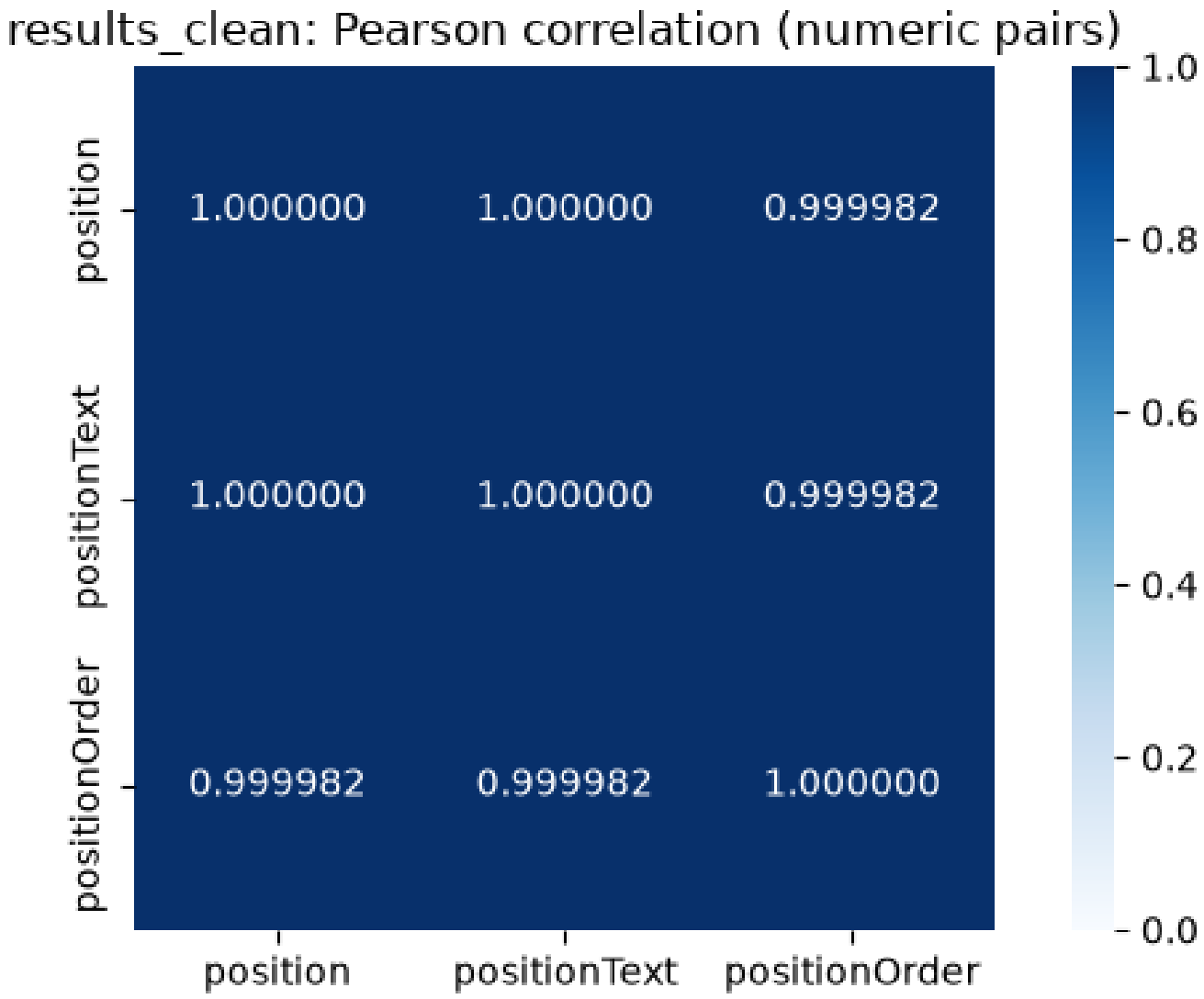
B46. 8. Final before-and-after checks

Notebook cell 361 | executed output | source: acl-ms1.ipynb



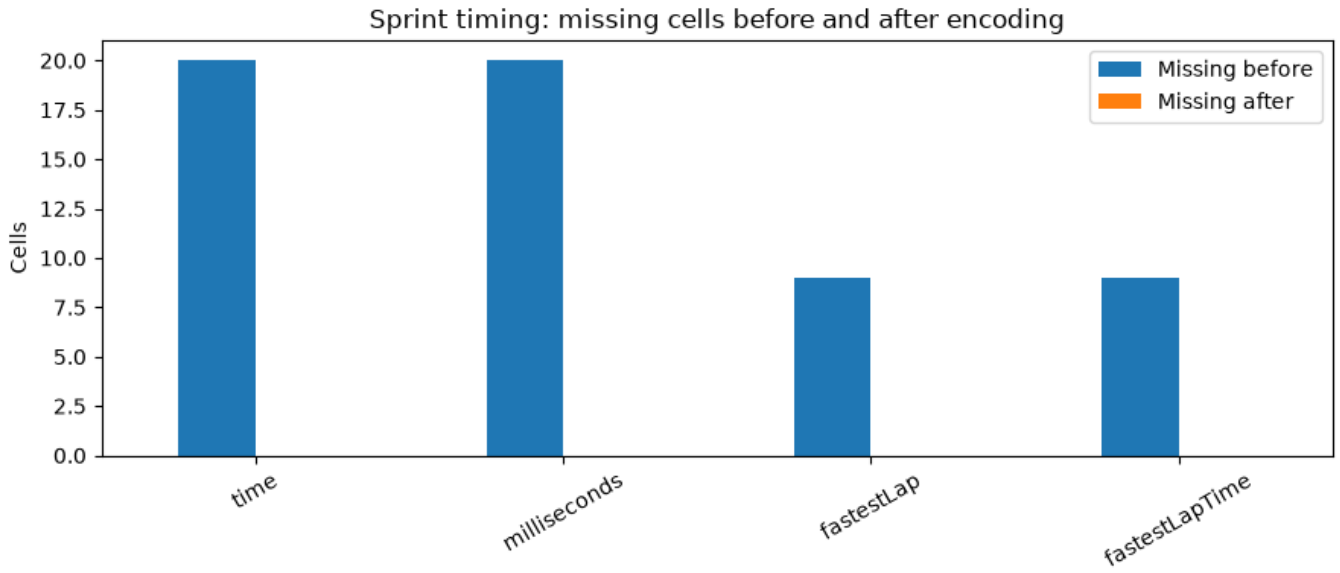
B47. Results: compare position columns and remove redundancy

Notebook cell 366 | executed output | source: acl-ms1.ipynb



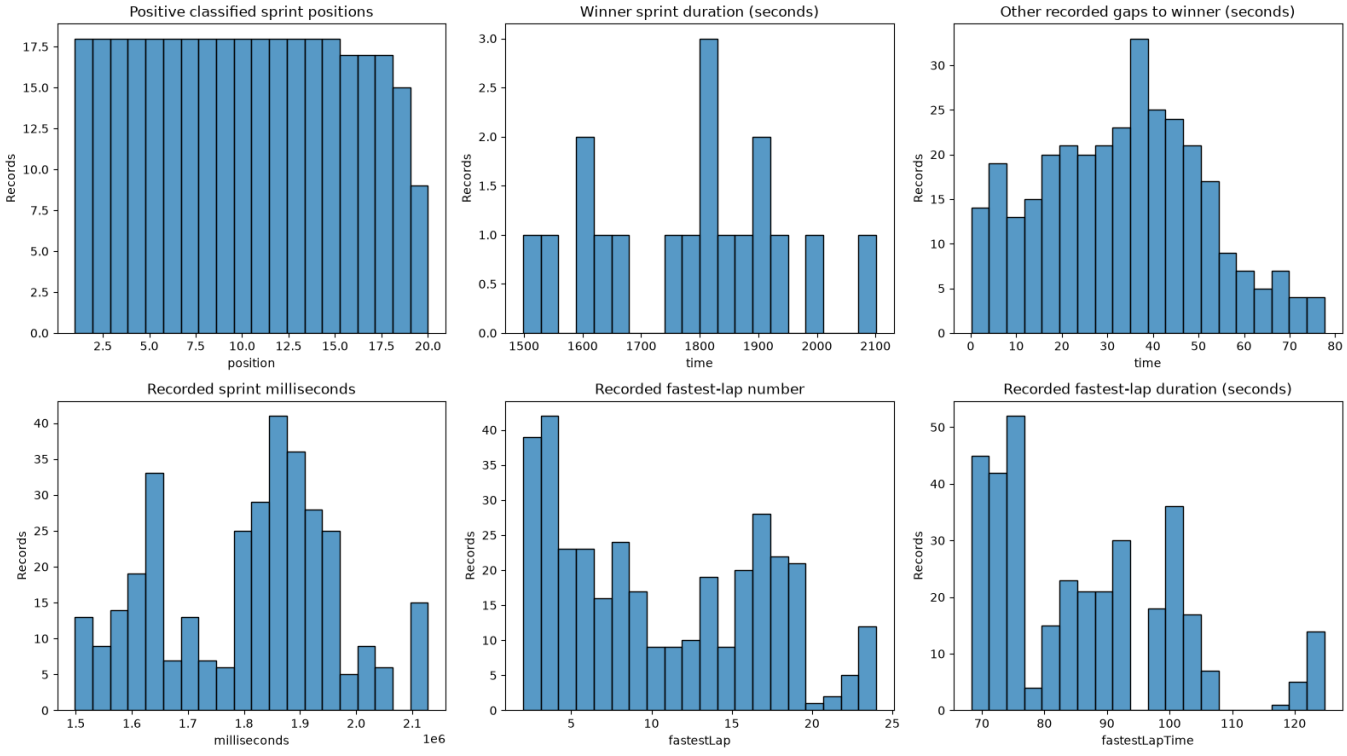
B48. Sprint timing: investigate missing values and fill with an unavailable code

Notebook cell 387 | executed output | source: acl-ms1.ipynb



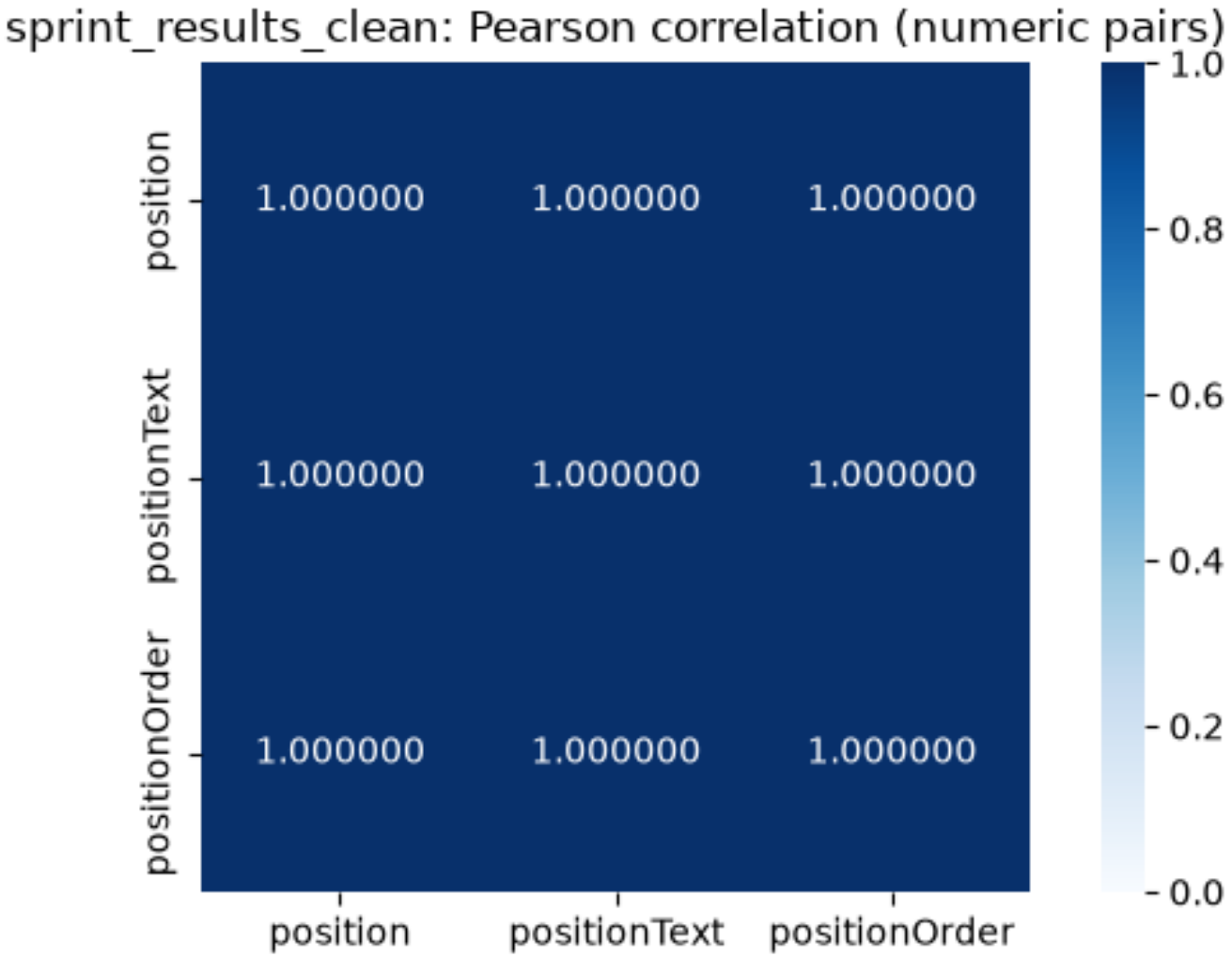
B49. 3.13.4 Feature Distribution Analysis

Notebook cell 399 | executed output | source: acl-ms1.ipynb



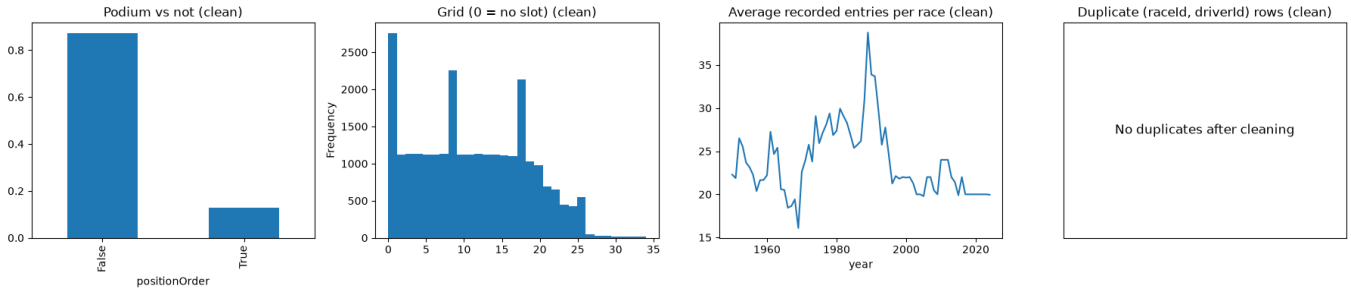
B50. Sprint results: compare position columns and remove redundancy

Notebook cell 401 | executed output | source: acl-ms1.ipynb



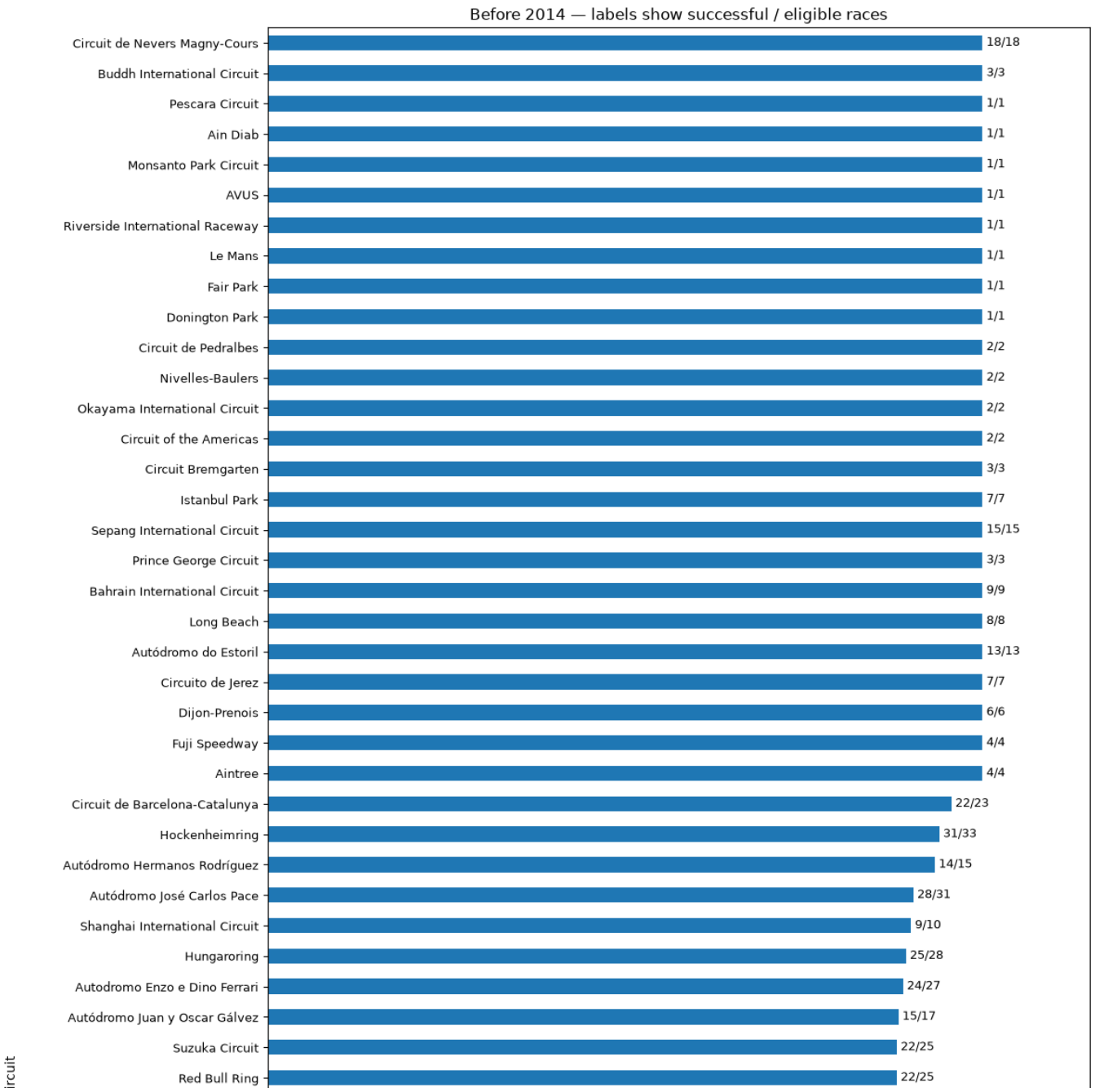
B51. 3.15.2 Target Class & Feature Sanity (Post-Cleaning)

Notebook cell 418 | executed output | source: acl-ms1.ipynb



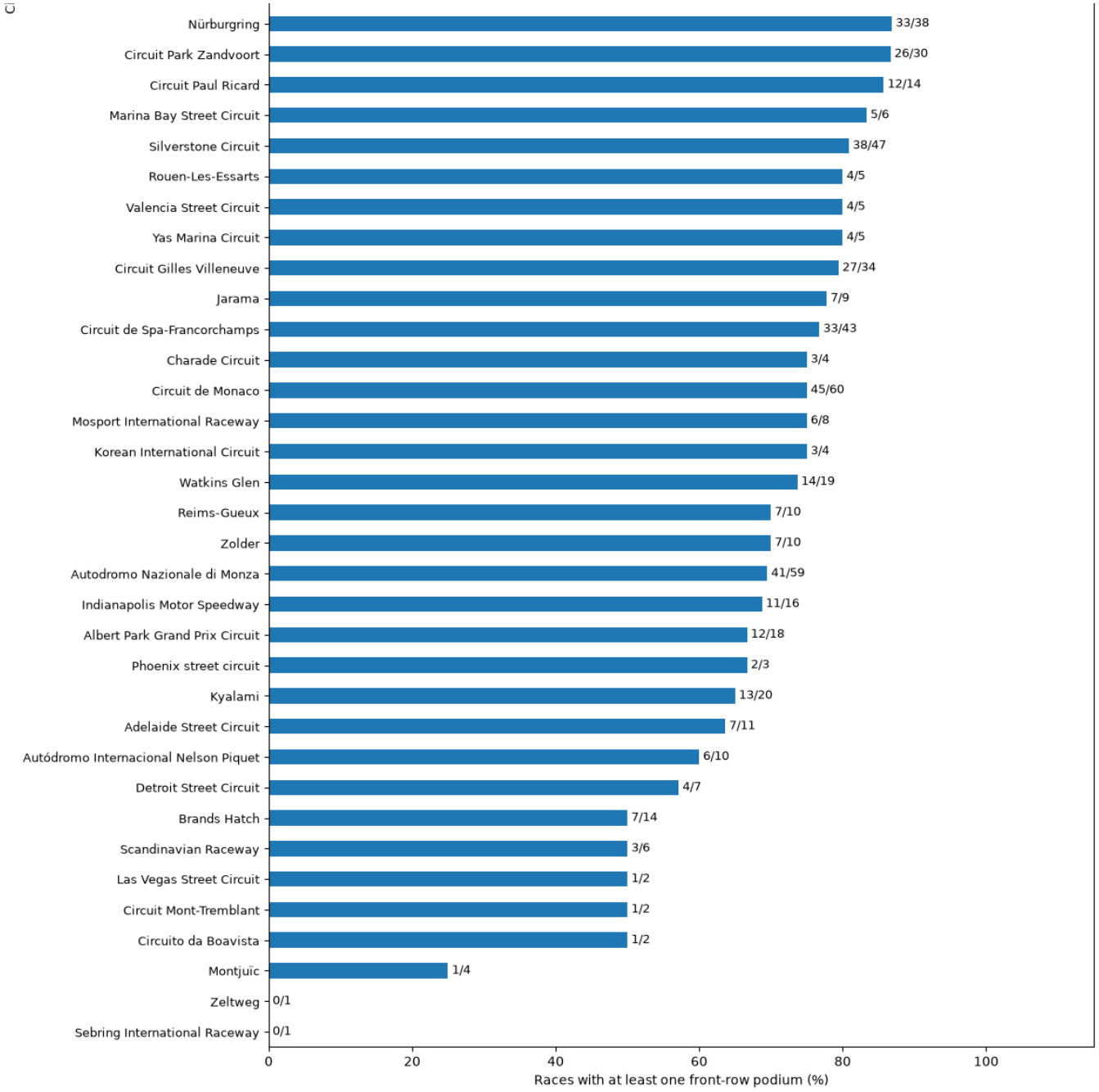
B52. Question 1 calculation: at least one front-row driver on the podium (part 1/2)

Notebook cell 437 | executed output | source: acl-ms1.ipynb



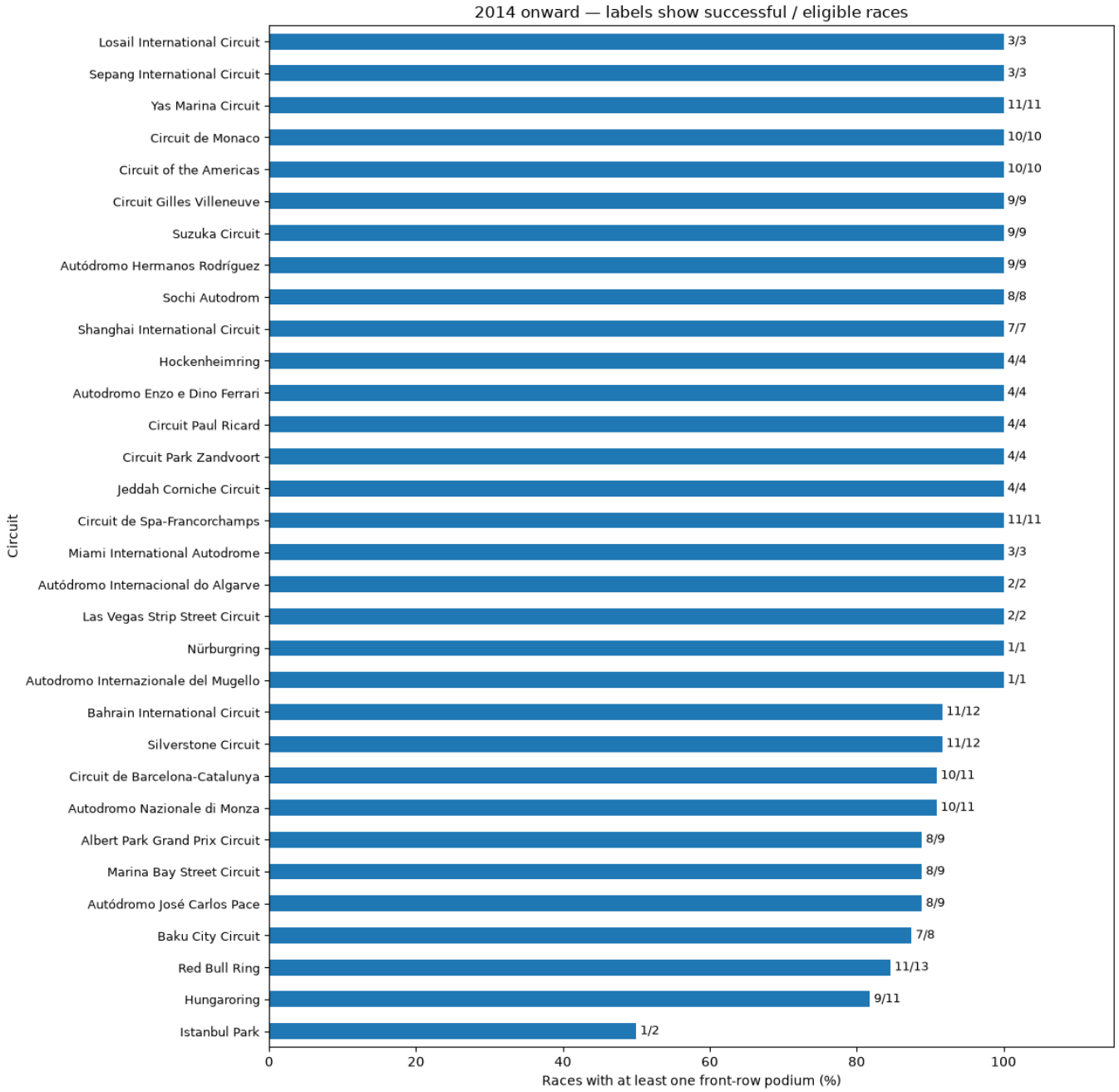
B52. Question 1 calculation: at least one front-row driver on the podium (part 2/2)

Notebook cell 437 | executed output | source: acl-ms1.ipynb



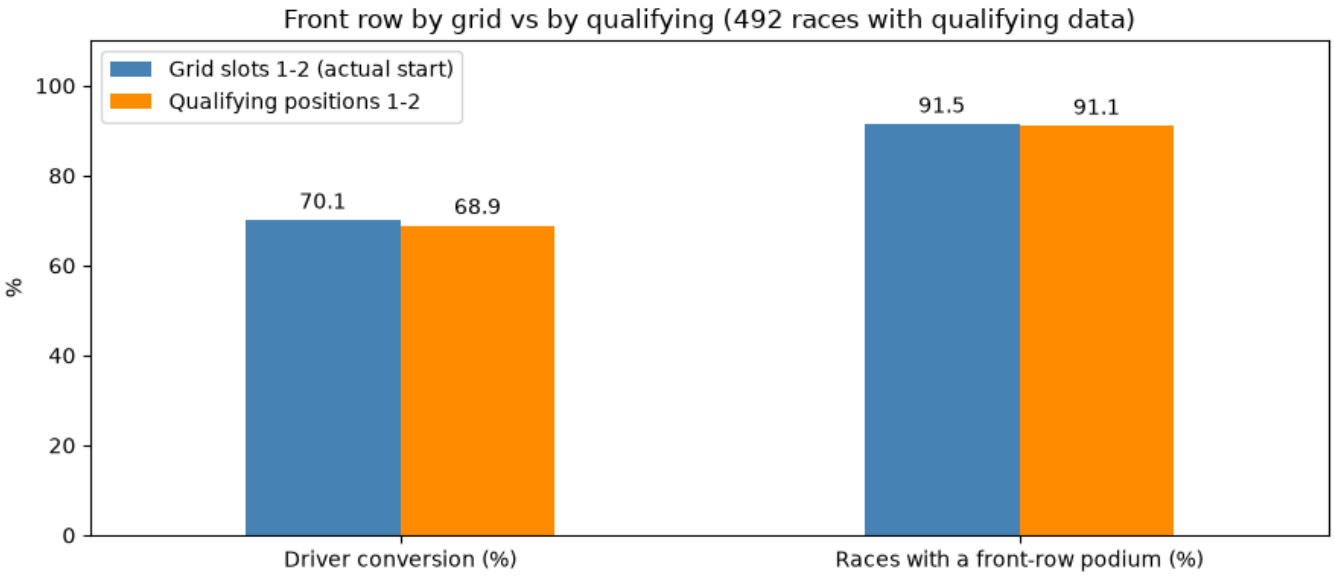
B53. Question 1 calculation: at least one front-row driver on the podium

Notebook cell 437 | executed output | source: acl-ms1.ipynb



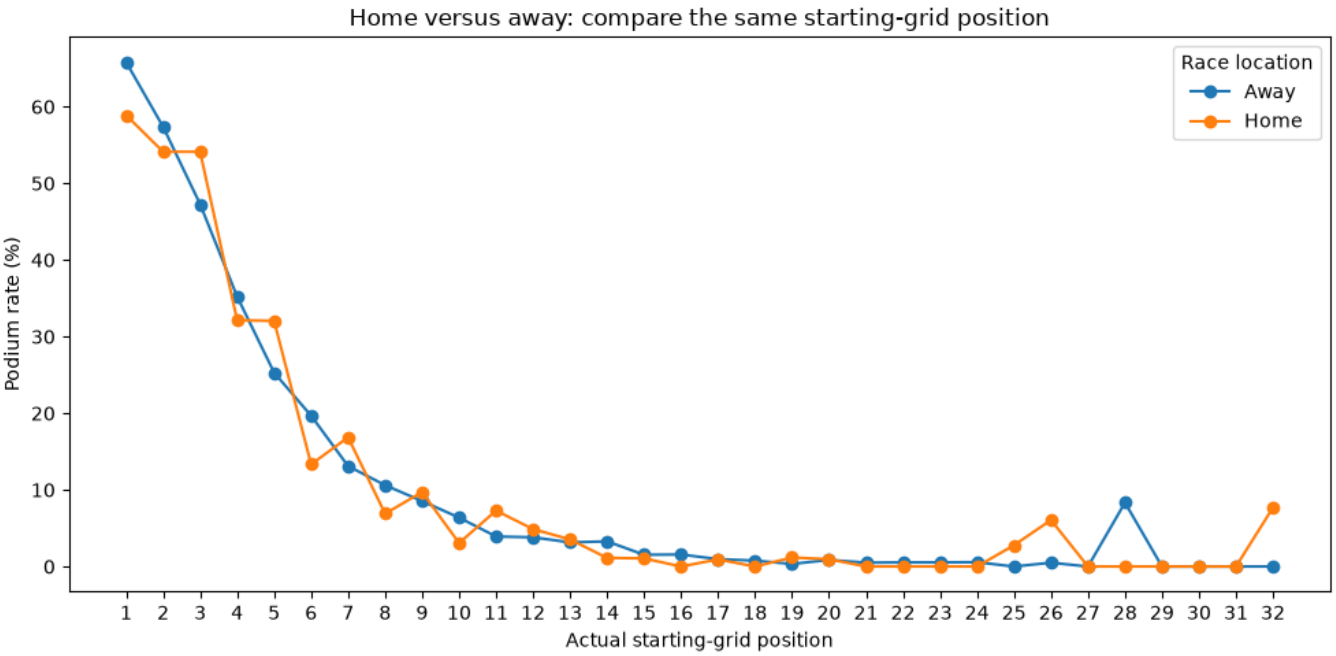
B54. Question 1 - Qualifying position versus starting grid

Notebook cell 440 | executed output | source: acl-ms1.ipynb



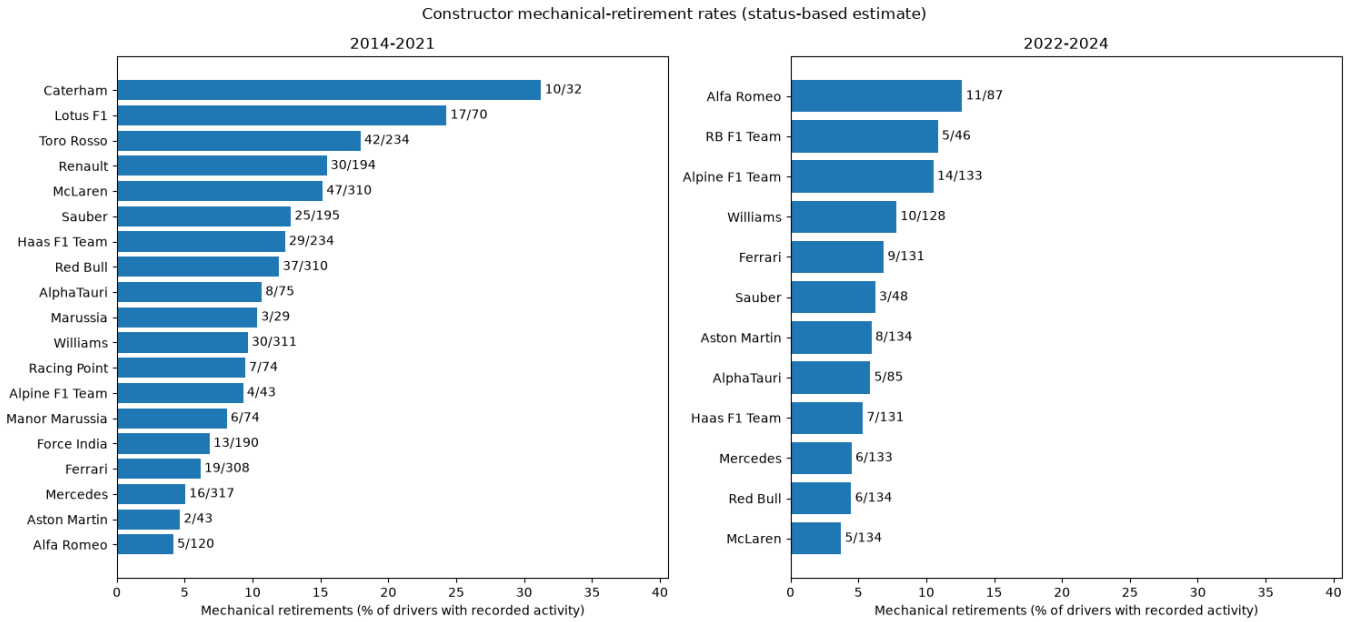
B55. Combine the comparisons fairly

Notebook cell 455 | executed output | source: acl-ms1.ipynb



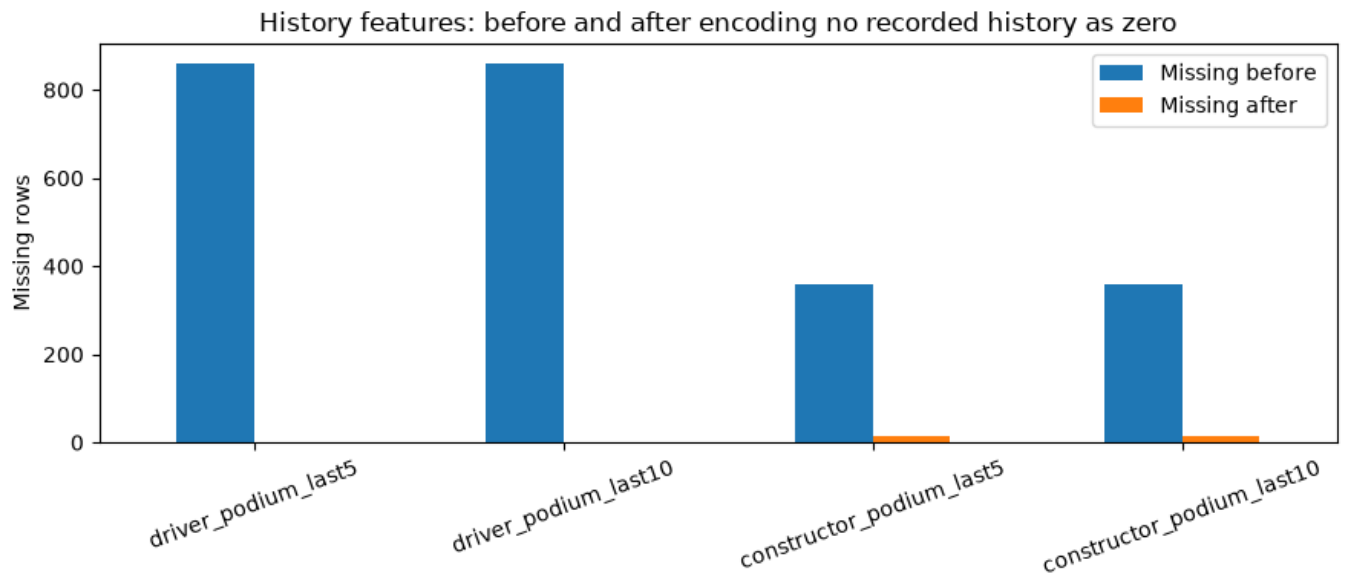
B56. Read the ranking and compare periods

Notebook cell 468 | executed output | source: acl-ms1.ipynb



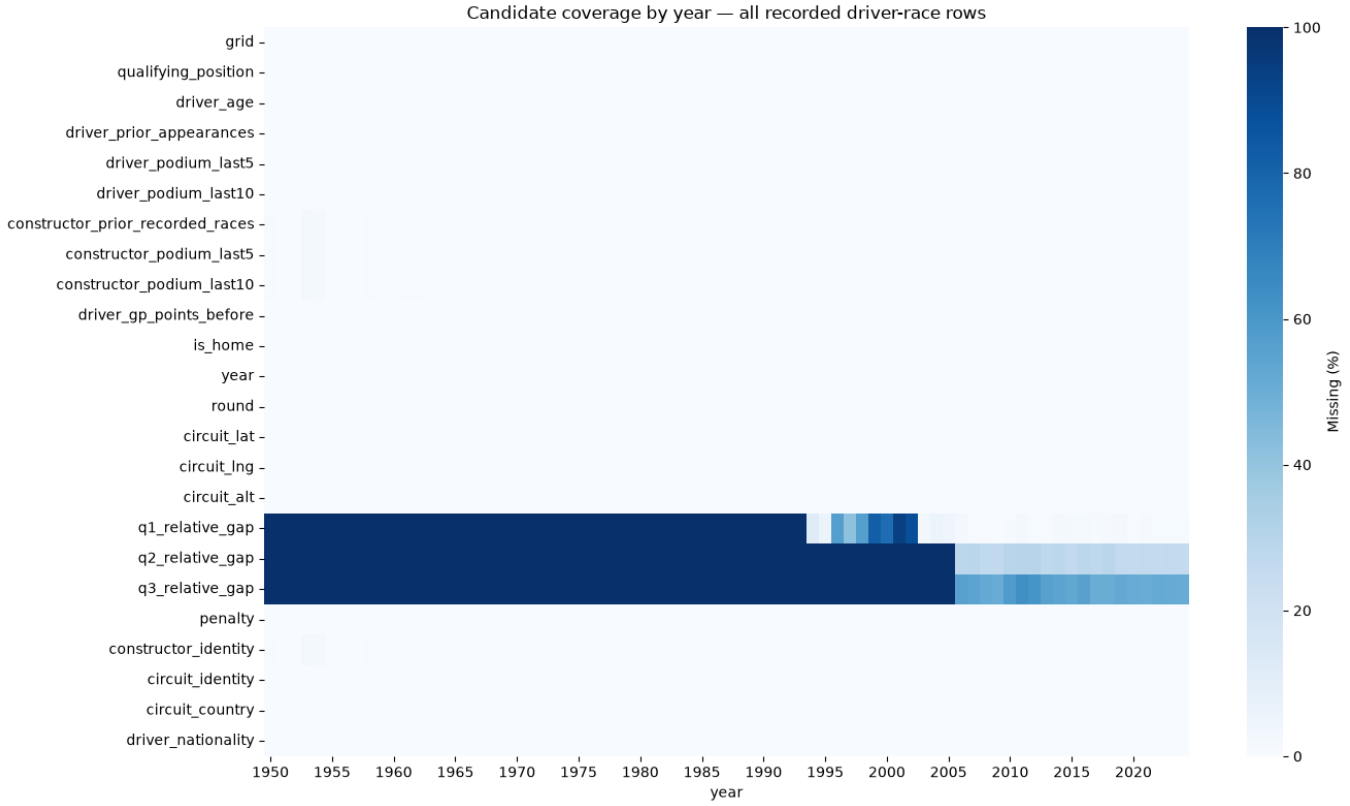
B57. Show how the history features work

Notebook cell 480 | executed output | source: acl-ms1.ipynb



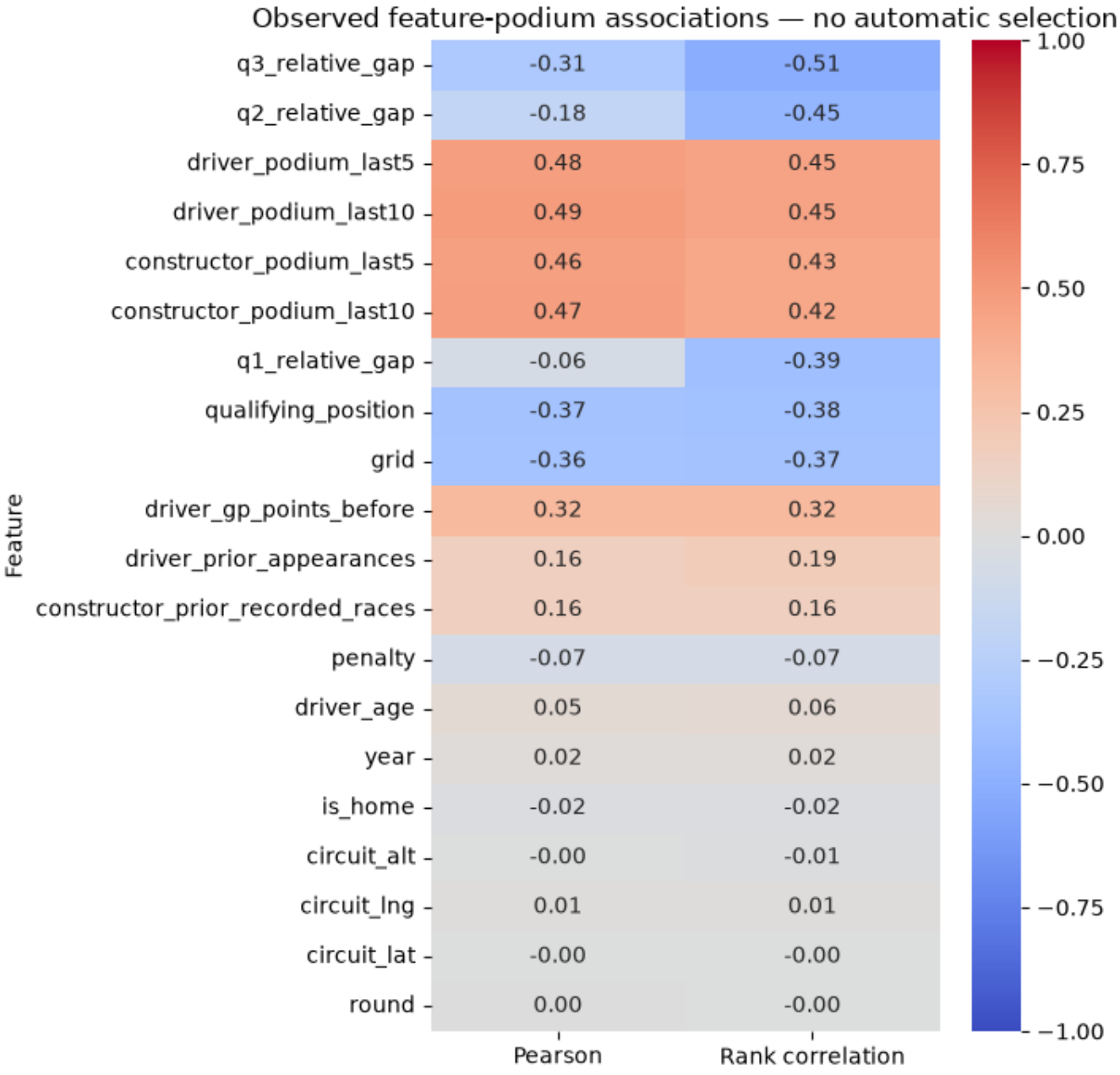
B58. 5.4 Coverage before relevance

Notebook cell 482 | executed output | source: acl-ms1.ipynb



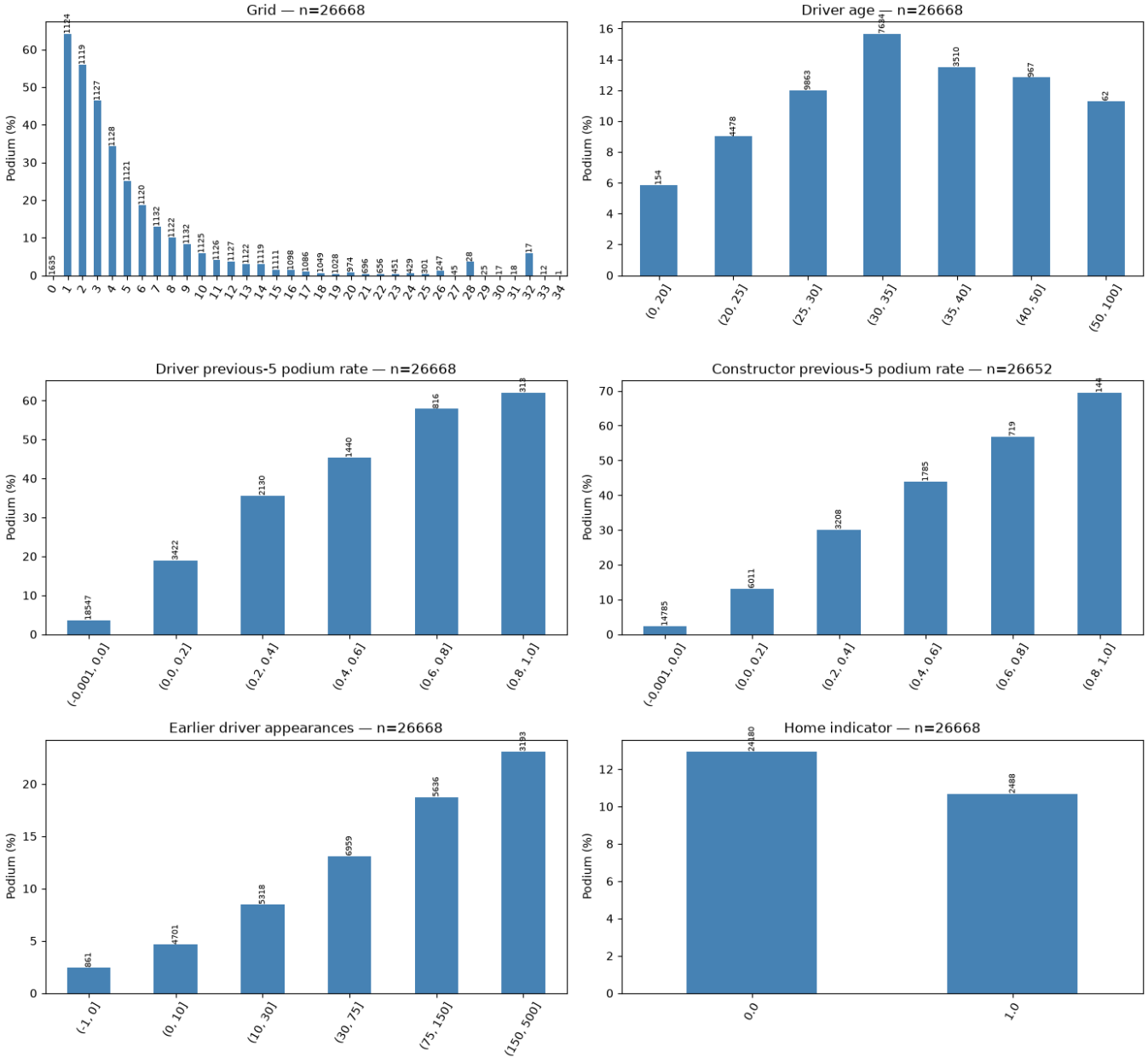
B59. 5.5 Relationships with podium - descriptive evidence

Notebook cell 484 | executed output | source: acl-ms1.ipynb



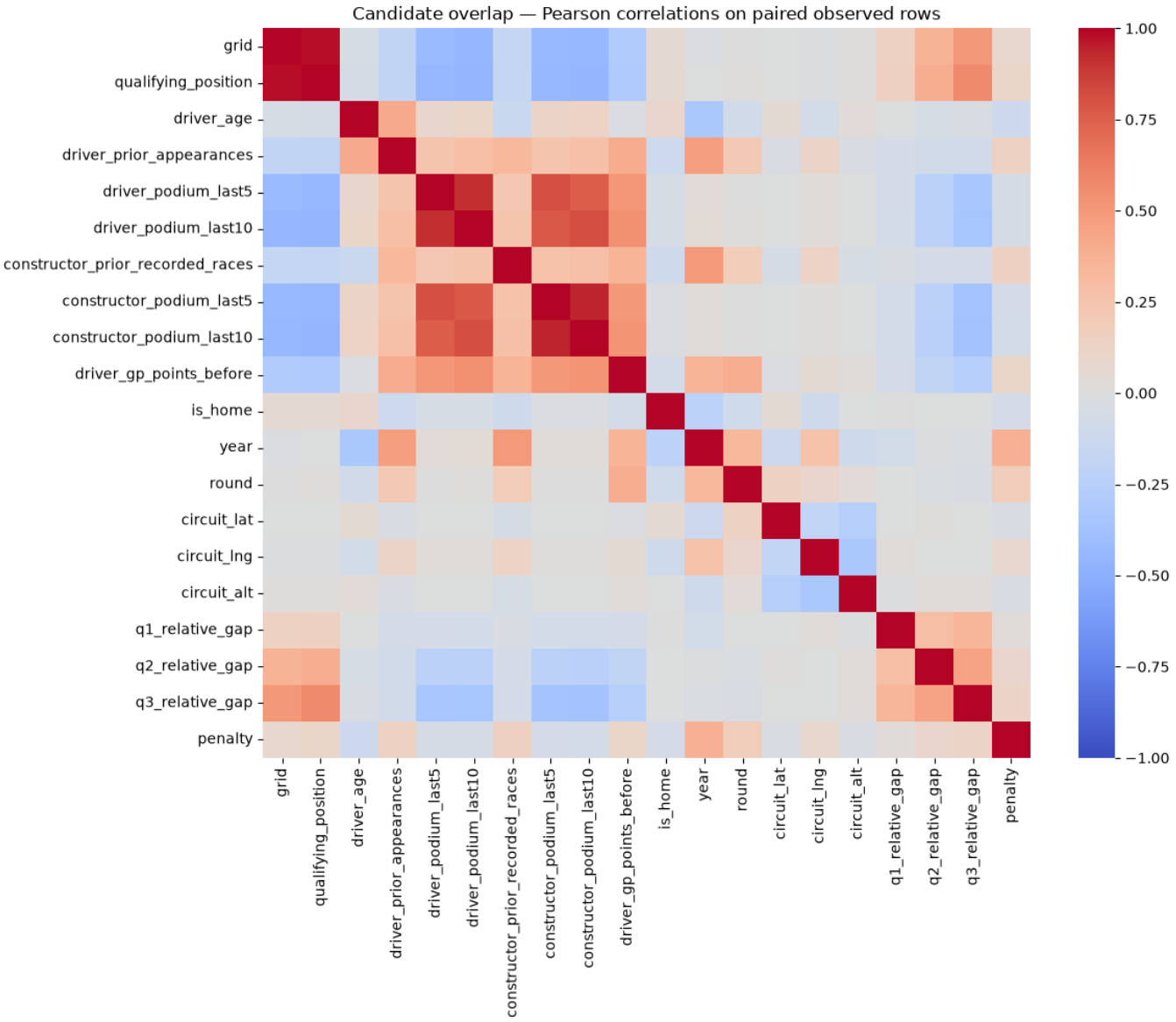
B60. 5.5 Relationships with podium - descriptive evidence

Notebook cell 485 | executed output | source: acl-ms1.ipynb



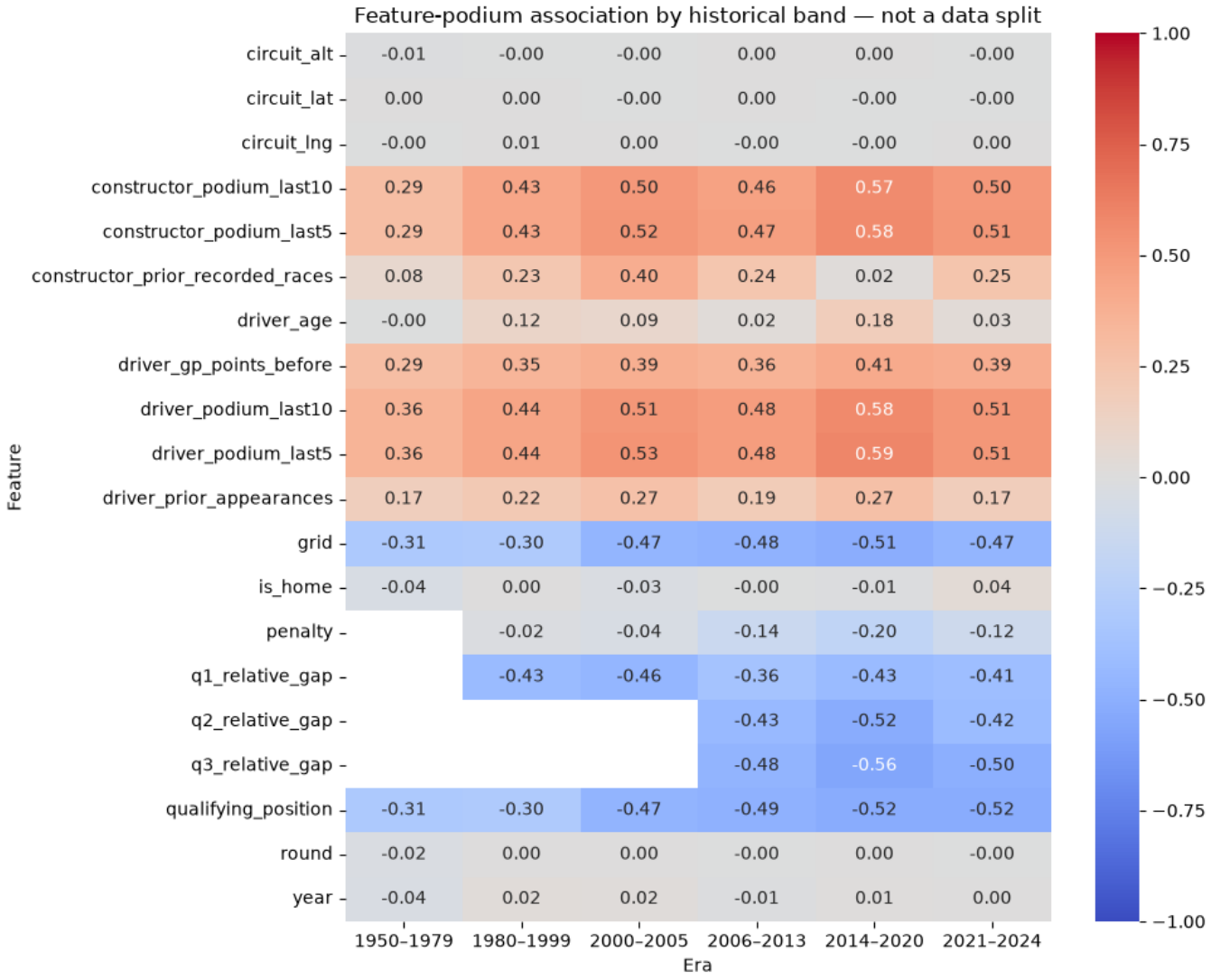
B61. 5.6 Check overlapping information and historical stability

Notebook cell 488 | executed output | source: acl-ms1.ipynb



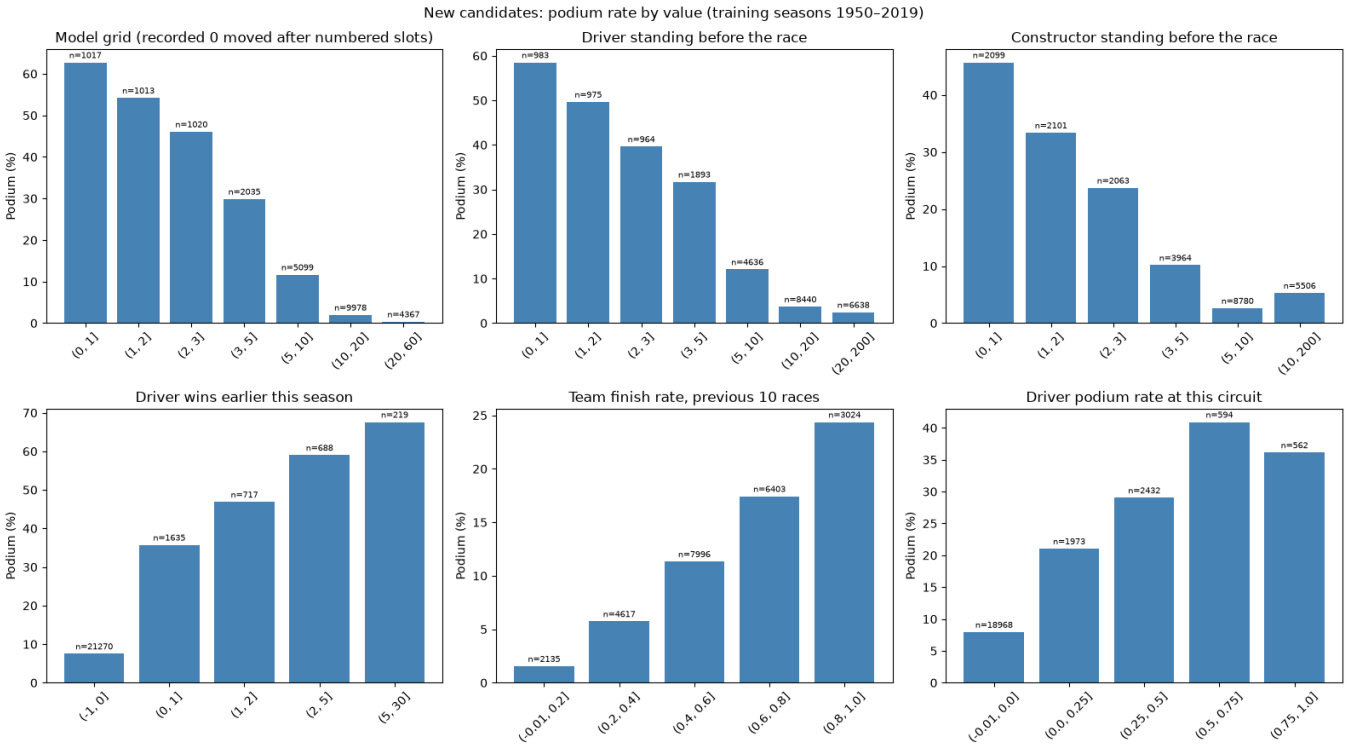
B62. 5.6 Check overlapping information and historical stability

Notebook cell 488 | executed output | source: acl-ms1.ipynb



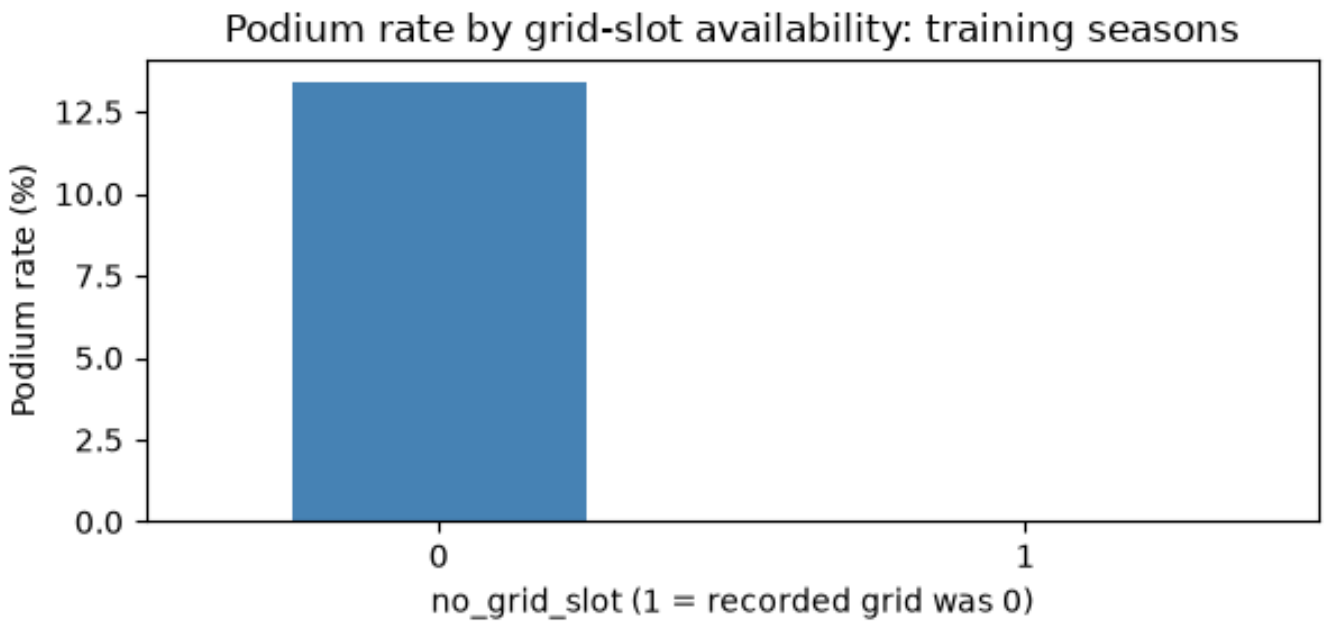
B63. Check the new columns

Notebook cell 502 | executed output | source: acl-ms1.ipynb



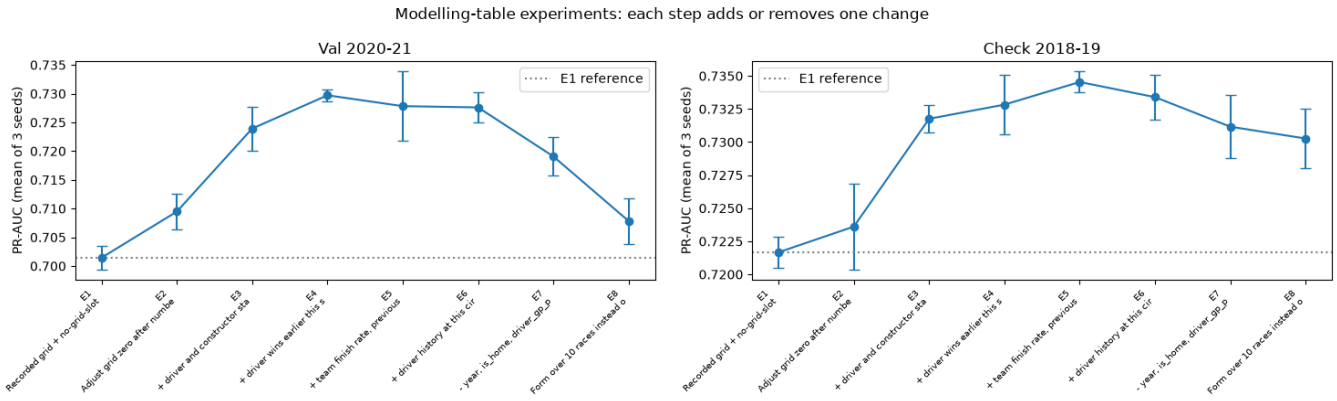
B64. Check the new columns

Notebook cell 502 | executed output | source: acl-ms1.ipynb



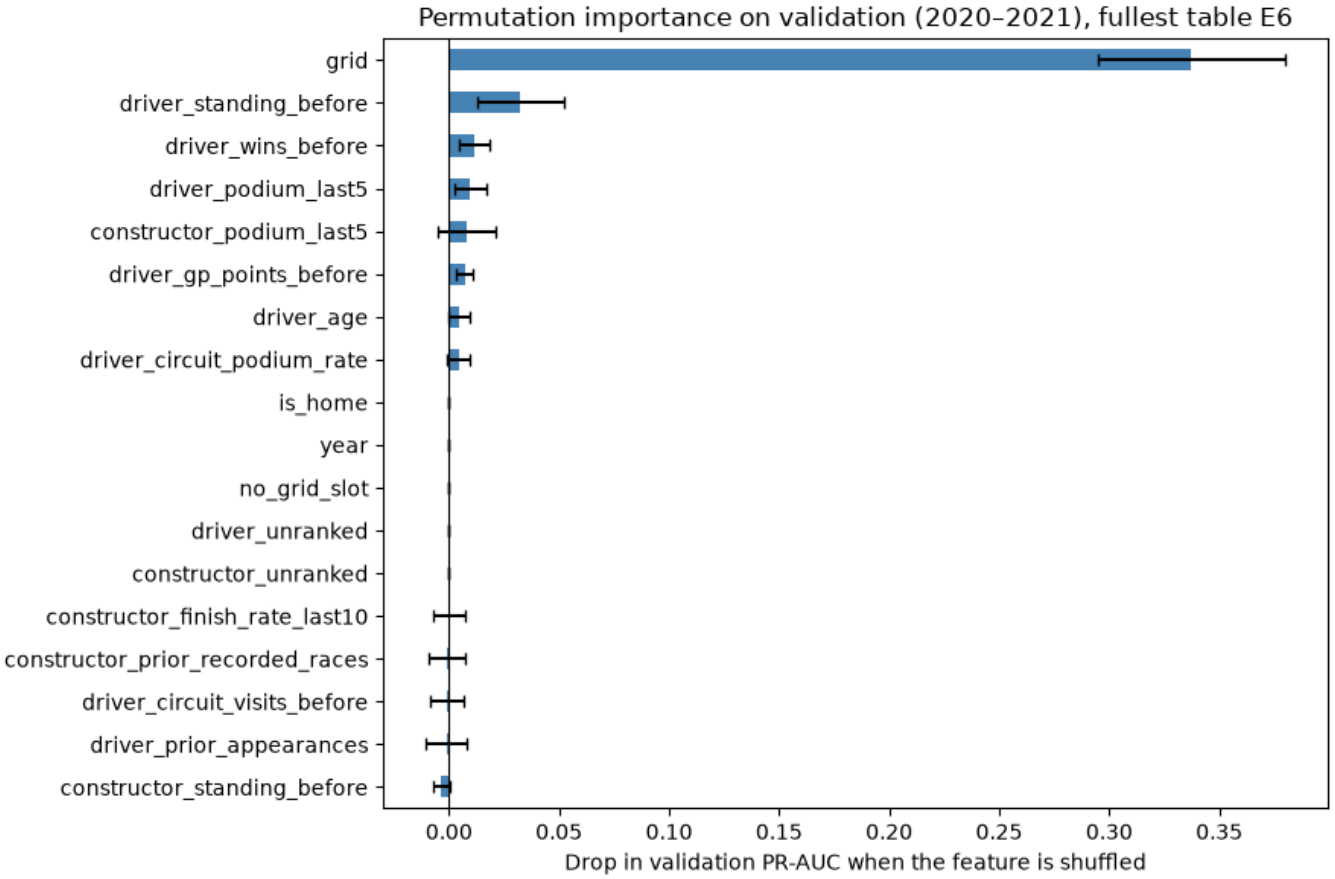
B65. 6.4 Modelling-table experiments: which features help?

Notebook cell 513 | executed output | source: acl-ms1.ipynb



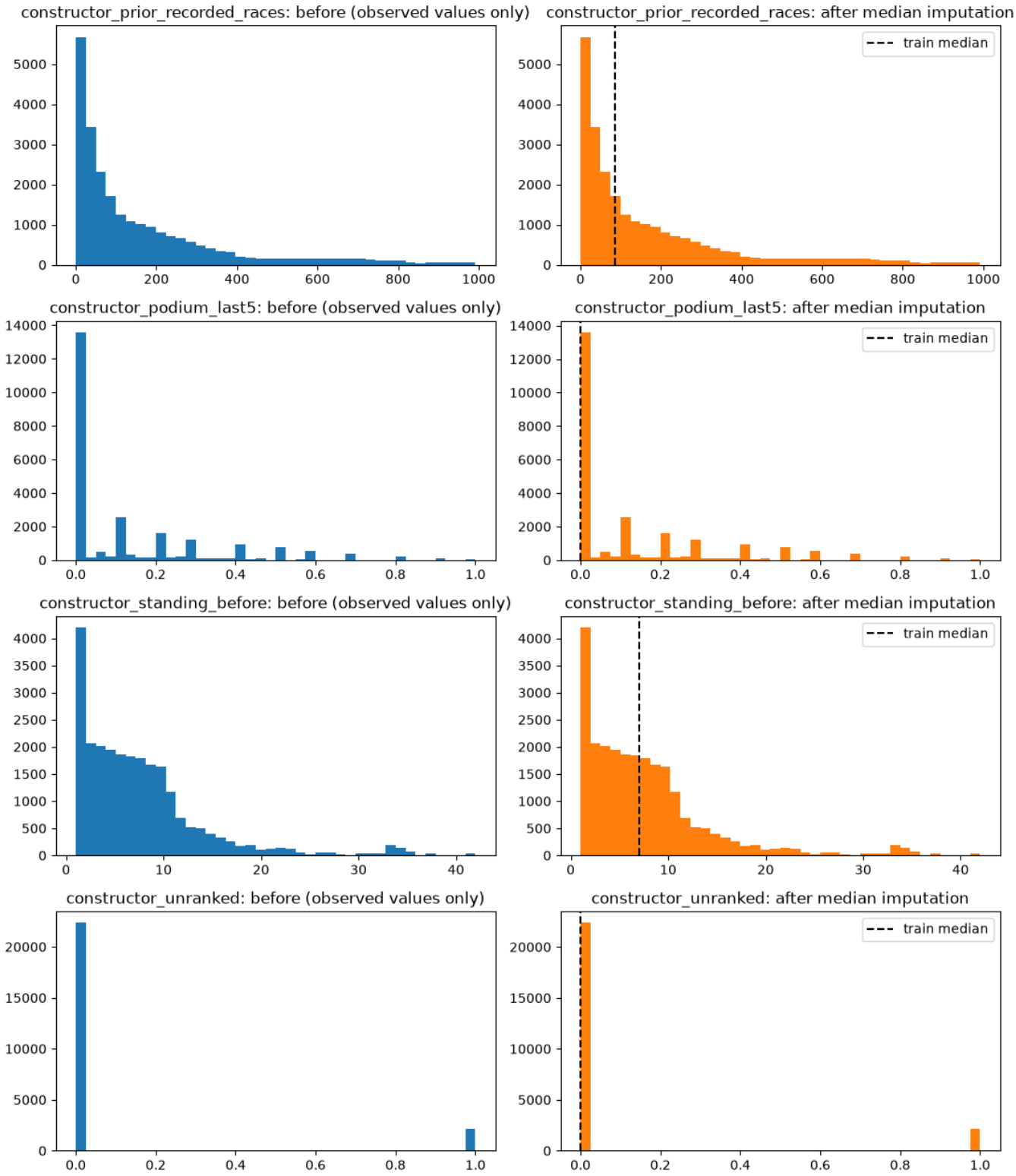
B66. 6.5 Which single features matter: permutation importance

Notebook cell 515 | executed output | source: acl-ms1.ipynb



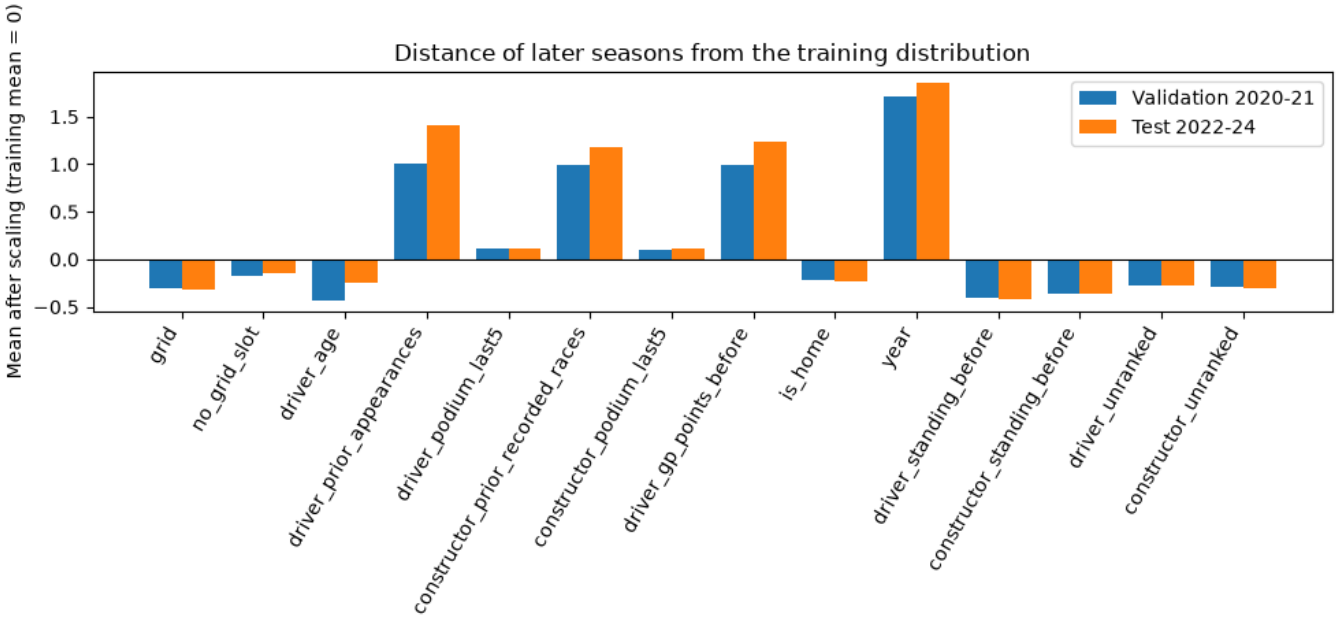
B67. 6.8.1 Imputation (median, fitted on training rows only)

Notebook cell 527 | executed output | source: acl-ms1.ipynb



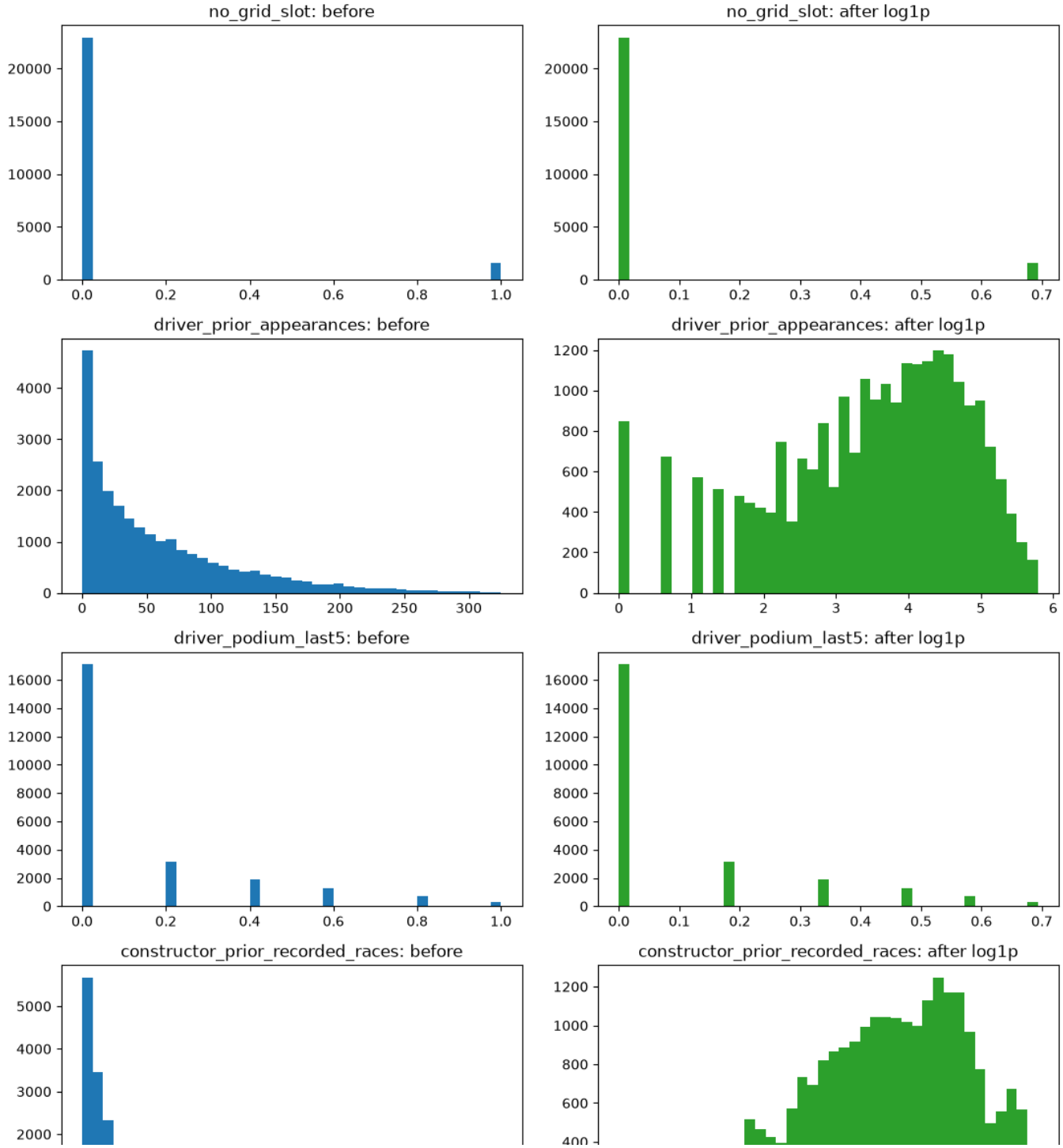
B68. 6.8.2 Scaling (`StandardScaler`, fitted on training rows only)

Notebook cell 531 | executed output | source: acl-ms1.ipynb



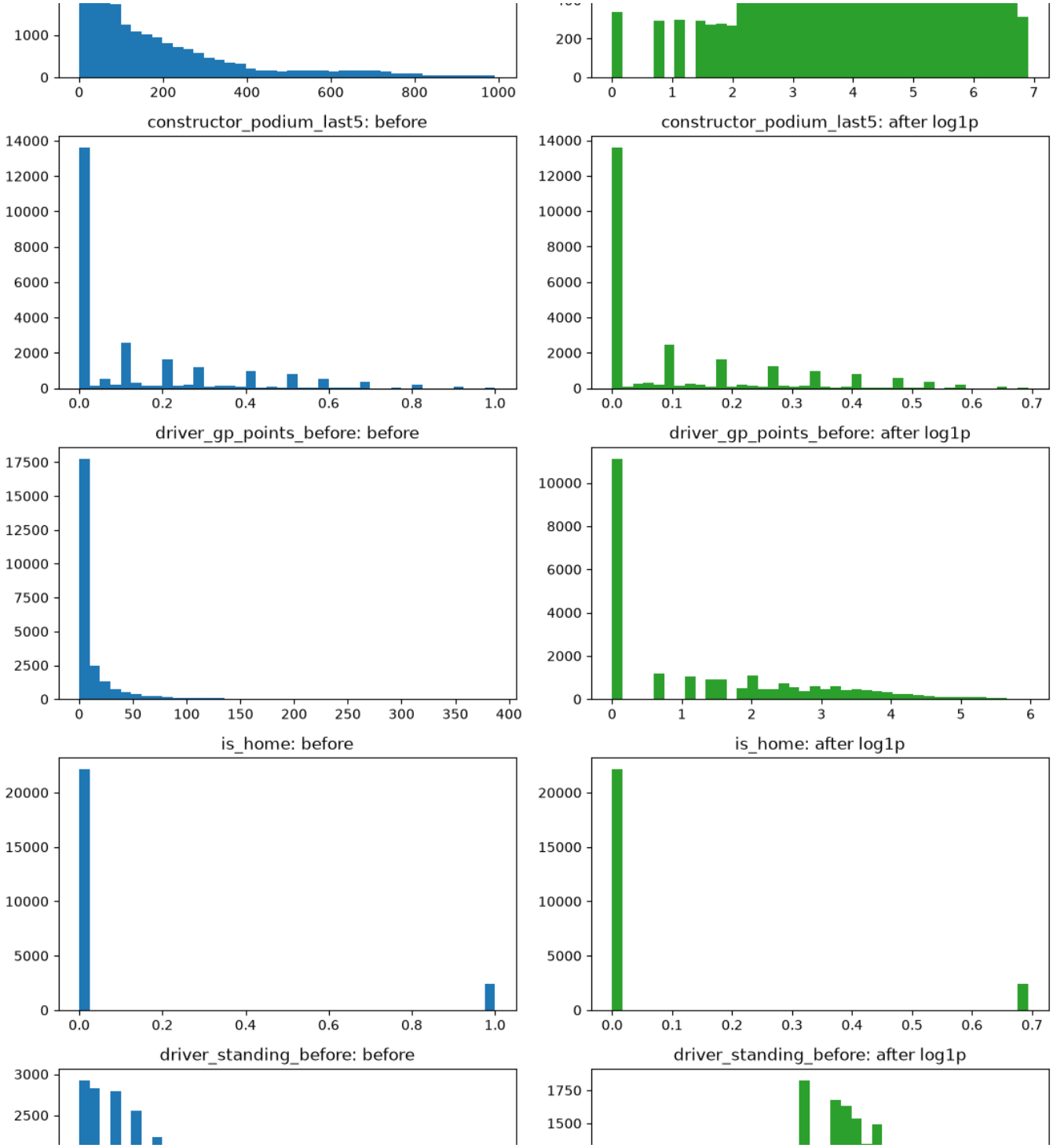
B69. 6.8.3 Skewness (part 1/3)

Notebook cell 534 | executed output | source: acl-ms1.ipynb



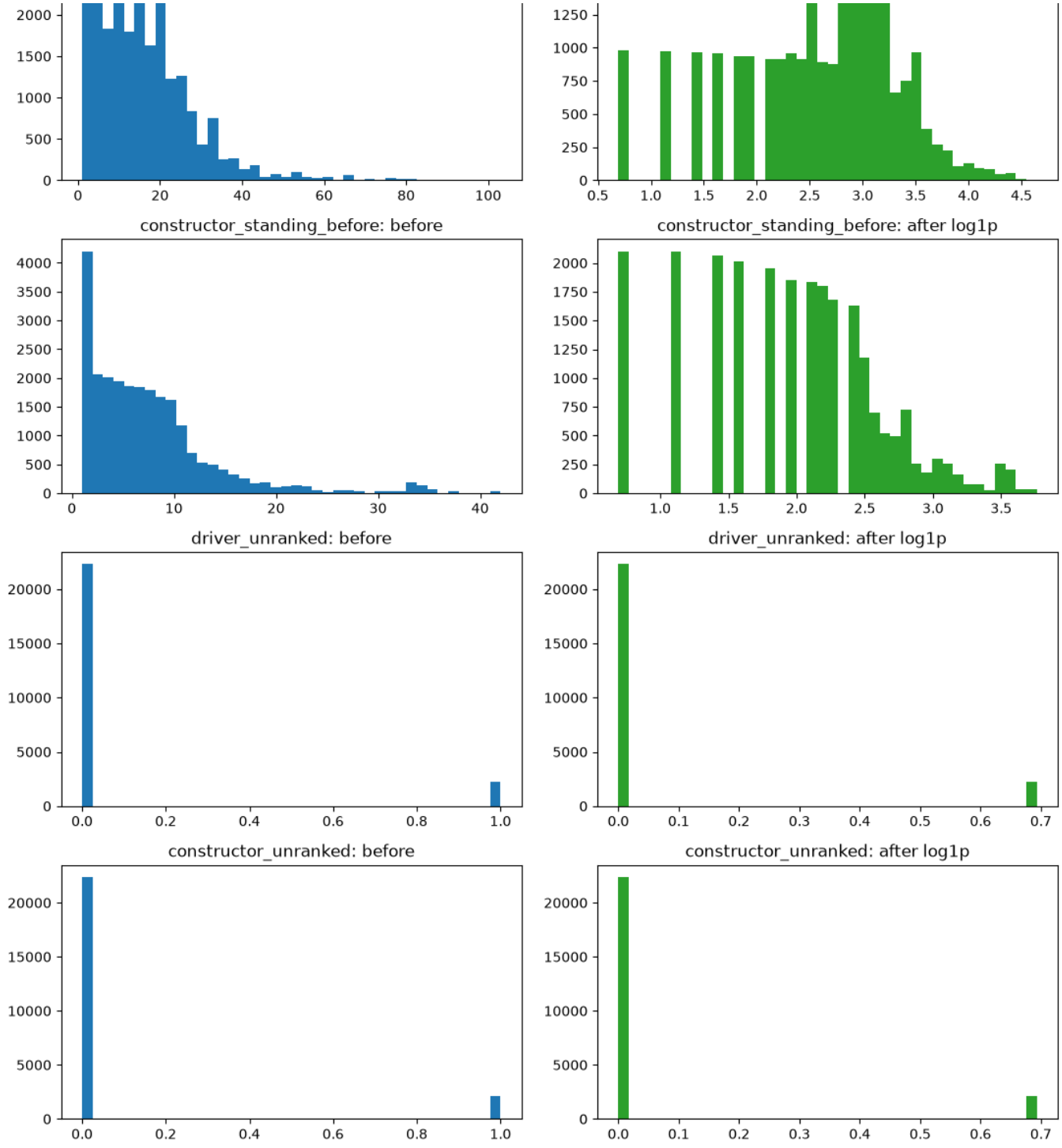
B69. 6.8.3 Skewness (part 2/3)

Notebook cell 534 | executed output | source: acl-ms1.ipynb



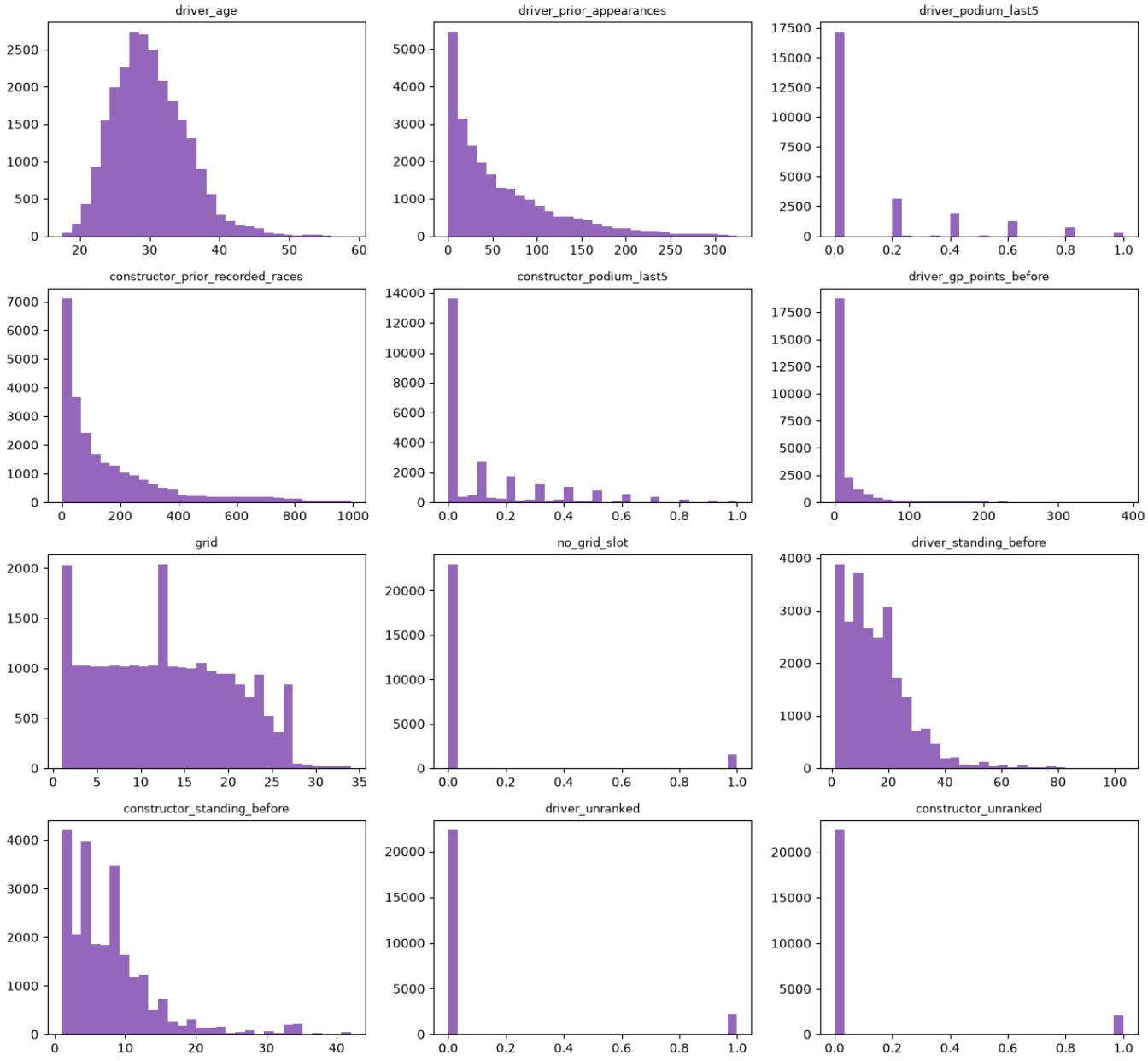
B69. 6.8.3 Skewness (part 3/3)

Notebook cell 534 | executed output | source: acl-ms1.ipynb



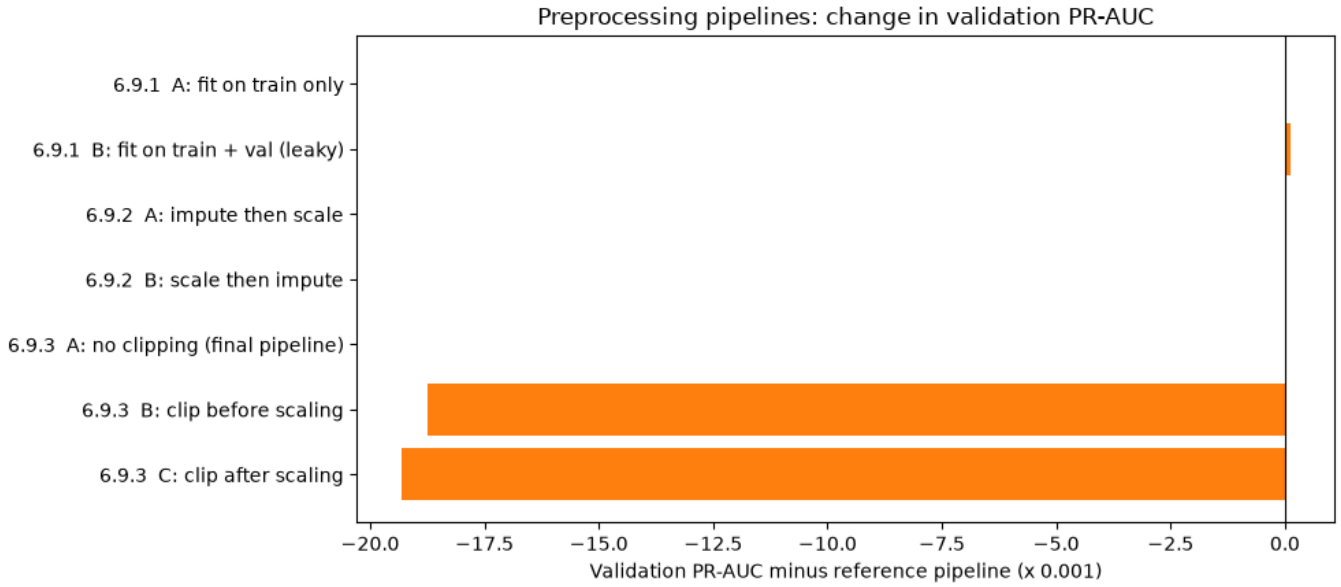
B70. 6.8.4 Engineered features: range, distribution and the past-only check

Notebook cell 537 | executed output | source: acl-ms1.ipynb



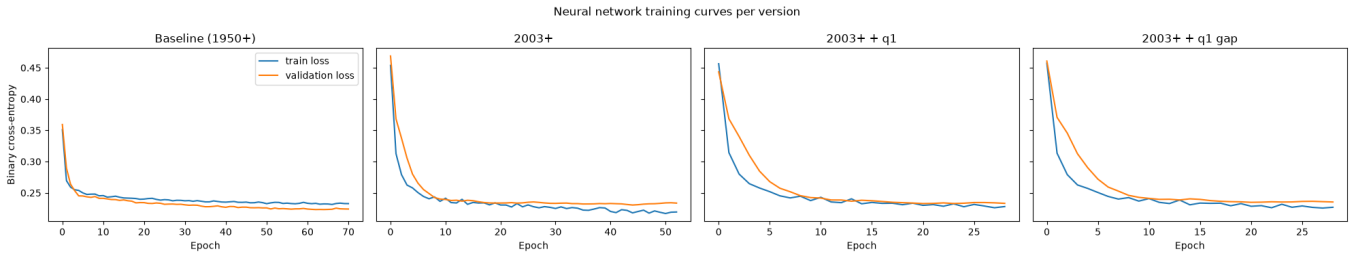
B71. 6.9.4 Summary and chosen order

Notebook cell 554 | executed output | source: acl-ms1.ipynb



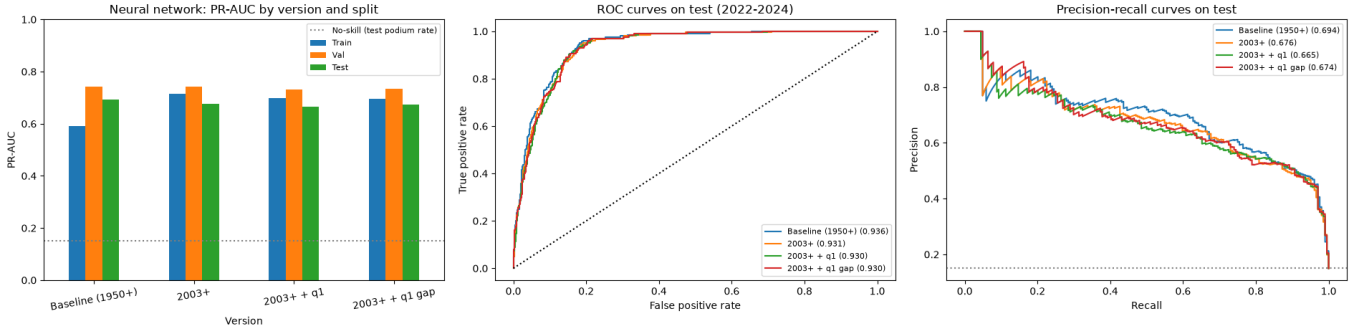
B72. 7.2 Train the four candidate versions for comparison

Notebook cell 560 | executed output | source: acl-ms1.ipynb



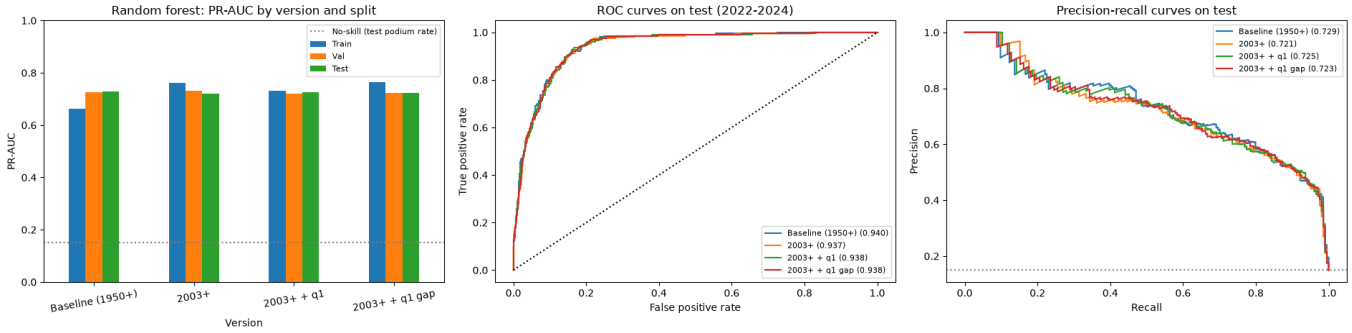
B73. 7.3 Compare the four versions

Notebook cell 562 | executed output | source: acl-ms1.ipynb



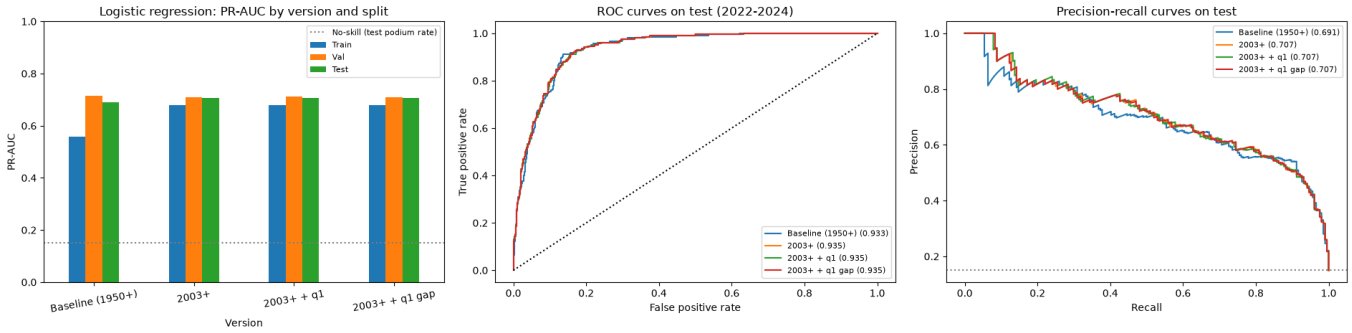
B74. 8.4 Fit the tuned forests and compare the experimental versions

Notebook cell 570 | executed output | source: acl-ms1.ipynb



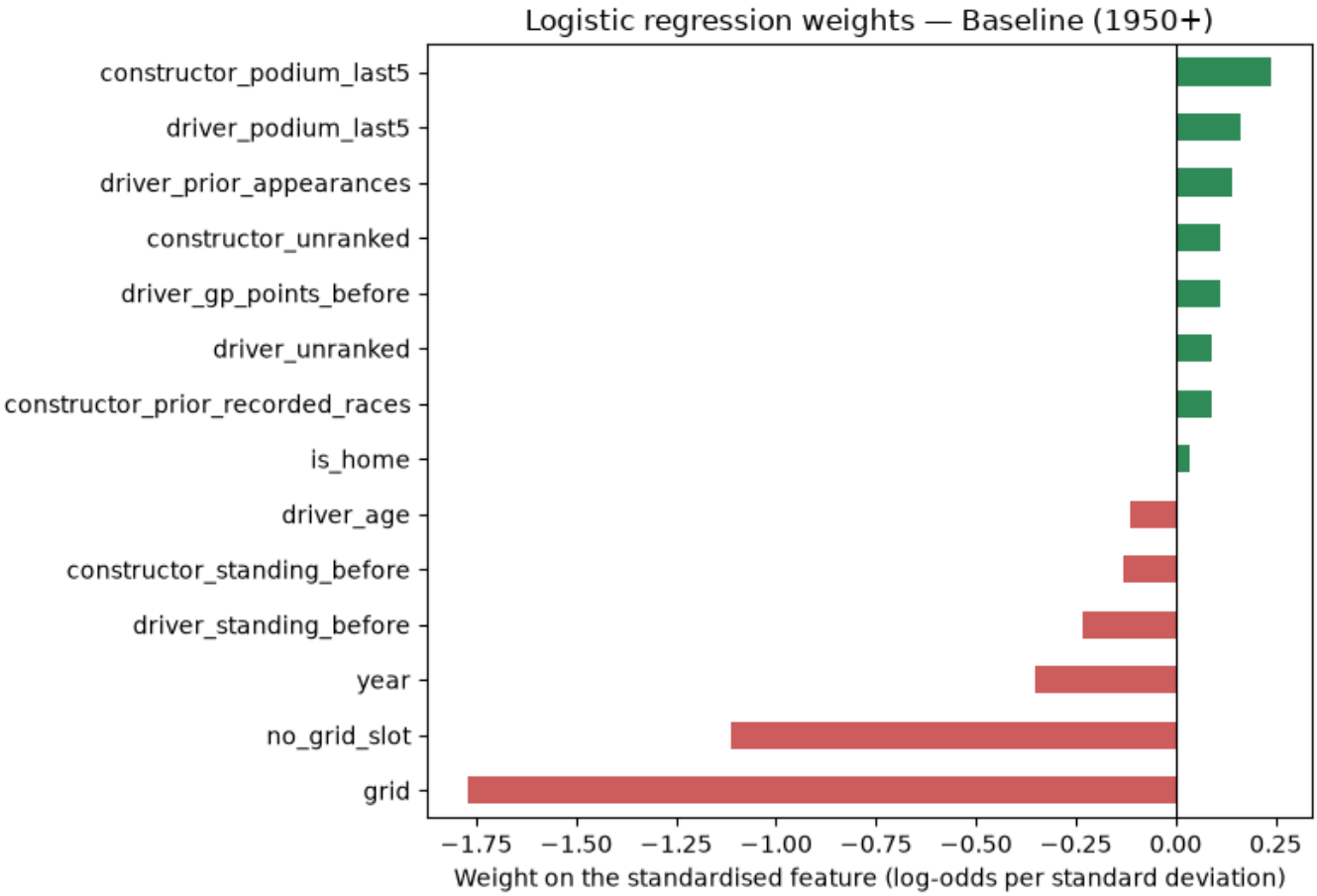
B75. 9.3 Fit the tuned models and compare the experimental versions

Notebook cell 576 | executed output | source: acl-ms1.ipynb



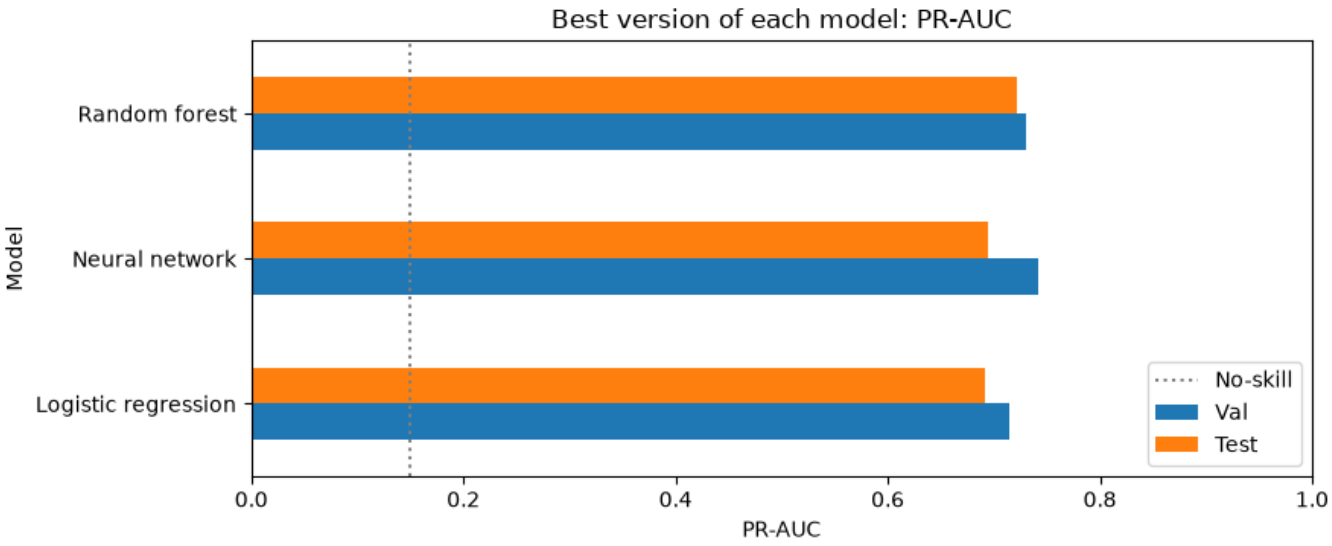
B76. 9.3 Fit the tuned models and compare the experimental versions

Notebook cell 577 | executed output | source: acl-ms1.ipynb



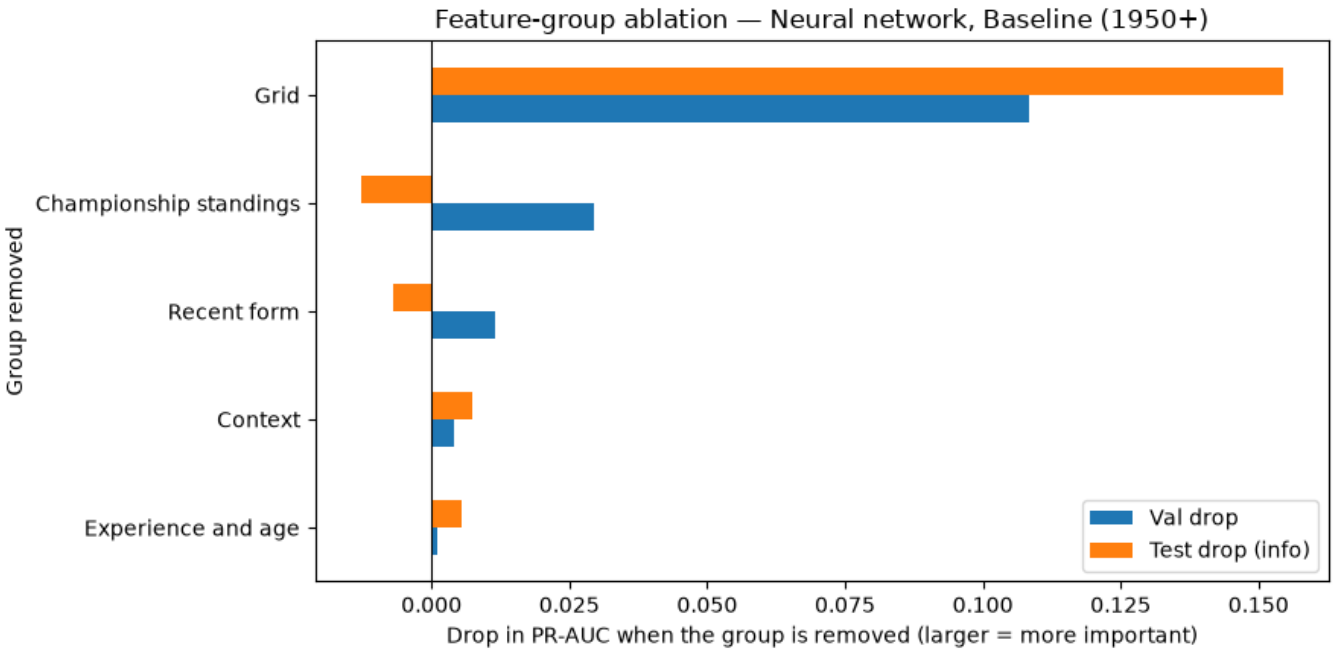
B77. 10.1 Best experimental version of each model

Notebook cell 581 | executed output | source: acl-ms1.ipynb



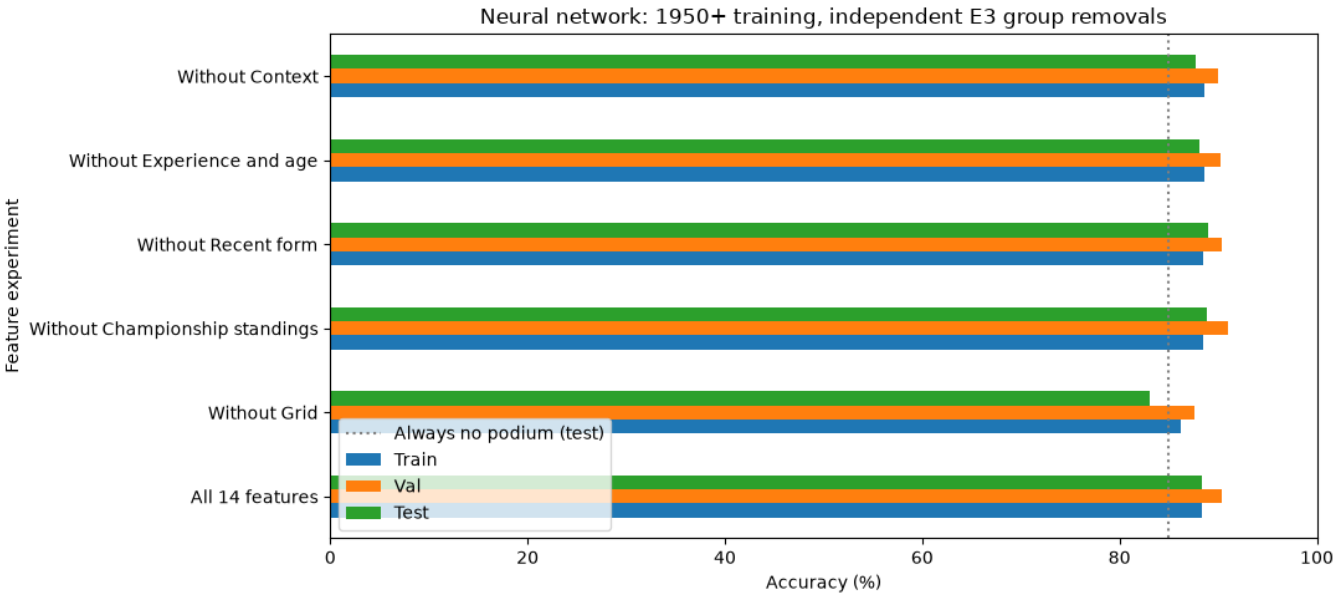
B78. 10.2 Feature-group ablation

Notebook cell 583 | executed output | source: acl-ms1.ipynb



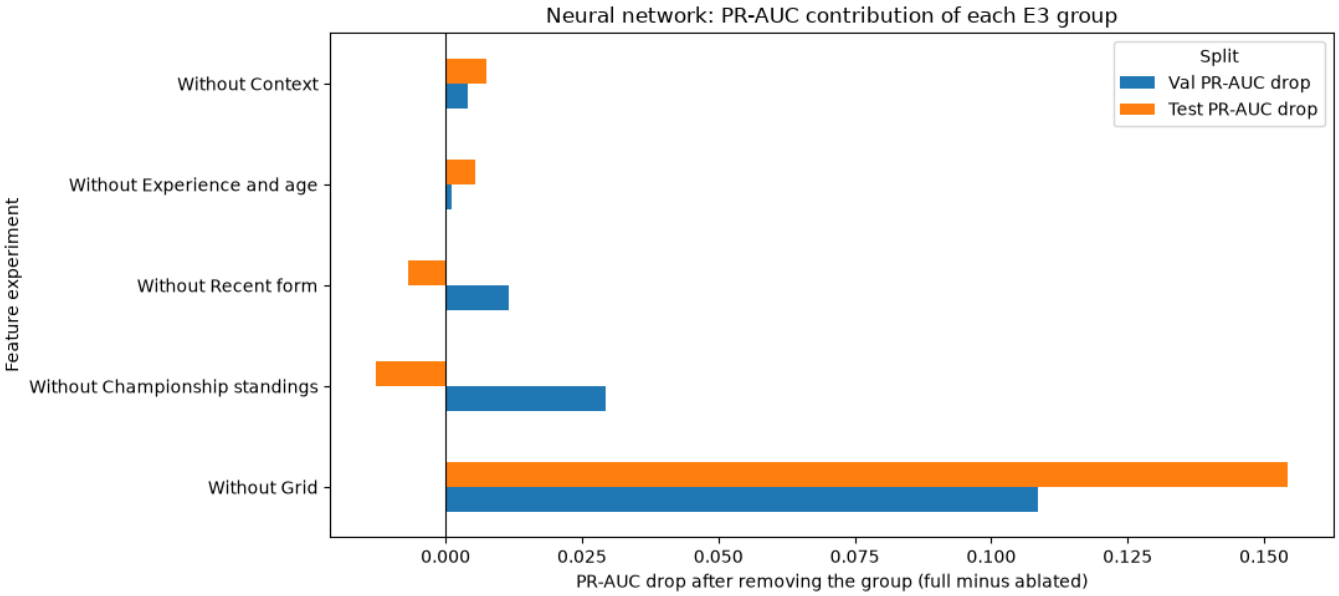
B79. 11.2 Neural network: final training and ablation

Notebook cell 590 | executed output | source: acl-ms1.ipynb



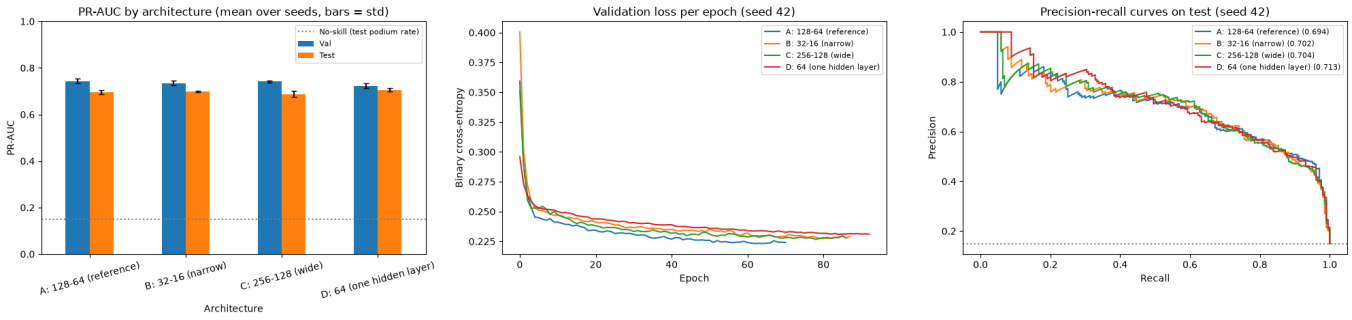
B80. 11.2 Neural network: final training and ablation

Notebook cell 590 | executed output | source: acl-ms1.ipynb



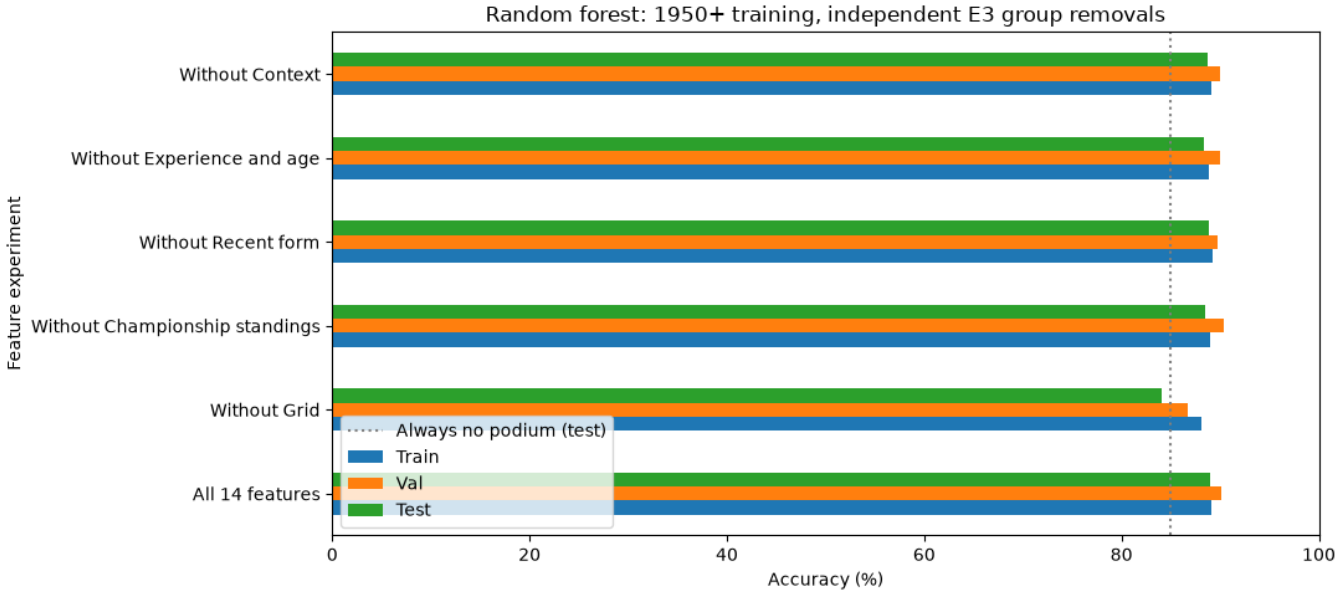
B81. 11.2.1 Neural network: comparing four architectures

Notebook cell 593 | executed output | source: acl-ms1.ipynb



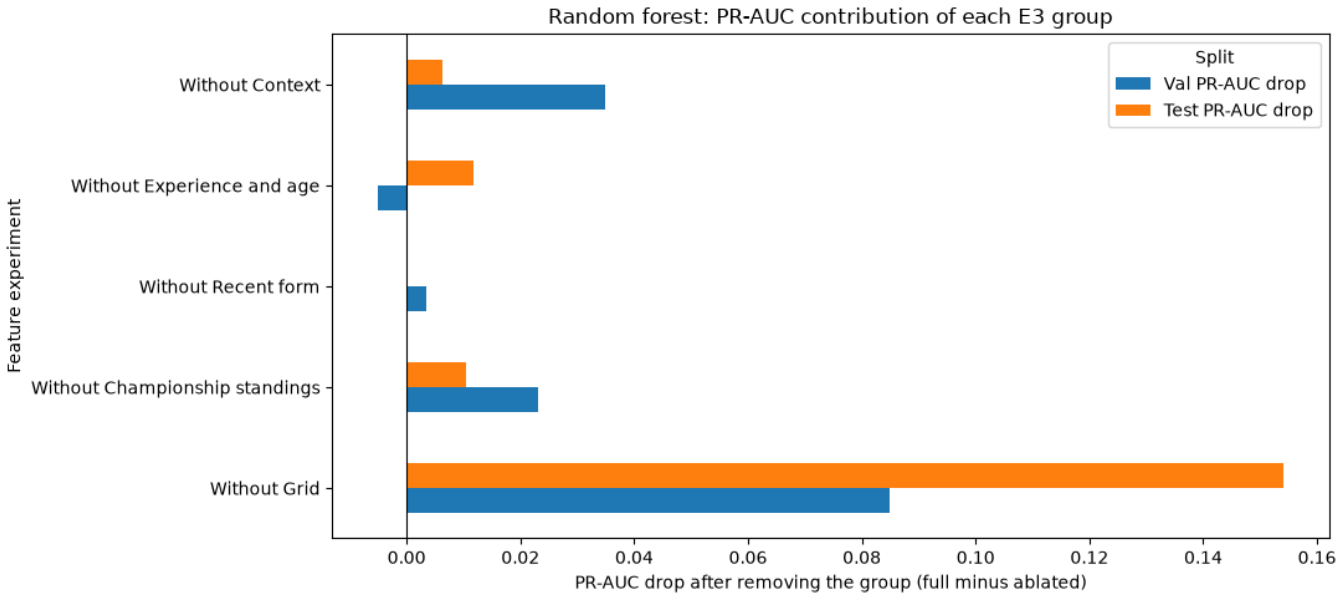
B82. 11.3 Random forest: final training and ablation

Notebook cell 598 | executed output | source: acl-ms1.ipynb



B83. 11.3 Random forest: final training and ablation

Notebook cell 598 | executed output | source: acl-ms1.ipynb



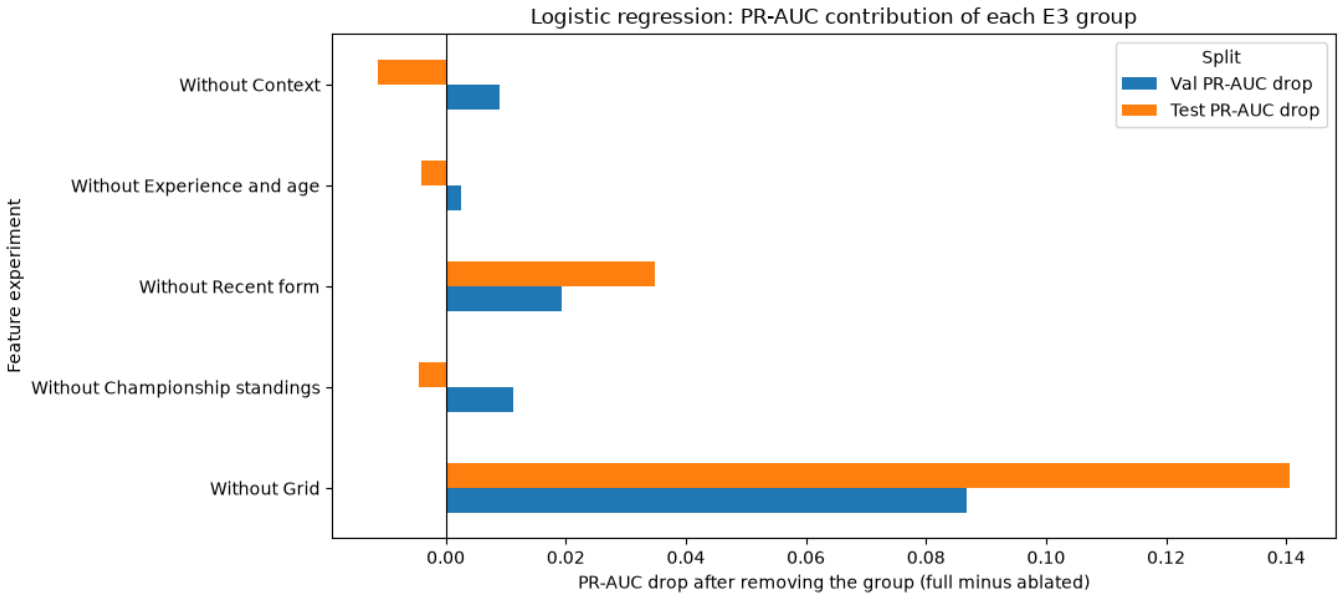
B84. 11.4 Logistic regression: final training and ablation

Notebook cell 601 | executed output | source: acl-ms1.ipynb



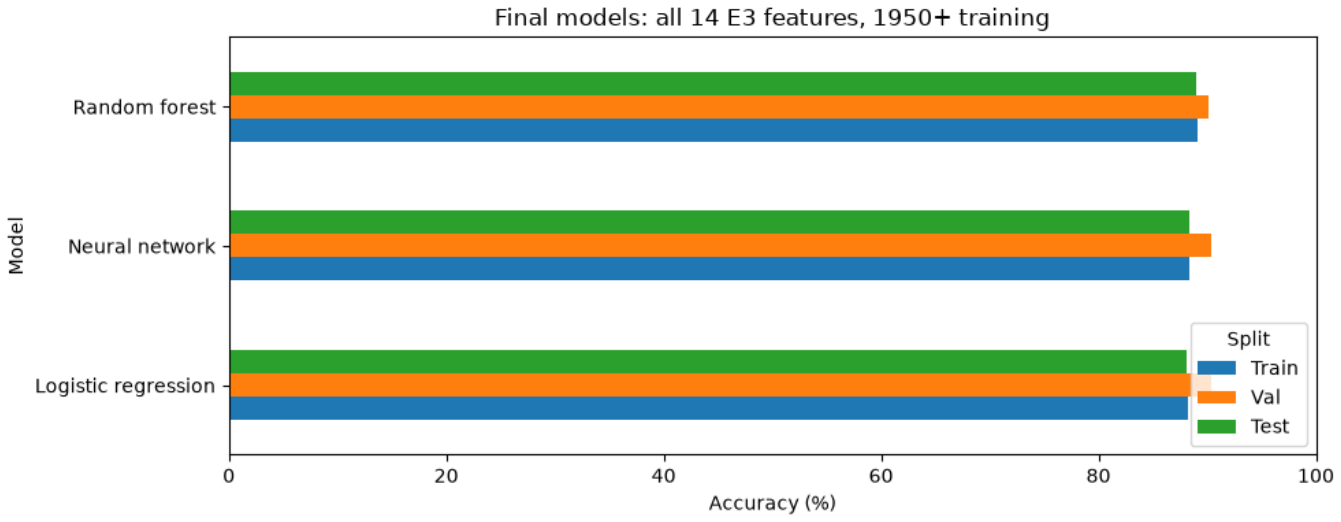
B85. 11.4 Logistic regression: final training and ablation

Notebook cell 601 | executed output | source: acl-ms1.ipynb



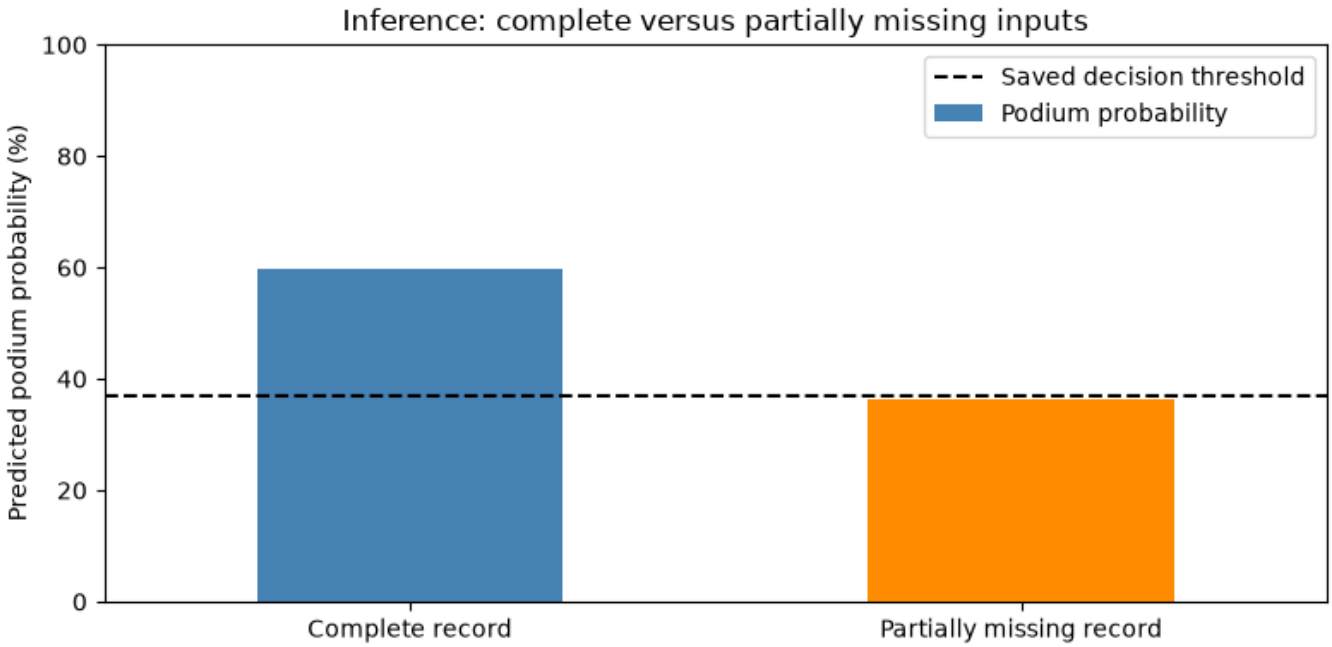
B86. 11.5 Compare the three full-feature final models

Notebook cell 603 | executed output | source: acl-ms1.ipynb



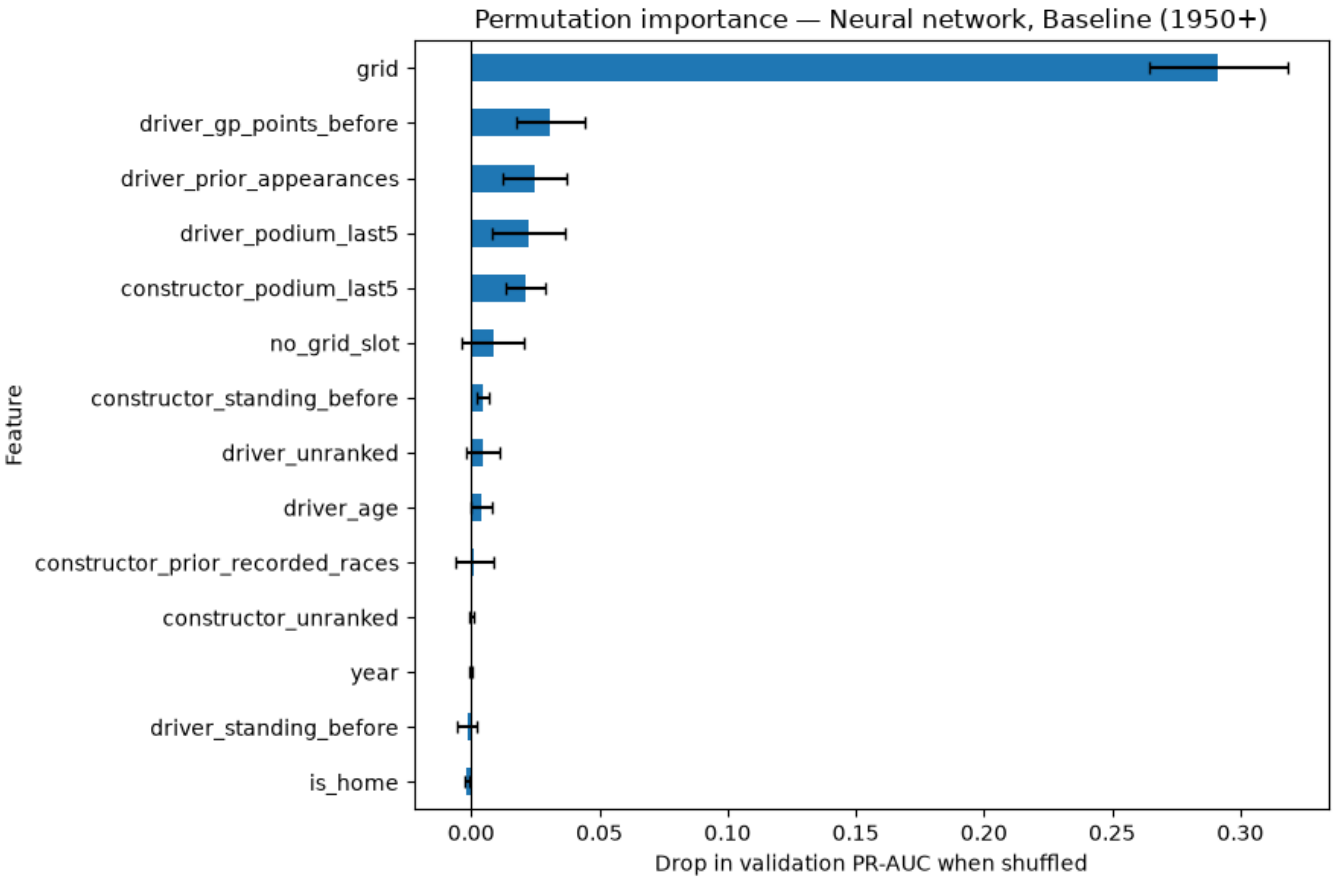
B87. Example B - the same record with two inputs missing

Notebook cell 609 | executed output | source: acl-ms1.ipynb



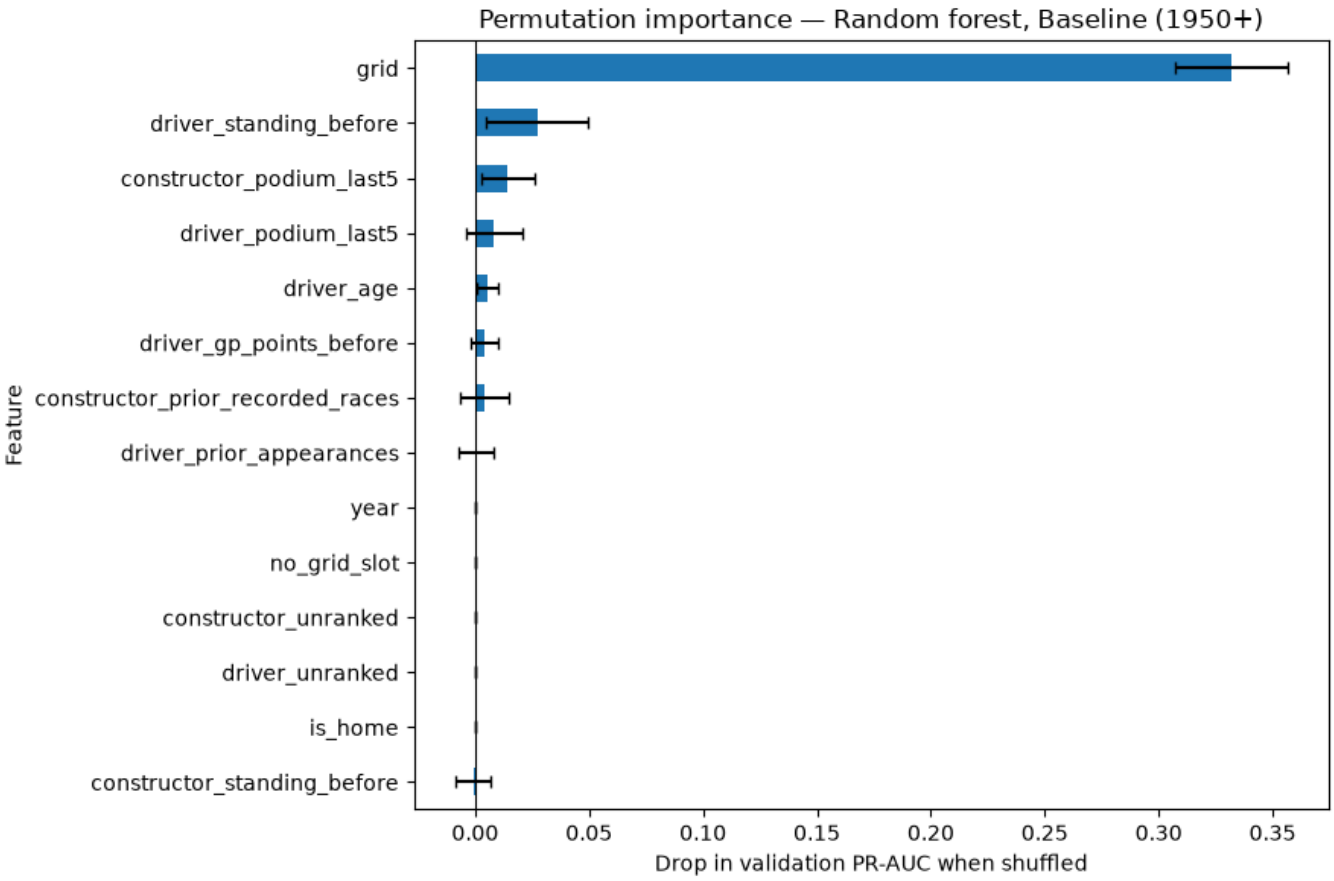
B88. 12.1 Global explanation: permutation importance for each model

Notebook cell 613 | executed output | source: acl-ms1.ipynb



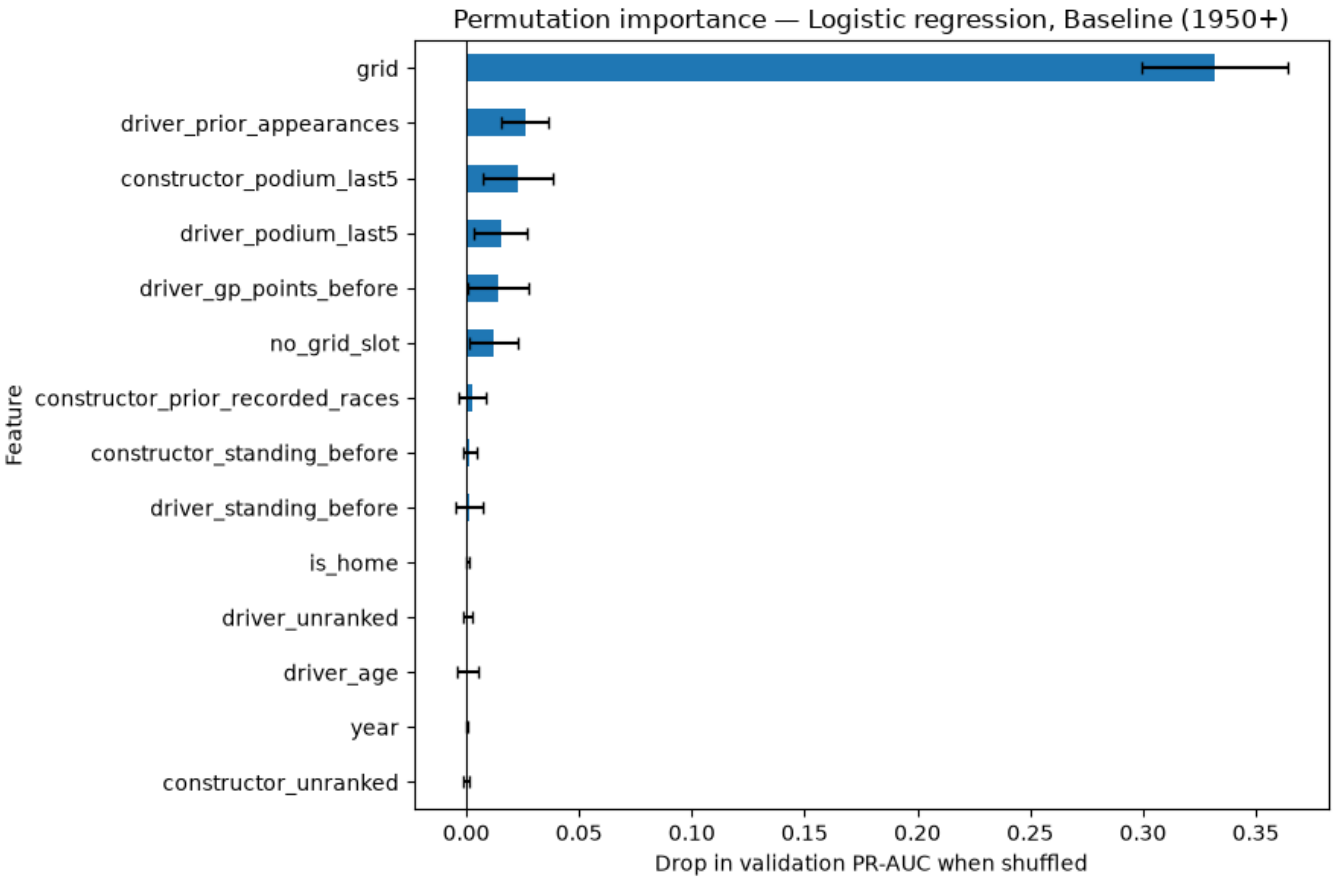
B89. 12.1 Global explanation: permutation importance for each model

Notebook cell 613 | executed output | source: acl-ms1.ipynb



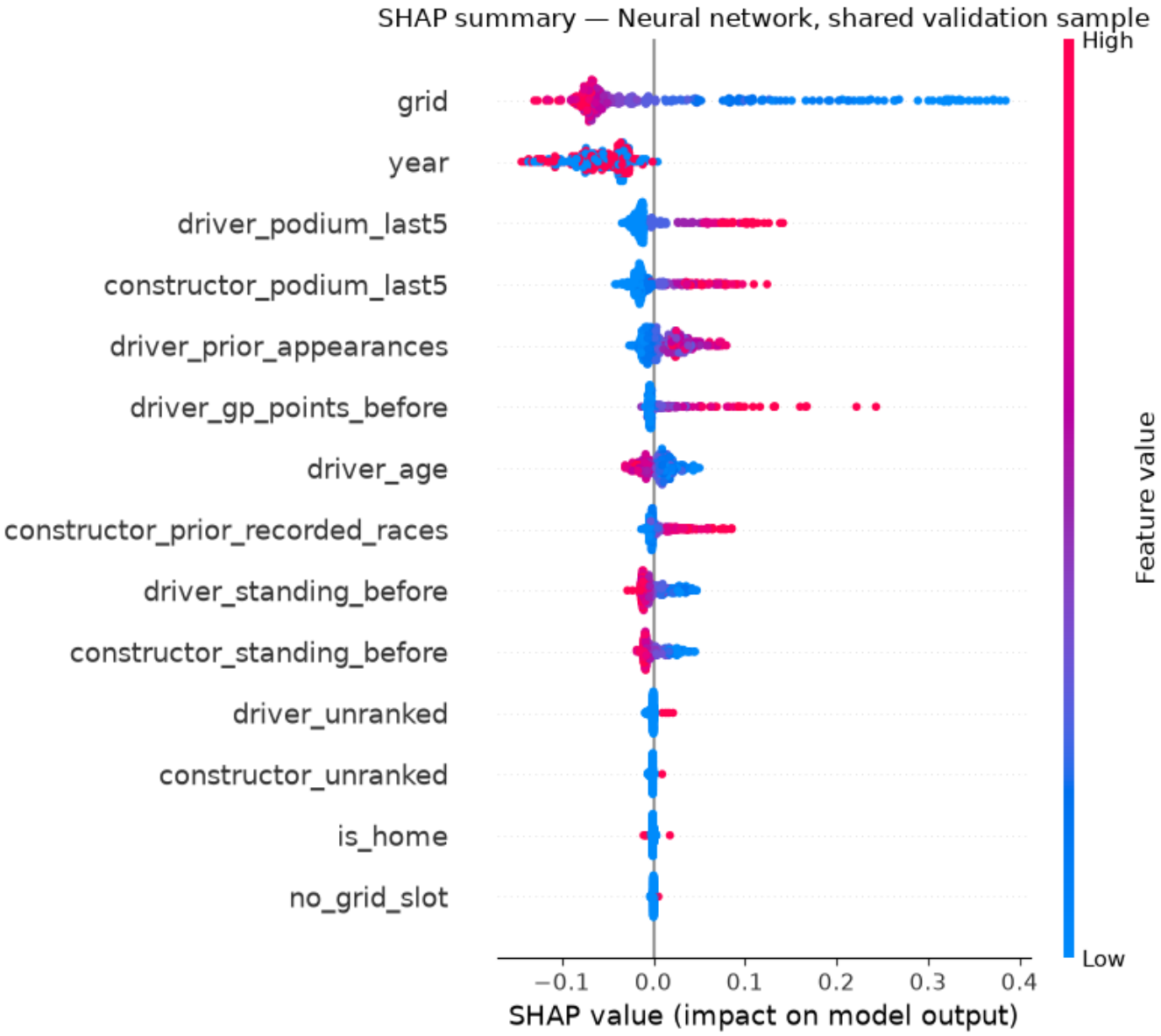
B90. 12.1 Global explanation: permutation importance for each model

Notebook cell 613 | executed output | source: acl-ms1.ipynb



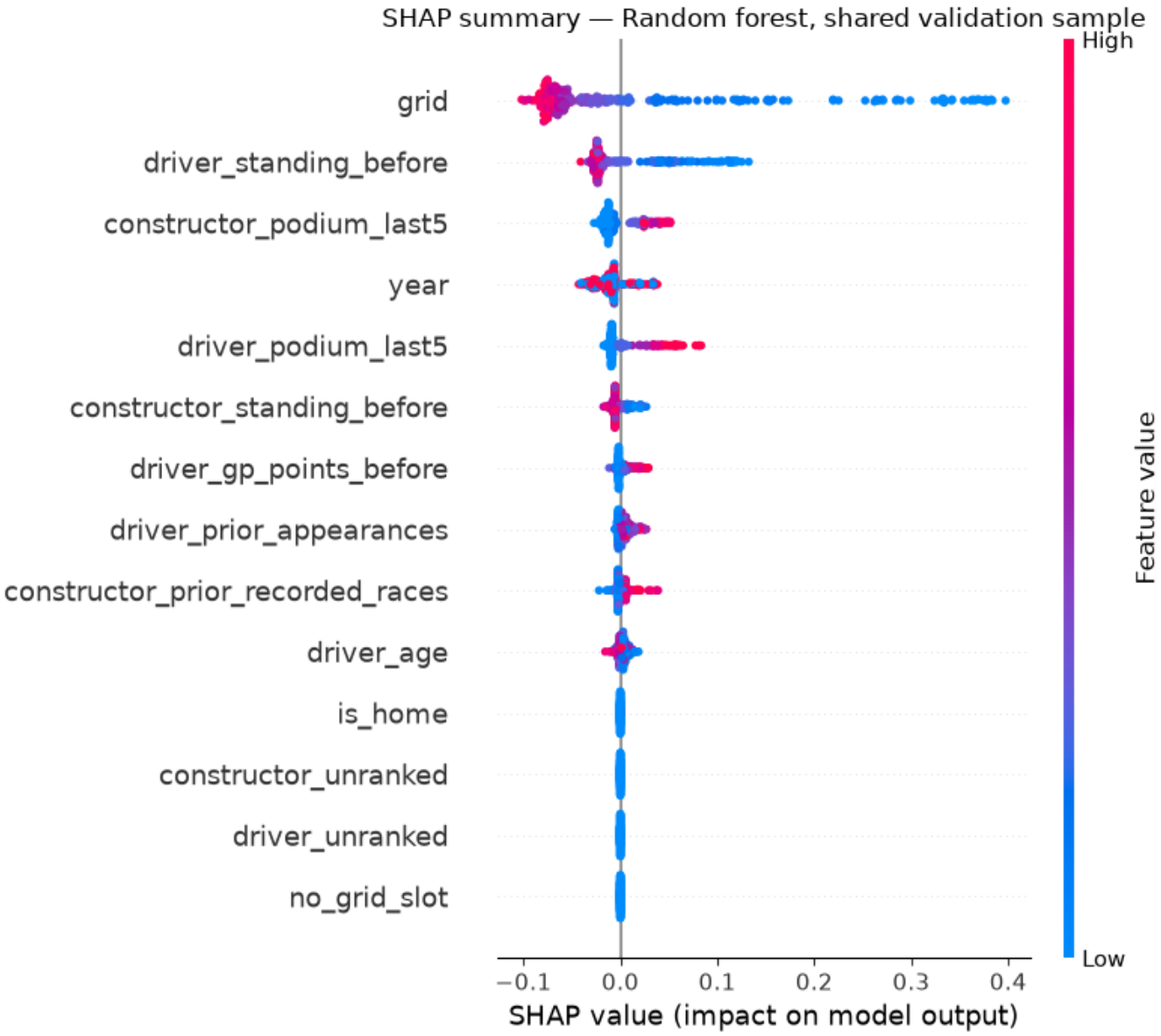
B91. 12.2 Global explanation: SHAP for each model

Notebook cell 615 | executed output | source: acl-ms1.ipynb



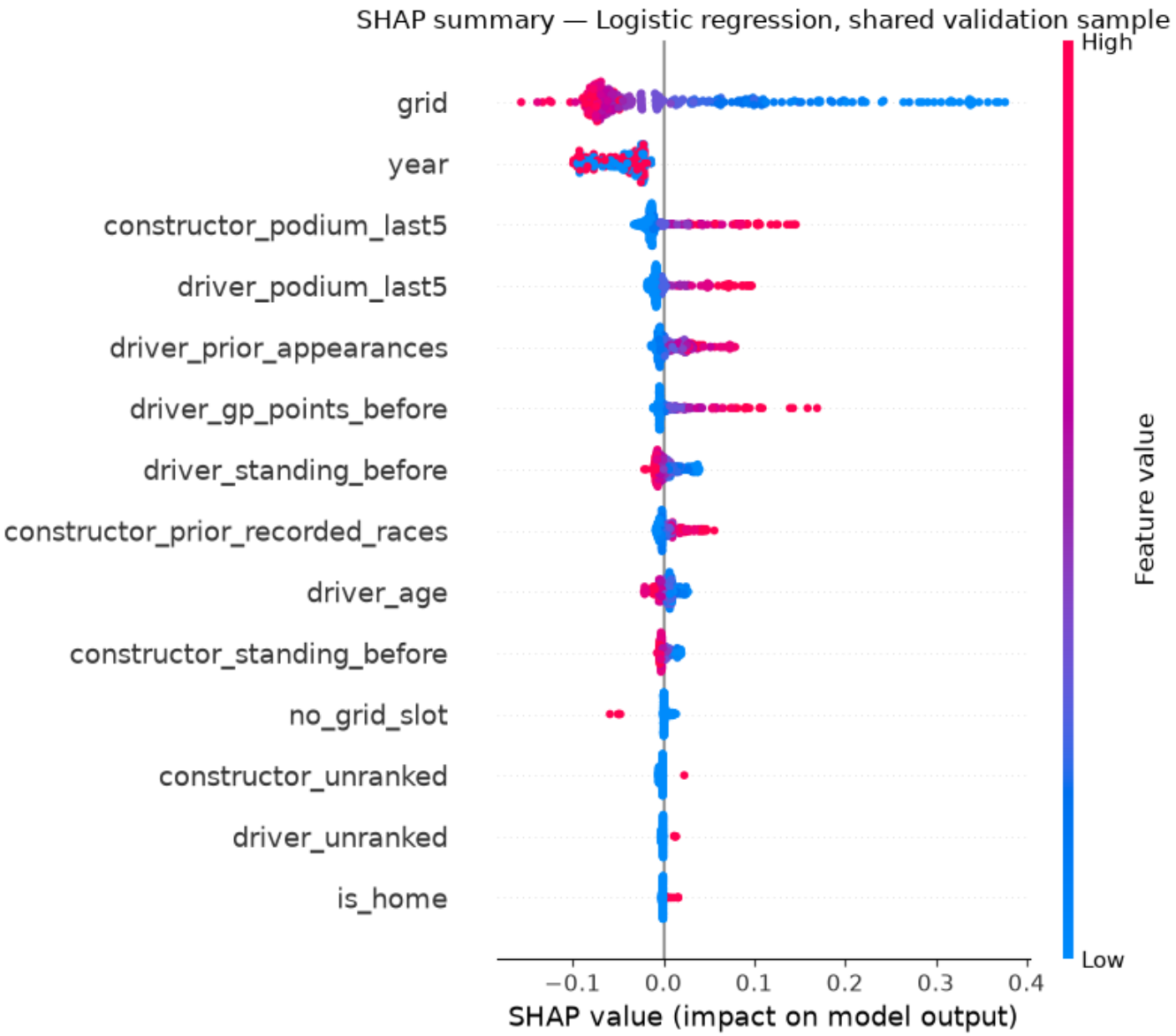
B92. 12.2 Global explanation: SHAP for each model

Notebook cell 615 | executed output | source: acl-ms1.ipynb



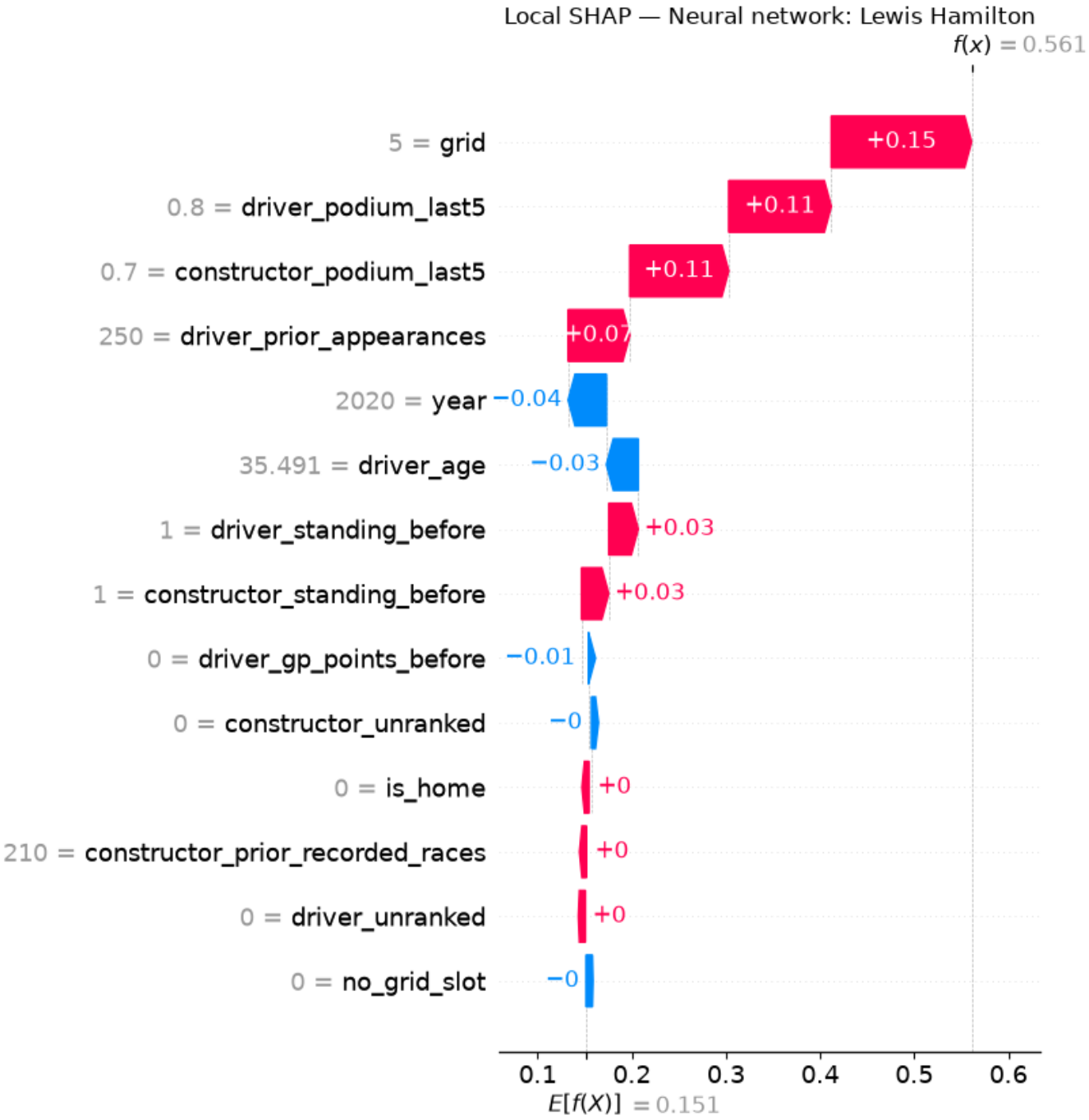
B93. 12.2 Global explanation: SHAP for each model

Notebook cell 615 | executed output | source: acl-ms1.ipynb



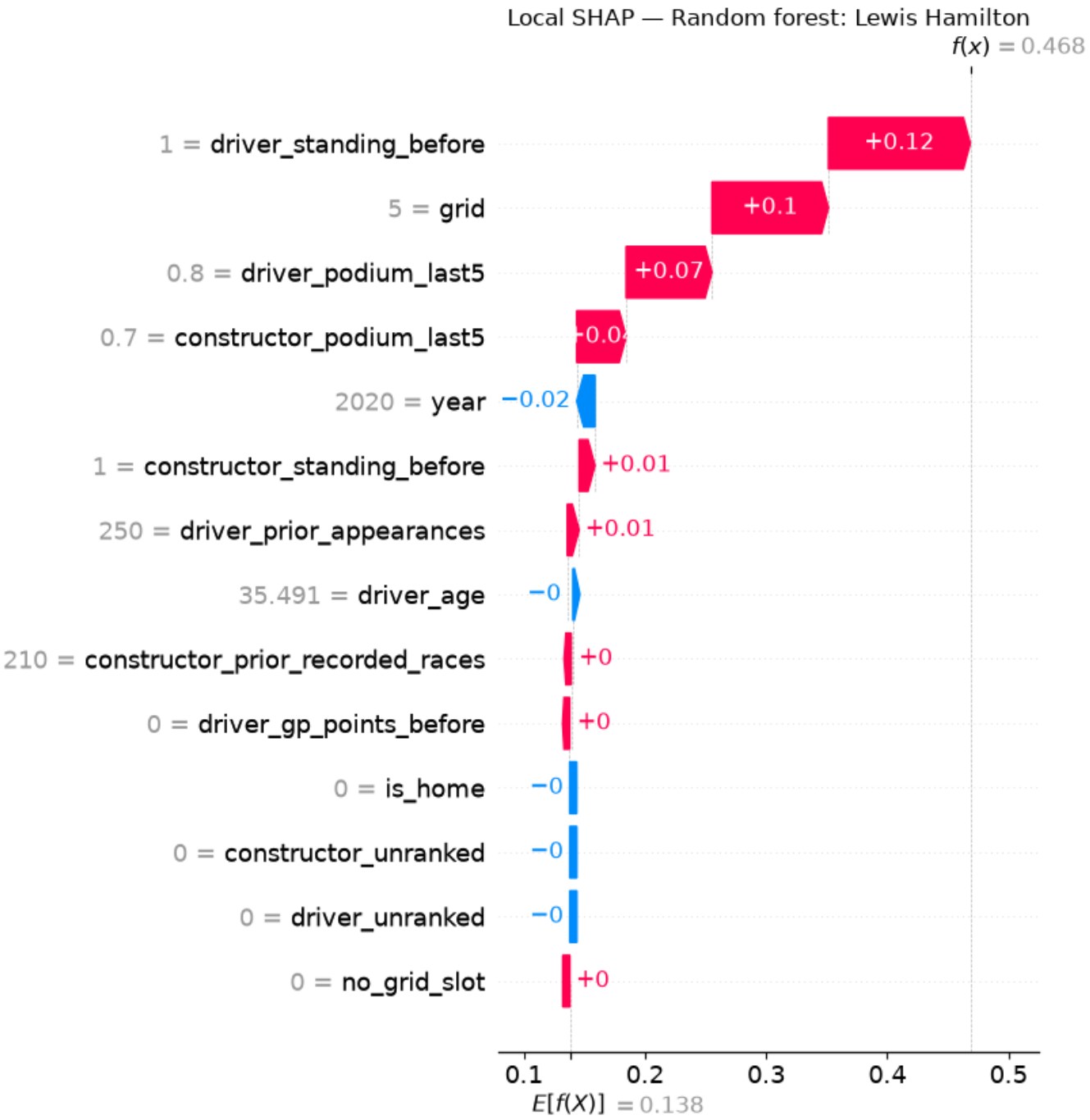
B94. 12.3 Local SHAP: the same driver-race prediction for all models

Notebook cell 617 | executed output | source: acl-ms1.ipynb



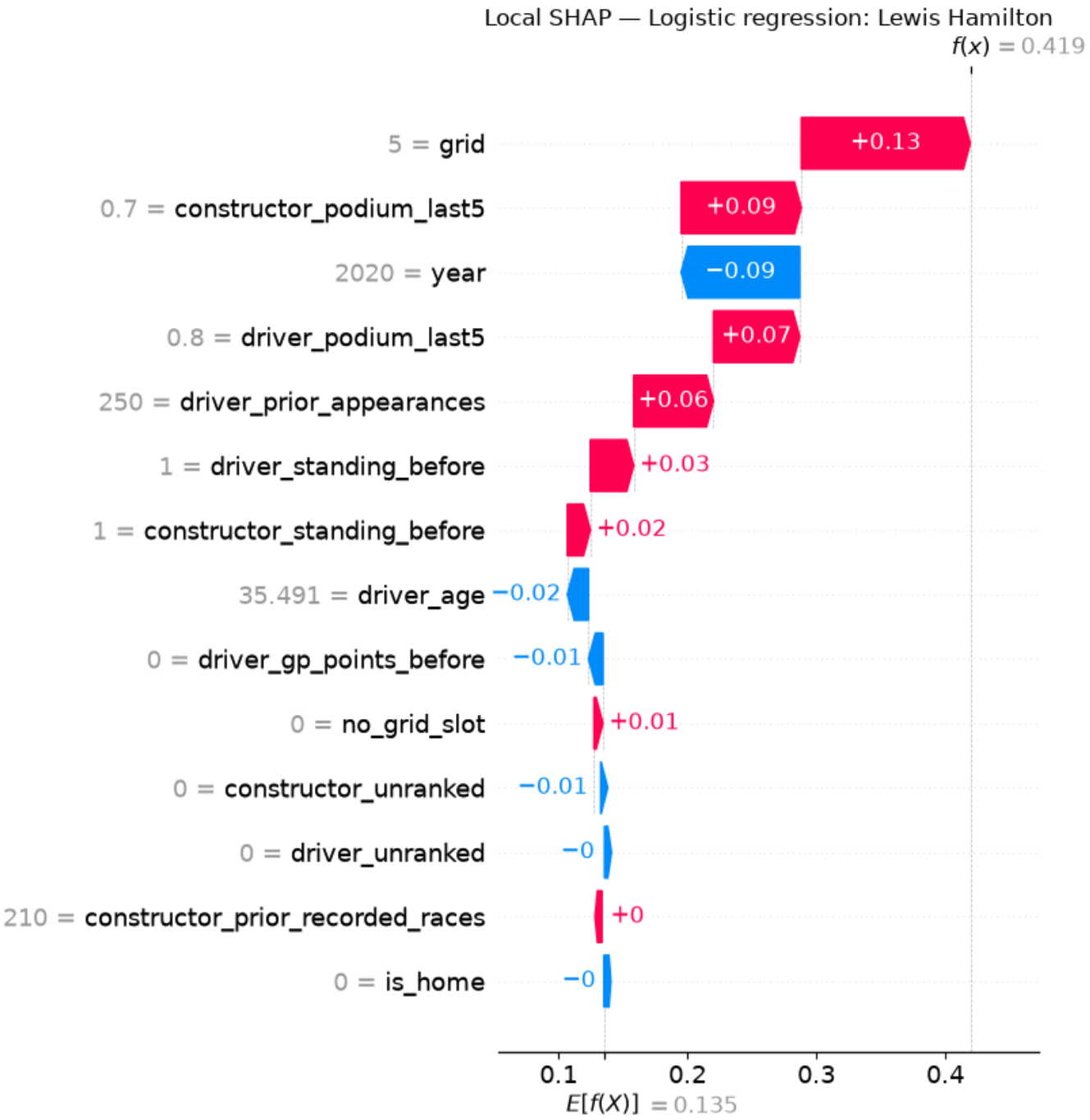
B95. 12.3 Local SHAP: the same driver-race prediction for all models

Notebook cell 617 | executed output | source: acl-ms1.ipynb



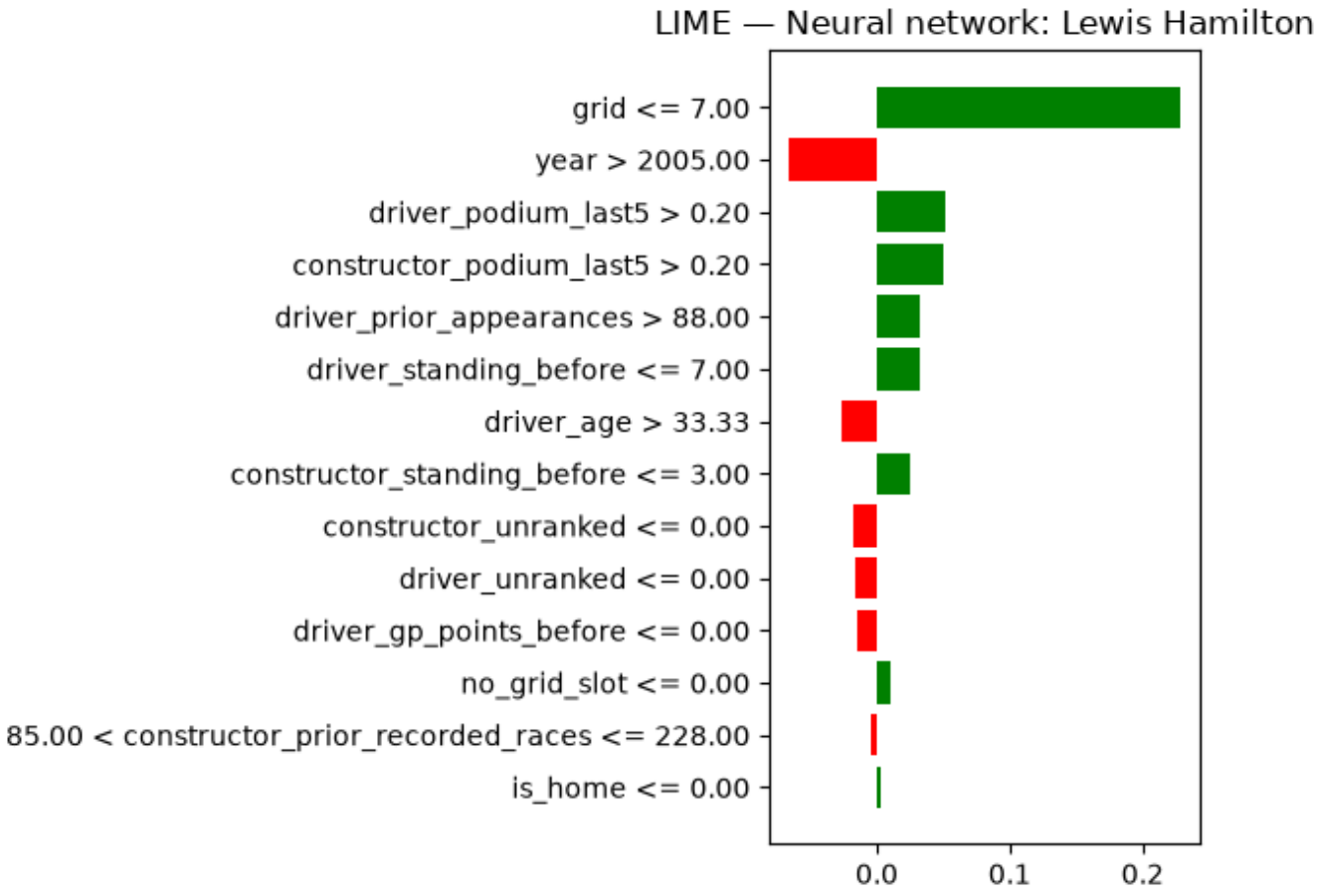
B96. 12.3 Local SHAP: the same driver-race prediction for all models

Notebook cell 617 | executed output | source: acl-ms1.ipynb



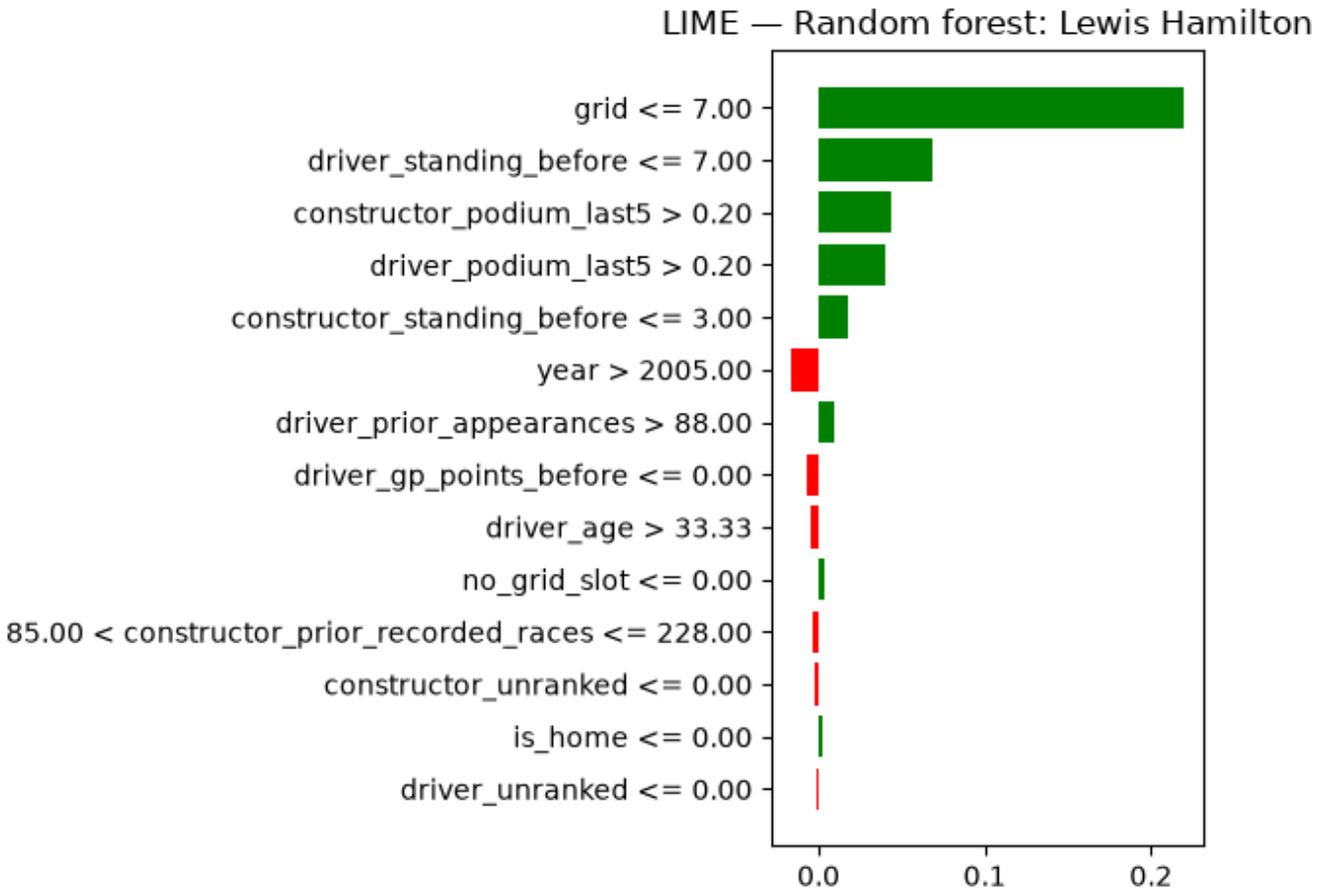
B97. 12.4 Local LIME: the same prediction for each model

Notebook cell 619 | executed output | source: acl-ms1.ipynb



B98. 12.4 Local LIME: the same prediction for each model

Notebook cell 619 | executed output | source: acl-ms1.ipynb



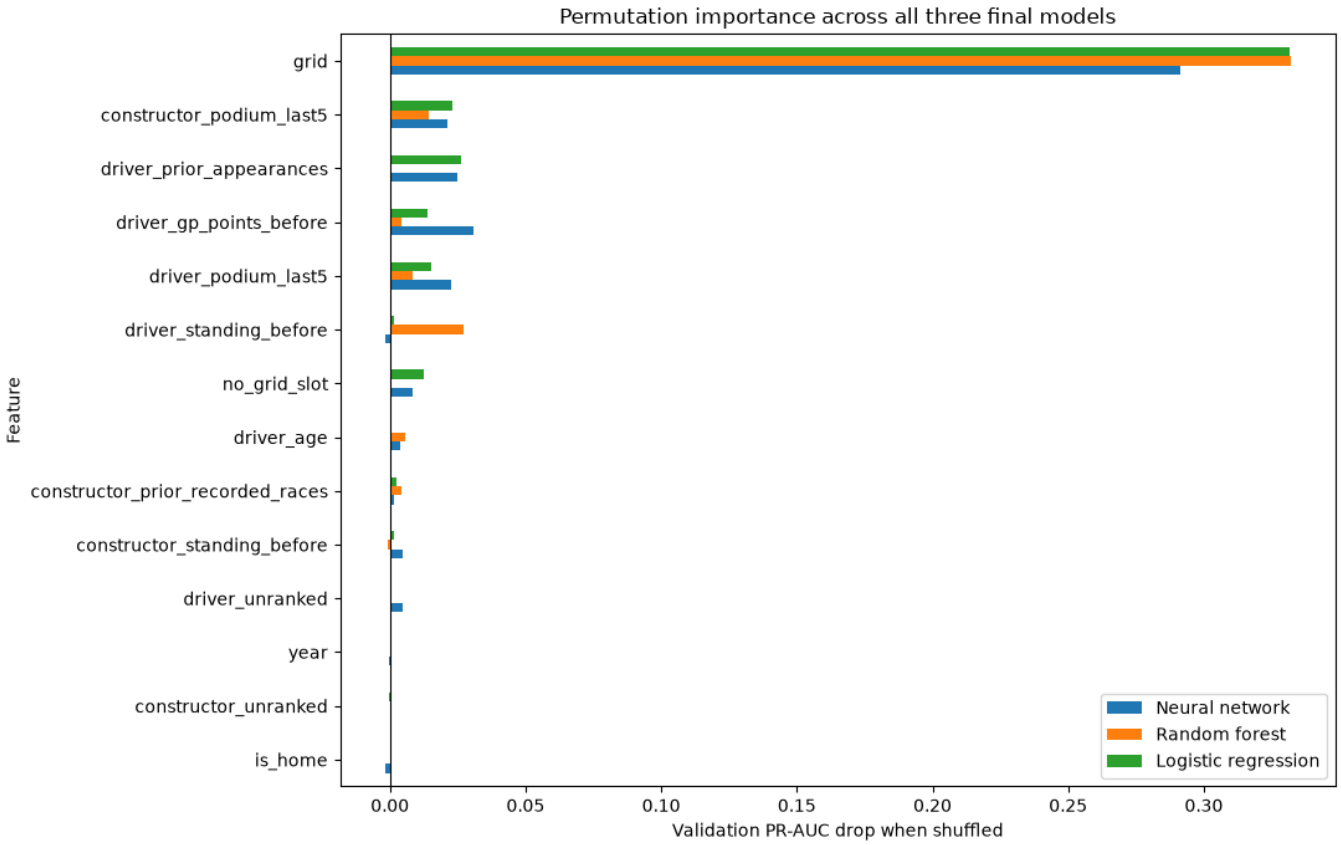
B99. 12.4 Local LIME: the same prediction for each model

Notebook cell 619 | executed output | source: acl-ms1.ipynb



B100. 12.5 Compare and interpret all three models

Notebook cell 621 | executed output | source: acl-ms1.ipynb



B101. 12.5 Compare and interpret all three models

Notebook cell 621 | executed output | source: acl-ms1.ipynb

